

Spatial scale space from hemigroup axioms: characterization of the kernels on the line

Daniel Fagerström

September 15, 2026 (v0.1)

Abstract

We characterize the spatial scale spaces on the line: the families of smoothing operators through which an uncommitted observer can measure a signal at every scale, under translation and reflection symmetry. The axioms are those of the extended class of Pauwels, Van Gool, Fiddelaers and Moons, with one weakened. Their recursivity, a one-parameter semigroup in scale, is stated as a two-parameter hemigroup $\Phi_{s,t}\Phi_{r,s} = \Phi_{r,t}$, which retains the composition of measurements and drops the tacit assumption that an increment depends only on the scale difference, the stationarity of the increments in the scale parameter. Two things then change. Without positivity no classification of the form proved here exists: the remaining axioms admit an infinite-dimensional family containing the Gaussian. With positivity the admissible kernels from the origin are exactly the symmetric self-decomposable laws. Each family is determined by a Gaussian coefficient and a nonincreasing function that sets the rate of displacements at every size. The Gaussian families form the one ray with a Gaussian coefficient alone, an extreme ray of the cone rather than the whole of it. Reimposing stationarity, in whichever scale parametrization the family has it, selects the symmetric stable family, the positive part of the extended class, with the bound $\alpha \leq 2$ proved from the representation. The class also contains members with all moments finite, the variance-gamma family among them, whose kernels are the Matérn covariance functions for smoothness above zero. The kernels from the origin are absolutely continuous and unimodal. The contribution is the derivation of this class from measurement axioms on the line; locality, non-creation of local extrema, recursive implementation and higher dimension are outside the paper. The characterization is machine-checked in Lean 4, resting on two cited analytic facts, Bochner's theorem and the symmetric Lévy–Khintchine representation; the regularity of the kernels rests on two more.

1 Introduction

This paper characterizes the spatial scale spaces on the line, the families of smoothing operators through which an uncommitted observer can measure a signal at every scale, under translation and reflection symmetry. A front end measures its input before anything is known about what the signal will contain. It must therefore hold records at every scale, and it must derive the form of those records from requirements on the measurement process rather than from a signal model. The classification says which apertures the axioms admit; whether an aperture creates structure across scale, the non-creation property of the Gaussian axiomatics, is a further requirement and is outside this paper. In the Gaussian axiomatics of scale-space theory [18, 38, 21, 3, 40, 24, 13] the requirements single out one family of records, the Gaussian smoothings, which solve the heat equation in scale. Each axiomatics that relaxes one of those requirements arrives at another class: the scale-invariant recursive filters of [28], the α -scale spaces of [6], among them the Poisson scale space of [11], and the Bessel scale space of [5]. Here too the answer is a class, arrived at by weakening one requirement, and the Gaussian is one ray of it.

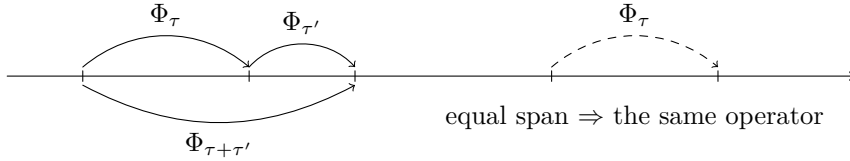
The assumption. The axioms are those of the extended class of Pauwels, Van Gool, Fiddelaers and Moons [28], with one change. Their recursivity, the semigroup form of the cascade

property $K_s K_t = K_{s+t}$, formalizes “a measurement of a measurement is a measurement” together with a second, tacit assumption: that the composite depends only on the total amount of scale crossed, so that an increment is determined by the scale difference alone. This second assumption is the *stationarity of the increments* in the scale parameter. It is not commutativity, which every family of convolutions has, and it is what this paper drops. The first assumption is kept, and the cascade is stated in its weakest defensible form, as a two-parameter *hemigroup*,

$$\Phi_{s,t} \Phi_{r,s} = \Phi_{r,t}, \quad 0 \leq r \leq s \leq t$$

(Figure 1). The name is that of the probability literature on groups, where a two-parameter family of measures composing in this way is a convolution hemigroup ([17], Def. 4.6.1, p. 308; [33], p. 365; [16], Def. 2.14.16, p. 380); Sato states the same requirements for a system of probability measures $\{\mu_{s,t}\}$ and constructs from it an additive process whose increment from s to t has law $\mu_{s,t}$ ([31], Thm. 9.7, p. 51). Every other axiom — linearity, translation covariance, reflection symmetry, positivity, unit mass, continuity, and scale covariance with the relabelling of the scale axis left free — is retained, restated as the two-parameter setting requires, with two smaller changes recorded in Remark 3.2: the kernels are measures rather than continuous functions, and positivity is stated as an axiom (§3). This is the same weakening that the companion article on temporal scale space [8], called the causal article in what follows, makes on the temporal axis, starting from the semigroup theory of [7]. The two axiom systems differ in exactly one axiom, causality there and reflection symmetry here, and that one axiom is what excludes the Gaussian aperture in time and admits it in space.

(a) semigroup: increments indexed by an *amount* of scale



(b) hemigroup: increments indexed by *intervals* of the scale axis

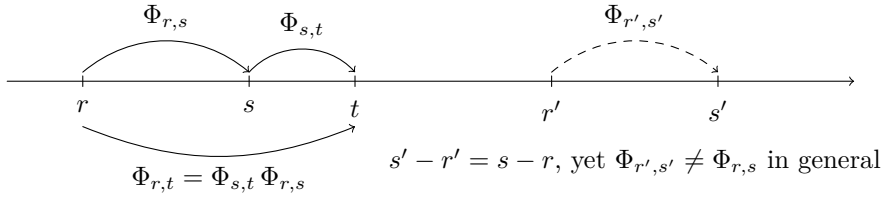


Figure 1: The two readings of the cascade property. (a) In the semigroup form the increment is indexed by an amount of scale: composition adds amounts, and increments of equal amount are the same operator wherever they act: the axis positions carry no labels because the operators cannot see them. (b) In the hemigroup form the increment is indexed by an interval of the scale axis: composition joins adjacent intervals, and increments of equal span at different positions need not coincide. The axioms retain what the two forms share (a measurement of a measurement is a measurement) and drop the rest.

What the class is. Pauwels et al. showed that recursivity and scale invariance alone do not single out the Gaussian: they admit the extended class $e^{-a|\omega|^\alpha t}$ for every $\alpha > 0$, with sign-changing kernels beyond $\alpha = 2$, and positivity enters their derivation only at its end, through the second-moment condition that bounds $\alpha \leq 2$. Under the hemigroup the admissible families are completely characterized (Theorem 7.3): in the canonical scale parametrization fixed in §6,

they are exactly the families whose kernel at scale t has transform $e^{-F(t\omega)}$ with

$$F(\omega) = a\omega^2 + \int_0^\infty (1 - \cos \omega x) \frac{k(x)}{x} dx, \quad a \geq 0, \quad k \geq 0 \text{ nonincreasing},$$

and the pair (gauge, exponent) is unique up to the normalization of the gauge, the gauge being the scale parametrization that covariance singles out (§6).

What the axioms leave free is therefore a number $a \geq 0$ and a nonincreasing function k , and both have a meaning at the kernel. The kernel at scale t is the law of a random displacement, and the representation says that the displacement has two independent parts: a Gaussian part of variance $2at^2$, and a sum of jumps whose intensity at scale t is a dilate of one function. Displacements of size about x occur at rate $k(x/t)/x$. So at unit scale $k(x)$ is the rate of jumps of size x weighted by their size, and the whole catalogue stretches with the scale rather than accumulating linearly in it. We call k the *displacement profile*. The condition on it, that larger displacements are never generated at a higher weighted rate than smaller ones, is what makes the family consistent across scales (Remark 2.6 develops the reading). In the vocabulary of probability the kernels are the *symmetric self-decomposable* laws — the displacements X that shrink into themselves, $X \stackrel{d}{=} cX + R_c$ with R_c independent, for every $c \in (0, 1)$ — and the family of their dilates is the marginal family of a self-similar additive process. The Gaussian scale space is the pair $(a, 0)$, the unique member with no jumps, and it is an extreme ray of the cone the admissible exponents form (Remark 7.7) rather than the whole answer.

When stationarity of the increments is reimposed, the recovered subclass is the symmetric stable family, $F(\omega) = |\omega|^\alpha$ with $0 < \alpha \leq 2$ (Corollary 7.4), which is the positive part of the extended class. The members below $\alpha = 2$ are heavy-tailed: their kernels decay like $|x|^{-1-\alpha}$ and have infinite variance. Those tails are therefore the price of that tacit assumption rather than of covariance. The class also contains members with all moments finite. The simplest is the variance-gamma family, with transform $(1 + t^2\omega^2)^{-\gamma}$. For $\gamma > \frac{1}{2}$ its kernels are the Matérn covariance functions of spatial statistics [37, 15, 29], of smoothness $\gamma - \frac{1}{2}$ (the smoothness parameter is positive in the standard form, [29], eq. (9)); for $\gamma \leq \frac{1}{2}$ they are unbounded at the origin and are densities but not covariance functions. Burgeth, Didas and Weickert [5] compose the same transforms into their Bessel scale space by varying the smoothness at fixed range, also in their scaled family (§ 2.4 there); here the cascade parameter is the range, at fixed smoothness. The family is exhibited in §3 beside the Gaussian as the second member on which the axioms are checked, and the two show what the class must contain: one has no jumps, one has jumps of every size.

Which assumption narrows. Positivity does. In the semigroup setting the class is one-parameter whether or not positivity is imposed, and positivity only trims the parameter range. In the hemigroup setting it is what makes a classification possible. Without it the remaining axioms admit an infinite-dimensional affine family through the Gaussian: the exponents $\omega^2 + \phi$ for every real, even ϕ that is the Fourier transform of an integrable function and vanishes at the origin. No description of that class by a Gaussian coefficient and a profile, in the form of Theorem 7.3, exists (Proposition 3.4; its construction is the one step of this paper not machine-checked, §1.1). With positivity, the class collapses to the cone above.

The route. The derivation runs in the order of the causal article: the operators are convolutions, the exponents add along the cascade, covariance forces a similarity form, and the classification is read off it. Fourier transforms stand in place of Laplace transforms, and the two theories part company at the transform. The Laplace transform of a probability measure on the half-line is positive and monotone; the Fourier transform of a symmetric one can vanish, and the zero set of its exponent can be a lattice (Lemma 2.11). So the exponent exists here only by a nonvanishing lemma that uses the cascade and continuity together (Lemma 4.2). The relabelling of the scale axis is determined not by the exponent itself but by a smoothed transmittance, the transform of the kernel averaged against a Gaussian window (Corollary 5.6).

And the analysis direction of the characterization is proved directly from a dilation identity (Lemma 7.1), so that the classical description of the self-decomposable Lévy measures is cited for orientation and never used. Each section records in one remark how its argument relates to the causal one. The kernels from the origin are absolutely continuous and unimodal about the origin (Proposition 7.9); the increments between two positive scales need not be, and can carry an atom. No truncation of the Gaussian is admissible, its transform changing sign where Lemma 4.2 requires positivity.

Relation to the scale-space axiomatics. The uniqueness of the Gaussian aperture is the oldest result of the subject, obtained from several axiom systems. Iijima derived it from linearity, translation and scale invariance, the semigroup and preservation of positivity [18, 35], together with a regularity assumption the derivation leaves implicit, that the transfer function is continuously differentiable at the origin; without it the Poisson kernel $e^{-t|\omega|}$ satisfies every listed condition, as Felsberg and Sommer point out ([10], § 2.1, Thm. 1 and p. 8). The one-dimensional theory of [38, 3, 40] and Koenderink’s causality [21] derive it from non-creation of structure across scale. Lindeberg derives it from non-enhancement of local extrema [23, 24, 25], and Florack from the scale-space differential structure [14, 13]. Each of these systems imposes, beyond the axioms of §3, either the semigroup or a requirement on how structure may change across scale. What this paper characterizes is the class from which such a requirement then selects. Within it the Gaussian is the boundary $\alpha = 2$ of the semigroup slice and the jump-free ray of the cone. The extended class of [28] and the α -scale spaces of [6] keep the semigroup and weaken other requirements, and they meet the present class in the symmetric stable family, its semigroup slice. The Bessel scale space of [5] has the transform of a member, the variance-gamma family, once it is composed along the range instead of along the smoothness index. The axiomatics of [2], which classify the nonlinear evolutions, are orthogonal to the question asked here.

Relation to probability theory. The class this paper arrives at is classical in probability theory. The self-decomposable laws are the class L of Lévy and Khintchine, the limit laws of normalized sums of independent variables, with a developed theory ([31], Ch. 3; [34]), and the marginal families of self-similar processes with independent increments are their dilates ([31], §16; the correspondence with its converse construction is [30], Thms. 3.1–3.2, pp. 293–294, in $\mathbb{R}^d$, and the notion of self-similarity is Lamperti’s [22]). The tools are classical: Bochner’s theorem, that a continuous positive definite function is the transform of a finite measure, and Schoenberg’s, that the exponents are exactly the functions ψ for which $e^{-\tau\psi}$ is such a transform for every $\tau > 0$ [4, 19]; the symmetric Lévy–Khintchine representation of those exponents [31]; Lévy’s continuity theorem [9]; and Yamazato’s unimodality theorem [39]. Each is stated with its source where it is used. The Matérn member is the variance-gamma law of [26], reviewed in [12]. What is new is the derivation of the class from measurement axioms, the reading of its two parameters as properties of an aperture, and the place the Gaussian takes in it. The probabilistic facts are cited, not reproven, and §1.1 says which of them the verified results use.

The contributions, each with the section that carries it:

- the characterization of the spatial scale spaces on the line under the hemigroup axioms: exactly the symmetric self-decomposable exponents, with the exponent unique and the gauge unique up to normalization (Theorem 7.3, §7);
- the recovery of the semigroup case from inside the hemigroup theory, the symmetric stable family with the bound $\alpha \leq 2$ proved from the representation (Corollary 7.4);
- the proposition that positivity is the narrowing axiom: without it the hemigroup axioms admit an infinite-dimensional affine family through the Gaussian (Proposition 3.4, §3; its construction clause rests on the printed proof, §1.1);
- the Fourier-side route — nonvanishing, the lattice zero sets, the smoothed transmittance,

the dilation argument — at the places where the line parts from the half-line (§§4–6);

- machine-checked proofs of the characterization and of everything it stands on, resting on two cited analytic facts (§1.1).

The paper is about the line, and about the classification. The geometry of the cone and its extreme rays, the named families inside it (Gaussian, Matérn, symmetric stable), the scale evolution in generator form, locality, non-creation of local extrema, and the exact implementation of the cascade by recursive filters are outside it, as are dimension d and the families covariant only under a discrete group of dilations. §8 names what is left open. Its contribution is mathematical foundations: no empirical claims about natural signals are made.

The sections follow the derivation in the order above. §2 fixes the notation, with the table shared with the causal article, and states the Fourier toolbox with its sources.

1.1 What is machine-checked

The characterization and everything it stands on have been formalized in the Lean 4 proof assistant, on top of its mathematical library Mathlib, and this subsection says which results, and what they rest on. It is self-contained: nothing below depends on reading the formal development.

What is proved. Theorem 7.3 is machine-checked in all three of its clauses: the construction of a family from an admissible exponent, the analysis that every family has one, and the uniqueness of the pair (gauge, exponent). The three are assembled in one declaration, so that the theorem is verified as it is printed. Everything §§4–6 contribute to it is verified with it. That is the convolution representation and the nonvanishing lemma, the additivity and infinite divisibility of the exponents with the truncation inequality that carries the null-array limit (the limit of a sum of many small independent displacements, Theorem 5.2), the exclusion of lattice kernels, the rigidity of the action and the gauge. Of §7, the self-decomposability lemma in all its directions, the semigroup case with its bound $\alpha \leq 2$, the convex cone, and the strict positivity and monotonicity of the admissible exponents are verified. So are the two toolbox facts that are proved rather than cited, uniqueness of the Fourier–Stieltjes transform and Lévy’s continuity theorem, from Mathlib’s characteristic-function development. Of §3, both members are verified in every axiom, including the continuity axiom at the operator, and the no-positivity proposition is verified in its positivity equivalence and in its example.

What it rests on. Two analytic facts are taken from the literature rather than reproved, and they are the only things the verified characterization assumes beyond the logical foundations of the proof assistant itself; each is cited with a page reference checked against the source.

1. *Bochner’s theorem, in the forward direction.* A real-valued continuous positive definite function with value 1 at the origin is the cosine transform of a symmetric probability measure on $\mathbb{R}$ ([31], Prop. 2.5, pp. 8–9). This is what turns an admissible exponent into actual kernels, and it is the only thing the constructive direction of Theorem 7.3 needs that is not proved here; the reverse direction is proved.
2. *The symmetric Lévy–Khintchine representation, at its converse and its uniqueness clauses.* Every pair (a, ν) with $a \geq 0$ and $\int (1 \wedge x^2) \nu(dx) < \infty$ defines a member of ND_s , the class of symmetric continuous negative definite functions, and the pair of a member is unique ([31], Thm. 8.1, pp. 37–38). The converse is used by the construction, the uniqueness by the analysis direction of Lemma 7.1 and by nothing else the theorem needs.

Table 1: The cited facts and their direct uses. A result not listed uses no cited fact directly; it may use one through a listed result, and the dependency graph of the repository shows which. Cited for orientation and used by no proof: Schoenberg’s theorem (Proposition 2.3(2)), the existence half of its clause (3) and its closure clause (4), the classical description of the self-decomposable Lévy measures (Proposition 2.13), and the closed form of the variance-gamma density.

cited fact	clause used	used directly by
Bochner’s theorem	forward direction	Thm. 5.2, last sentence; Thm. 7.3, construction
symmetric Lévy–Khintchine representation	converse	the same two
	uniqueness	Thm. 5.2, the pair; Lemma 7.1, analysis
absolute continuity in class L	as stated, $d = 1$	Prop. 7.9
Yamazato’s unimodality theorem	symmetric case	Prop. 7.9

Table 1 lists every cited fact of the paper with the results that use it directly; the dependency graph of the repository carries the transitive picture.

The second fact carries one step beyond its pages: its converse is admitted for Lévy measures already folded onto $(0, \infty)$, where the source states the representation on the punctured line with a centering term, and the folding reduction of §2, elementary and written out there, sits inside the admitted statement rather than beside it. The first fact carries nothing beyond its pages; its admitted hypotheses are stronger than the source’s.

The regularity of the kernels, Proposition 7.9, rests on two further cited facts, used by that proposition alone and by nothing the characterization needs: that a nondegenerate self-decomposable law is absolutely continuous ([31], Thm. 27.13, p. 181) and that a law of class L is unimodal ([39], Thm. 1, p. 523). The one admitted statement that carries them adds two steps beyond the pages, the symmetric rider that places the mode at the origin and the choice of a named density whose monotonicity holds pointwise on the open half-line. The hypothesis check, that the kernels from the origin are nondegenerate and self-decomposable, is proved. Nothing is claimed about the density’s value at the origin, where it may be unbounded. The two members of §3 use no cited fact: the moments of the Matérn member are verified by the Gaussian-mixture argument printed in the appendix, on Lean core.

Why these and not more. The classical route to the characterization passes through the closure of ND_s under pointwise limits and the existence half of the Lévy–Khintchine representation, and neither is used: the null-array limit extracts the representation of an increment directly, before any limit is taken, and the profile form is proved from the dilation identity where the half-line development cites it. So the trust base is narrower than the naive route by two clauses of the toolbox, and narrower than the causal article’s at the one place where that article carries the classical self-decomposability theorem. The same policy runs in the other direction: classical results of structural rather than analytic character are proved, not cited: the convolution representation of §4, classically Wendel’s theorem, is proved from scratch, and the quadratic growth bound that gives $\alpha \leq 2$ is three lines from the truncation split rather than a cited estimate.

What is not formalized. One thing in the scope of this paper, and the text names it where it occurs. The first clause of Proposition 3.4, the construction of the signed family and the verification of the axioms for it, rests on the printed proof: the library used carries the L^1 Fourier transform but not the algebra of such transforms as a Banach algebra, and a route through the convolution operators has been estimated and not carried out. Every other numbered statement is machine-checked, including both members of §3 at every axiom, the variance and the moments. The closed form of the Matérn member’s kernel is cited there and

is not a statement of this paper.

Where to look. The development and its axiom ledger, with page-verified citations for every cited interface, are at the repository accompanying this paper, <https://github.com/danielfagerstrom/spatial-hemigroup-scale-space-kernels>: the Lean modules that the statements of this paper rest on, the trust boundary and the ledger, and the sources of the blueprint chapters behind §2–§7, in which the declaration behind each numbered statement is named in a `\lean` tag on that statement. It is an export, at the revision named in its changelog, from the development repository, which also holds the material of two further articles. The toolchain is pinned in-tree — Lean 4 v4.31.0, Mathlib v4.31.0, and the shared library `ScaleSpaceCore` at its tag v0.2.0 — so the verification is reproducible with one command, `lake build in Formalization/`. A second, `lake env lean CIAxiomGuard.lean`, prints the axiom usage of every named declaration, to be read against the trust-boundary file beside it. On the declaration `SpatialLine.main_characterization` it prints Lean’s three logical axioms and the three names of the two cited facts above, and nothing else:

```
'SpatialLine.main_characterization' depends on axioms: [propext,
Classical.choice, Quot.sound,
SpatialLine.fourier_toolbox_bochner_symm,
SpatialLine.fourier_toolbox_levy_converse,
SpatialLine.fourier_toolbox_levy_unique]
```

2 Preliminaries and notation

This section fixes the notation and states the Fourier facts and the profile representation that the later sections use. The axioms themselves begin in §3.

Letters. Space runs over $\mathbb{R}$ with coordinate x . The transform variable is the wavenumber ω . Scale is t , the evolution variable of the diffusion equation in the scale-space literature. At first t is only an ordered label of the records; in the canonical parametrization of §6 it has the dimension of length, and the Gaussian member’s diffusion parameter is proportional to t^2 . A stage is indexed by an ordered pair of scales $s \leq t$, with a third letter $r \leq s \leq t$ when three are needed. This is the convention for evolution families, $U(t, s)U(s, r) = U(t, r)$, and it matches the indexing of additive processes: the kernel $\mu_{s, t}$ is the law of an increment $X_t - X_s$. The dilation ratio is λ .

The causal article [8] used other letters: the time coordinate t , the Laplace variable s , scales $x \leq y \leq z$ and the ratio σ . Table 2 records the correspondence, together with the symbols defined in this section, and it is used whenever a statement of that article is cited. No formula holds both meanings of s or t at once, since the Laplace variable and the time coordinate do not occur here.

Two letters carry two roles each, always locally. The letter a is the Gaussian coefficient from (2.2) on, and the amount of a translation in T_a . The letter c is the contraction factor of Definition 2.12, the orbit coordinate $c(\lambda)$ of §6, and a generic constant.

Spaces. We write $L^1 = L^1(\mathbb{R}, dx)$, real, with positive cone L^1_+ and $\int f := \int_{\mathbb{R}} f(x) dx$. Its dual is $L^\infty(\mathbb{R})$. $M_+(\mathbb{R})$ denotes the finite positive Borel measures on $\mathbb{R}$, of total mass $\|\mu\| = \mu(\mathbb{R})$.

Actions. Translations are $(T_a f)(x) = f(x - a)$, $a \in \mathbb{R}$. The reflection is $(Rf)(x) = f(-x)$, extended to measures as the pushforward under $x \mapsto -x$, and a measure is *symmetric* when $R\mu = \mu$. Dilations act on functions by $(D_\lambda f)(x) = \lambda^{-1}f(x/\lambda)$, $\lambda > 0$, which preserves mass, and on measures by the pushforward under $x \mapsto \lambda x$.

Convolution. $(\mu * f)(x) = \int f(x - y) \mu(dy)$ for $\mu \in M_+(\mathbb{R})$ and $f \in L^1$. Then $\|\mu * f\|_1 \leq \|\mu\| \|f\|_1$, and both actions are compatible with convolution:

$$D_\lambda(\mu * f) = (D_\lambda \mu) * (D_\lambda f), \quad R(\mu * f) = (R\mu) * (Rf).$$

Table 2: Notation: the symbol, where it is defined, and the letter the causal article uses for the same object.

object	here	the causal article
signal coordinate	x (space)	t (time)
transform variable	ω (Fourier)	s (Laplace)
scales in a stage	$r \leq s \leq t$	$x \leq y \leq z$
stage operator and its kernel	$\Phi_{s,t}, \mu_{s,t}$	$\Phi_{x,y}, \mu_{x,y}$
exponent of a stage	$g_{s,t}(\omega) = -\log \hat{\mu}_{s,t}(\omega)$ (Lemma 4.2)	$g_{x,y}(s)$
kernel from the origin, accumulated exponent	$\mu_{0,t}, G(t, \omega) = g_{0,t}(\omega)$	$\mu_{0,x}, G(x, s)$
smoothed transmittance	$\Theta(t)$ (Corollary 5.6)	—
canonical gauge	$\chi(t)$ (Proposition 6.6)	$\chi(x)$
the family's exponent and its pair	$F(\omega)$; Gaussian coefficient a , profile k	$F(s)$; drift b_0 , profile k
Lévy measure, tail measure of the profile	ν ((2.2)), $\varpi = -dk$ (Lemma 7.1)	ν, ϖ
symbol of the exponent	$B(\omega) = \omega F'(\omega)$ (Lemma 7.1)	$B(s) = sF'(s)$
dilation ratio, action, orbit coordinate	$\lambda, S_\lambda, c(\lambda)$	$\sigma, S_\sigma, c(\sigma)$
random displacement, random delay	$X_{s,t}$	$T_{x,y}$
function space	$L^1 = L^1(\mathbb{R}, dx)$	$X = L^1(\mathbb{R}, dt)$
translation, reflection	T_a, R	$\tau_a, \text{—}$
exponents by definition, by representation	ND_s (Def. 2.2), LE_s (Def. 2.5)	BF_0, LE

Transforms. For $\mu \in M_+(\mathbb{R})$ the Fourier transform is

$$\hat{\mu}(\omega) = \int_{\mathbb{R}} e^{-i\omega x} \mu(dx), \quad \omega \in \mathbb{R}, \quad (2.1)$$

continuous, with $|\hat{\mu}| \leq \|\mu\|$ and $\hat{\mu}(0) = \|\mu\|$; for symmetric μ it is real and even, $\hat{\mu}(\omega) = \int \cos(\omega x) \mu(dx)$. Under dilation $\widehat{D_\lambda \mu}(\omega) = \hat{\mu}(\lambda\omega)$, and $\widehat{\mu * \nu} = \hat{\mu} \hat{\nu}$. Unlike the Laplace transform of a measure carried by a half-line, $\hat{\mu}$ need not be positive. That the axioms exclude a zero is Lemma 4.2.

Two classes of functions on $\mathbb{R}$ carry the analysis: the positive definite functions, which are the transforms of the kernels, and the symmetric continuous negative definite functions, which are their exponents.

Definition 2.1 (positive definite). A function $\varphi : \mathbb{R} \rightarrow \mathbb{C}$ is *positive definite* if

$$\sum_{j,k} c_j \overline{c_k} \varphi(\omega_j - \omega_k) \geq 0 \quad (\text{in particular real})$$

for every finite family $(\omega_j) \subset \mathbb{R}$ and every finite family $(c_j) \subset \mathbb{C}$.

The words in parentheses belong to the definition. The double sum is required to be a nonnegative real number, and for a complex-valued φ that is a genuine condition. It forces φ to be Hermitian, $\varphi(-\omega) = \overline{\varphi(\omega)}$, and it is the reading under which Proposition 2.3(1) holds.

Definition 2.2 (symmetric continuous negative definite function). A function $\psi : \mathbb{R} \rightarrow [0, \infty)$ is a *symmetric continuous negative definite function* if ψ is continuous and even, $\psi(0) = 0$, and $e^{-\tau\psi}$ is positive definite for every $\tau > 0$. We write $\psi \in \text{ND}_s$.

Definition 2.2 defines ND_s by the exponential form, that $e^{-\tau\psi}$ is positive definite for every $\tau > 0$, and not by the quadratic-form inequality in ψ that the sources call the kernel form (clause (2) of the next proposition). The exponential form is the one the cascade asks for, and a proof that a given ψ lies in ND_s then has to exhibit the transforms $e^{-\tau\psi}$ and nothing else. Schoenberg's theorem, the equivalence of the two forms, is needed only to import results that the sources state in the kernel form. That is why clause (2) is a cited interface rather than a definition.

The toolbox that connects the two classes to transforms is classical and is cited rather than reproven. The sources, clause by clause, follow the statement.

Proposition 2.3 (the Fourier toolbox). 1. (Bochner.) φ is the Fourier transform of a probability measure on $\mathbb{R}$ if and only if φ is continuous, positive definite and $\varphi(0) = 1$; it is the transform of a symmetric probability measure if and only if moreover φ is real, equivalently even.

2. (Schoenberg.) For continuous even $\psi \geq 0$ with $\psi(0) = 0$: $\psi \in \text{ND}_s$ if and only if ψ is negative definite in the kernel sense,

$$\sum_{j,k} (\psi(\omega_j) + \psi(\omega_k) - \psi(\omega_j - \omega_k)) c_j \overline{c_k} \geq 0$$

for every finite family; equivalently, by clause (1), if and only if $e^{-\tau\psi}$ is the transform of a symmetric probability measure for every $\tau > 0$.

3. (Lévy–Khintchine, symmetric form.) Every $\psi \in \text{ND}_s$ has the representation

$$\psi(\omega) = a\omega^2 + \int_{(0,\infty)} (1 - \cos \omega x) \nu(dx), \quad a \geq 0, \quad \int_{(0,\infty)} (1 \wedge x^2) \nu(dx) < \infty, \quad (2.2)$$

the pair (a, ν) being unique; and conversely every such pair defines a member of ND_s .

4. (Closure.) ND_s is a convex cone, and in the kernel form of clause (2) the class is closed under pointwise limits with no proviso. If $\psi_n \in \text{ND}_s$ and $\psi_n \rightarrow \psi$ pointwise with ψ continuous at 0, then $\psi \in \text{ND}_s$.

Sources. Clause (1) is [31], Prop. 2.5, pp. 8–9, whose parts are numbered (i)–(xii): part (i) is Bochner’s theorem, and parts (ii) and (v) add that the measure is unique and that it is symmetric exactly when the transform is real. Feller corroborates it ([9], Vol. 2, § XIX.2, Theorem (Bochner), p. 622). Clause (2) is [32], Def. 4.3 and Prop. 4.4, p. 36; also [4], Thm. 7.8, p. 41, and [19], Def. 3.6.5, p. 122, Def. 3.6.6, p. 123, and Thm. 3.6.11, p. 125. Clause (3) is [31], Thm. 8.1, pp. 37–38, whose parts (i), (ii) and (iii) are existence, uniqueness and the converse, with (8.2) there for the condition on the Lévy measure. Clause (4) is [19], Lemma 3.6.7A, p. 123, for the kernel form, and [31], Prop. 2.5, part (viii), p. 9, for the proviso. Sato’s characteristic function is $\int e^{i\langle z, x \rangle} \mu(dx)$, the opposite sign to (2.1), which is immaterial for symmetric laws.

Remark 2.4 (what the two classes are, and why the transform sees them). The prototype of a positive definite function is a single *character*, $\omega \mapsto e^{-i\omega x}$ for a fixed x : the double sum of Definition 2.1 collapses to $|\sum_j c_j e^{-i\omega_j x}|^2 \geq 0$. The condition is linear in φ and survives pointwise limits, so every positive mixture of characters, that is every Fourier transform of a positive measure, is positive definite for free. Bochner’s theorem is the converse: a continuous positive definite function is such a mixture, and the mixing measure is unique. In the language of convex cones, the characters are the extreme rays of the cone of positive definite functions, and every continuous member is the barycentre of the rays, with the measure as the weights. In the symmetric case the rays are $\cos \omega x$, $x \geq 0$.

The negative definite functions are the same picture one level up. Ask which ψ make $e^{-\tau\psi}$ a transform for every $\tau > 0$, which is what the cascade of §5 will ask. The answer is again a cone described twice: by the checkable kernel-form inequality of clause (2), and by its extreme rays, which is the Lévy–Khintchine representation (2.2). The rays are ω^2 and $1 - \cos \omega x$ for $x > 0$, and there is no constant ray because $\psi(0) = 0$. Each ray has a probabilistic identity, given in Remark 2.6 below.

The mechanism behind Bochner’s theorem is the one this article reuses. In Feller’s proof the transform is inverted against an even window and the general case is reached through the convergence factor $e^{-\varepsilon\omega^2}$ ([9], Vol. 2, § XIX.2, Lemma 1, p. 621, and (2.4)); positive definiteness is what makes the windowed inverse transforms nonnegative, and the finite-family inequality is the discrete form of that, the Toeplitz form (2.6)–(2.7) on p. 622. The same device, a transform

read against a Gaussian window, is the test function of §4 and the smoothed transmittance of §5. Two facts used throughout are also Feller’s (Lemma 3, p. 622): $|\varphi| \leq \varphi(0)$ and $\varphi(-\omega) = \overline{\varphi(\omega)}$.

The contrast with the half-line is one of kind. Complete monotonicity, the transform-free description of “comes from positive mass” for the Laplace transform, is a pointwise test: countably many derivative signs at each point. Positive definiteness is a global quadratic-form condition, and there is no local test for it. That is the first sign that the half-line theory is the more rigid of the two, and the remark at the end of this section lists what the difference costs.

The folding convention. The sources state clause (3) on $\mathbb{R}$, with a centering term and a Lévy measure on $\mathbb{R} \setminus \{0\}$. The reduction to (2.2) is ours and elementary. A symmetric Lévy measure ν_2 on $\mathbb{R} \setminus \{0\}$ enters (2.2) as its image ν under $x \mapsto |x|$. So a two-sided Lévy density written $h(|x|)/|x|$, the form the probabilistic sources use, has folded profile $k = 2h$ in (2.3) below: the factor is the two directions of a displacement of a given size, and it is the one place where a constant is easily lost in carrying a statement from a source into this article. The centering term of the general representation vanishes by symmetry, read on the compensated integrand: the odd part of $e^{-i\omega x} - 1 + i\omega x \mathbf{1}_{|x|<1}$ is $-i(\sin \omega x - \omega x \mathbf{1}_{|x|<1})$, which is $O(x^3)$ at the origin and bounded, hence ν_2 -integrable, and odd, so its integral against a symmetric ν_2 is 0. The uncompensated $\int \sin(\omega x) \nu_2(dx)$ need not converge.

Which vocabulary does the work. The later sections name the class by its representation.

Definition 2.5 (symmetric Lévy exponents). LE_s is the set of functions $\psi : \mathbb{R} \rightarrow [0, \infty)$ of the form (2.2), for some $a \geq 0$ and some measure ν on $(0, \infty)$ with $\int (1 \wedge x^2) \nu(dx) < \infty$. By Proposition 2.3(3), $\text{LE}_s = \text{ND}_s$.

The representation is the object the proofs manipulate, and LE_s is the name the later sections use. Definition 2.2 itself enters only where the class ND_s is the subject, and in one leg of Lemma 7.1. The machine-checked development defines LE_s alone. What that development takes from Proposition 2.3 is less than the proposition states, and the difference is part of the trust base stated in §1.1. It takes the forward implication of clause (1) (a real-valued continuous positive definite function with value 1 at the origin is the cosine transform of a symmetric probability measure) and the converse and uniqueness clauses of (3). Clause (2), the existence half of (3) and clause (4) are stated because they are the toolbox a reader meets, and they are used by no proof: the arguments below extract the representation of a limit directly (Theorem 5.2), so the closure clause is never needed.

Remark 2.6 (the displacement dictionary). A symmetric kernel is the law of a *random displacement*: $\Phi_{s,t}f$ reports the signal at a randomly displaced position $x - X_{s,t}$ with $X_{s,t} \sim \mu_{s,t}$. Its transform $\hat{\mu}_{s,t}(\omega) = \mathbb{E} \cos(\omega X_{s,t})$ is the contrast that a sinusoidal probe of wavenumber ω retains after the random displacement, the stage’s modulation transfer function (Remark 4.3). The exponent $\psi = -\log \hat{\mu}$ adds along the cascade. In (2.2) the Gaussian coefficient a is the *diffusive part*, a Gaussian displacement of variance $2a$, the one term the time-causal axis deletes and the only one Gaussian scale space uses. The measure ν is a *catalogue of jump displacements*. Displacements of size about x arrive as a Poisson stream, that is at independent random moments with a constant rate, and $\nu(dx)$ is that rate; the streams of different sizes are independent, and each event of size x costs the probe the phase deficit $1 - \cos \omega x$. So a symmetric infinitely divisible displacement is a Gaussian part plus an independent sum of jumps of all sizes, and the representation says that every such displacement decomposes in this way.

Two further classical facts are used repeatedly and are recorded as statements so that each later use can cite one. Both are parts of Sato’s Prop. 2.5 ([31]): uniqueness is its part (ii), p. 8, and the continuity theorem its parts (vii)–(viii), p. 9. Neither is taken on trust: each is proved, in the form the article uses, from a result in Mathlib, and the proof is given.

Proposition 2.7 (uniqueness of the Fourier–Stieltjes transform). *A finite Borel measure on $\mathbb{R}$ is determined by its Fourier transform: if $\hat{\mu}(\omega) = \hat{\rho}(\omega)$ for every $\omega \in \mathbb{R}$, then $\mu = \rho$.*

Proof. The functions $e_\omega(x) = e^{-i\omega x}$, $\omega \in \mathbb{R}$, span a subalgebra $\mathcal{A}$ of the bounded continuous functions on $\mathbb{R}$: it contains the constants, since $e_0 = 1$; it is closed under conjugation, since $\overline{e_\omega} = e_{-\omega}$; and it separates points, since $e_\omega(x) = e_\omega(y)$ for every ω forces $x = y$. The hypothesis says that μ and ρ give equal integrals to every e_ω , hence, by linearity, to every member of $\mathcal{A}$. Two finite Borel measures on a complete separable metric space that give equal integrals to every member of such an algebra are equal: on each compact set the Stone–Weierstrass theorem makes $\mathcal{A}$ uniformly dense in the continuous functions, and a finite Borel measure on a Polish space is inner regular, so the equality extends first to the bounded continuous functions and then, by the portmanteau theorem, to the Borel sets. This is the argument Mathlib’s `MeasureTheory.Measure.ext_of_charFun` runs, in the stated generality; every step of it is machine-checked, and the article’s declaration is its specialisation to $\mathbb{R}$. $\square$

The statement is about finite measures on $\mathbb{R}$ and claims nothing for infinite ones.

Proposition 2.8 (Lévy’s continuity theorem). *Let μ_n, μ be probability measures on $\mathbb{R}$. If $\hat{\mu}_n(\omega) \rightarrow \hat{\mu}(\omega)$ for every $\omega \in \mathbb{R}$, then $\mu_n \rightarrow \mu$ weakly.*

Proof. Since $\hat{\mu}$ is continuous at 0 with $\hat{\mu}(0) = 1$, the sequence is tight. Indeed, for a probability measure ρ on $\mathbb{R}$ and $r > 0$, Fubini’s theorem gives

$$\frac{1}{r} \int_{-r}^r (1 - \hat{\rho}(\omega)) d\omega = 2 \int_{\mathbb{R}} \left(1 - \frac{\sin rx}{rx}\right) \rho(dx) \geq \rho(\{|x| \geq 2/r\}),$$

because the integrand is nonnegative and is at least $\frac{1}{2}$ where $|rx| \geq 2$. Applying this to μ_n and letting $n \rightarrow \infty$ — the integrands are bounded by 2, so bounded convergence applies — bounds the upper limit of $\mu_n(\{|x| \geq 2/r\})$ by $\frac{1}{r} \int_{-r}^r (1 - \hat{\mu})$, which tends to 0 as $r \downarrow 0$ by continuity of $\hat{\mu}$ at 0. So $\{\mu_n\}$ is tight. By Prokhorov’s theorem every subsequence has a further subsequence converging weakly to some probability measure ρ ; along it $\hat{\rho} = \lim \hat{\mu}_n = \hat{\mu}$, since $x \mapsto e^{-i\omega x}$ is bounded and continuous, so $\rho = \mu$ by Proposition 2.7. Every subsequence therefore has a further subsequence converging weakly to μ , which is convergence of the whole sequence. The statement is machine-checked: Mathlib’s `MeasureTheory.ProbabilityMeasure.tendsto_of_tendsto_charFun` proves it with the same tightness estimate (`isTightMeasureSet_of_tendsto_charFun`) and, in place of the subsequence extraction, a direct density argument in the algebra of characters. $\square$

Weak convergence is read as convergence of $\int g d\mu_n$ for every bounded continuous $g : \mathbb{R} \rightarrow \mathbb{R}$, which is the form the $\varepsilon/3$ argument in the proof of Theorem 7.3 uses. What is not part of the statement is the passage from weak convergence to convergence of the convolution operators in L^1 . That is the $\varepsilon/3$ argument itself, and it is proved where it is used.

Remark 2.9 (the general form of the continuity theorem). The classical statement is more general: if $\hat{\mu}_n \rightarrow \varphi$ pointwise with φ continuous at 0, then φ is the transform of a probability measure μ and $\mu_n \rightarrow \mu$ weakly. The tightness estimate in the proof above uses only continuity of the limit function at the origin, so it applies unchanged. What the general form needs in addition is that the limit along a Prokhorov subsequence has transform φ , which identifies φ as a transform. The continuity proviso is what the line costs against the half-line, where pointwise limits of Laplace exponents on $(0, \infty)$ suffice — and it is why Proposition 2.3(4) carries it too. No statement of this article uses the general form: every use of Proposition 2.8 has the limit law already in hand.

The next lemma is a growth bound on the exponents. The literature states it with an unspecified constant ([19], Lemma 3.6.22); here it is proved with an explicit one, in three lines from a split of the integral, and it is used twice later. The constant C is finite exactly because $\int (1 \wedge x^2) \nu < \infty$: the two integrals inside it are the two halves of that condition.

Lemma 2.10 (continuous negative definite functions grow at most quadratically). *Let $\psi \in \text{ND}_s$ have the pair (a, ν) of (2.2). Then*

$$\psi(\omega) \leq C(1 + \omega^2), \quad \omega \in \mathbb{R}, \quad C := a + \frac{1}{2} \int_{(0,1]} x^2 \nu(dx) + 2\nu((1, \infty)) < \infty.$$

In particular $\psi(\omega) = O(\omega^2)$ as $|\omega| \rightarrow \infty$.

Proof. Split the integral in (2.2) at $x = 1$. For $x \leq 1$ use $1 - \cos \omega x \leq \frac{1}{2} \omega^2 x^2$; for $x > 1$ use $1 - \cos \omega x \leq 2$. This gives $\psi(\omega) \leq a\omega^2 + \frac{1}{2} \omega^2 \int_{(0,1]} x^2 \nu(dx) + 2\nu((1, \infty)) \leq C(1 + \omega^2)$. Finiteness of the two integrals is $\int (1 \wedge x^2) \nu(dx) < \infty$. $\square$

The exponent of a kernel is $-\log$ of its transform, so it vanishes at exactly the frequencies where the transform equals 1, where the kernel lets a probe wave through untouched. Three later proofs turn on where that can happen: the nonvanishing lemma of §4, the exclusion of lattice kernels in §6, and the strict monotonicity of the accumulated exponent in §7. The next statement says what the set of such frequencies can be. On a half-line a Laplace exponent that vanishes at one interior point vanishes identically. On the line the set is closed under addition and negation and is closed as a set, so it is a closed subgroup of $\mathbb{R}$, and a closed subgroup of $\mathbb{R}$ is $\{0\}$, a lattice $c\mathbb{Z}$ with $c > 0$, or all of $\mathbb{R}$. The lattice case does occur, and it means that the kernel is carried by a lattice.

Lemma 2.11 (the zero set of a symmetric exponent is a closed subgroup). *Let μ be a probability measure on $\mathbb{R}$ and put $N(\mu) := \{\omega \in \mathbb{R} : \hat{\mu}(\omega) = 1\}$. Then $N(\mu)$ is a closed subgroup of $\mathbb{R}$, hence $\{0\}$, a lattice $c\mathbb{Z}$ with $c > 0$, or $\mathbb{R}$; the last case holds if and only if $\mu = \delta_0$. For $\omega_0 \neq 0$,*

$$\omega_0 \in N(\mu) \iff \cos(\omega_0 x) = 1 \text{ for } \mu\text{-almost every } x \iff \mu \text{ is carried by the lattice } \frac{2\pi}{|\omega_0|} \mathbb{Z}.$$

If μ is symmetric with $\hat{\mu} = e^{-\psi}$ for some $\psi \in \text{ND}_s$, then $N(\mu) = \{\psi = 0\}$: a symmetric exponent vanishes at a nonzero frequency exactly when its law is a lattice law. Lattice laws can be infinitely divisible — the symmetric compound Poisson law on $\mathbb{Z}$, of exponent $c(1 - \cos \omega)$, is one — so membership of ND_s does not exclude them.

Proof. Since $\text{Re}(1 - e^{-i\omega x}) = 1 - \cos \omega x \geq 0$, the identity $\hat{\mu}(\omega) = 1$ forces $\int (1 - \cos \omega x) \mu(dx) = 0$, hence $\cos(\omega x) = 1$ for μ -almost every x , hence $e^{-i\omega x} = 1$ almost everywhere and $\hat{\mu}(\omega) = 1$; the three conditions are therefore equivalent. So $N(\mu)$ is closed under sums and negation, and closed as a set because $\hat{\mu}$ is continuous: a closed subgroup of $\mathbb{R}$, hence $\{0\}$, some $c\mathbb{Z}$ with $c > 0$, or $\mathbb{R}$. If $N(\mu) = \mathbb{R}$ then $\hat{\mu} \equiv 1 = \hat{\delta}_0$, so $\mu = \delta_0$ by Proposition 2.7; the converse is immediate. For $\omega_0 \neq 0$, $\cos(\omega_0 x) = 1$ says $x \in \frac{2\pi}{|\omega_0|} \mathbb{Z}$, so the almost-everywhere clause says exactly that μ is carried by that lattice. When $\hat{\mu} = e^{-\psi}$ with ψ real, $\hat{\mu}(\omega) = 1$ if and only if $\psi(\omega) = 0$. For the example, $c(1 - \cos \omega)$ is of the form (2.2) with $a = 0$ and $\nu = c\delta_1$, hence lies in ND_s by Proposition 2.3(3); its law is carried by $\mathbb{Z}$, and it is infinitely divisible because $c(1 - \cos \omega)/n$ is again of that form. $\square$

The last sentence of the lemma bears on what follows: the exclusion of lattices cannot come from the function class, so it must come from elsewhere. It comes from covariance, Lemma 6.2, and the gauge argument of §6 is arranged so as not to need the exclusion at all.

The class this article characterizes is the class L of [31], Def. 15.1, p. 90. Its members are infinitely divisible (Prop. 15.5, p. 93). Read at the displacement, the definition says that every contraction cX of the displacement X is a component of X itself: what is left after shrinking is again a displacement, independent of the shrunk part.

Definition 2.12 (self-decomposable; class L). A probability measure μ on $\mathbb{R}$ is *self-decomposable* if for every $b > 1$ there is a probability measure ρ_b on $\mathbb{R}$ with $\hat{\mu}(\omega) = \hat{\mu}(\omega/b) \hat{\rho}_b(\omega)$ for every ω ; equivalently, if $X \stackrel{d}{=} cX + R_c$ with R_c independent of X , for every $c \in (0, 1)$, where X has law μ .

The classical description of the class is by its Lévy measures: [31], Cor. 15.11, p. 95, the one-dimensional form of Thm. 15.10 there, which also records that self-decomposability imposes no restriction on the Gaussian part. The corollary is stated for a general law, with k increasing on $(-\infty, 0)$ and decreasing on $(0, \infty)$. The symmetric specialisation to a profile on $(0, \infty)$ alone is immediate.

Proposition 2.13 (symmetric self-decomposable exponents). *A symmetric infinitely divisible law with pair (a, ν) is self-decomposable if and only if its Lévy measure has the form $\nu(dx) = k(x)x^{-1}dx$ on $(0, \infty)$ with $k : (0, \infty) \rightarrow [0, \infty)$ nonincreasing, so that its exponent is*

$$\psi(\omega) = a\omega^2 + \int_0^\infty (1 - \cos \omega x) \frac{k(x)}{x} dx, \quad a \geq 0, \quad k \geq 0 \text{ nonincreasing.} \quad (2.3)$$

Self-decomposability imposes no restriction on the Gaussian coefficient a .

Proposition 2.13 is where a reader meets the class, and it is stated for orientation: it lies on no proof path. The profile form the development needs is proved in Lemma 7.1, from the dilation identity, the uniqueness clause of Proposition 2.3(3) and a monotone difference-quotient argument in the log-displacement coordinate, and every later statement cites that lemma and not this one.

Remark 2.14 (the profile in log-displacement). In the log-displacement coordinate $\theta = \log x$ the density shape of (2.3) is the profile $\tilde{k}(\theta) = k(e^\theta)$ per unit log-displacement, dilation acts by translation in θ , and self-decomposability is the profile being nonincreasing: zooming out may only add displacement intensity. The Gaussian is the profile-free member; the symmetric stable exponents $|\omega|^\alpha$ are the exactly scale-free profiles $\tilde{k} = ce^{-\alpha\theta}$. The extreme rays of the cone are the Gaussian ray ω^2 and the unit steps, whose exponents are the Cin functions of Lemma A.1 (Appendix A).

The integrability conditions in (2.3) are the admissibility test every family in this article is put through, so they are recorded as a statement. Its two halves are the symmetric forms of the causal criterion: near the origin the integrand is $\asymp \omega^2 x^2 k(x)/x$, so $\int_0^1 x k$ replaces $\int_0^1 k$, while at infinity $1 - \cos \leq 2$ gives the same condition. The criterion takes k measurable and nonnegative and nothing else, in particular not monotone, because it is later applied to increment densities $k(u/t) - k(u/s)$, which are not.

Lemma 2.15 (the integrability criterion for a profile). *Let $k : (0, \infty) \rightarrow [0, \infty)$ be measurable and put $\nu(dx) := k(x)x^{-1}dx$. Then*

$$\int_{(0, \infty)} (1 \wedge x^2) \nu(dx) < \infty \quad \Longleftrightarrow \quad \int_0^1 x k(x) dx < \infty \quad \text{and} \quad \int_1^\infty \frac{k(x)}{x} dx < \infty.$$

Consequently, by Proposition 2.3(3), the right-hand pair of conditions is necessary and sufficient for (2.3) to define a member of ND_s .

Proof. Both terms below are nonnegative, so

$$\int_{(0, \infty)} (1 \wedge x^2) \nu(dx) = \int_0^1 x^2 k(x) \frac{dx}{x} + \int_1^\infty k(x) \frac{dx}{x} = \int_0^1 x k(x) dx + \int_1^\infty \frac{k(x)}{x} dx,$$

and the left side is finite exactly when both terms are. The second sentence is Proposition 2.3(3) applied to the pair (a, ν) . $\square$

Admissible. A pair (a, k) with $a \geq 0$, $k : (0, \infty) \rightarrow [0, \infty)$ nonincreasing and the two integrals of the criterion finite is called *admissible*, and so is the exponent (2.3) it defines; profiles equal almost everywhere define the same exponent. The zero exponent, $a = 0$ and $k = 0$, is admissible as an exponent, and it is the one exponent that (ND) excludes as a family (Theorem 7.3).

Classes referred to later. The *symmetric stable* laws have exponent $c|\omega|^\alpha$ with $0 < \alpha \leq 2$ ([31], Thm. 14.14, p. 86); §7 recovers them as the semigroup case. The *variance-gamma* laws have exponent $\gamma \log(1 + \omega^2/2)$ up to scale, and for $\gamma > \frac{1}{2}$ the Matérn kernels as densities ([26]; [12], § 2.3, for their self-decomposability); they are the second member of §3. Both classes appear as the answer to “which laws have this property”, never as tools.

Remark 2.16 (relation to the causal theory). Every object of this section has a twin in the causal article [8], and the correspondence of Table 2 is exact at the level of the classes. ND_s stands where the Bernstein class BF_0 stood, with Bochner’s theorem in the role of Bernstein–Widder and Schoenberg’s theorem in the role of “ $e^{-\tau g}$ is completely monotone for every τ ”. The representation (2.2) is the Lévy–Khintchine form of a Laplace exponent with the drift slot $b_0 s$ become the Gaussian slot $a\omega^2$ and the same profile k in (2.3). The criterion of Lemma 2.15 has $\int_0^1 x k$ where the causal criterion has $\int_0^1 k$. The vanishing lemma and the closure under limits do not transfer, and on both the line costs more than the half-line. The causal vanishing lemma, that a Laplace exponent vanishing at one interior point vanishes identically, has no Fourier analogue (Lemma 2.11), and the exclusion of lattices, free there, is here the work of covariance. Closure under pointwise limits, which holds for the Bernstein functions on $(0, \infty)$ (though not for the normalized subclass on $[0, \infty)$, since $1 - e^{-ns}$ converges to the indicator of $s > 0$), is unconditional here only in the kernel form. For the functions themselves it needs continuity of the limit at the origin, Proposition 2.3(4) and Remark 2.9. The classical characterization of self-decomposable Lévy measures transfers in the opposite direction. The causal development cites it and uses it in the analysis direction of its main theorem; here it is Proposition 2.13, cited and used nowhere, since the profile form is proved. At that point the trust base of this article is the narrower of the two.

3 Spatial measurement: the axioms

This section assembles the axiom system of the paper: a list of requirements on the measurement of a signal on the line, each motivated in turn and collected in Definition 3.1. The substantive change from [28] is one axiom, the cascade property of §3.4, and it is the subject of the paper. The smaller differences are recorded in Remark 3.2. Positivity is stated in its place in the list, but the section is arranged so that its role stays visible: §3.8 says what the other axioms admit without it, and §7 what it cuts that down to.

It is impossible to measure the value of a signal at a point: a measurement collects input from a region of nonvanishing extent, an *extended point* [13]. What an observer holds is therefore a record of the signal at some resolution, never the signal itself, and a multi-scale front end maintains a family of such records, indexed by a scale. The front end is *uncommitted*: it is built before anything is known about what the signals will contain, so the family of records cannot be derived from a signal model and must be derived from requirements on the measurement process itself. Those requirements are what this section states.

The requirements are those of Pauwels, Van Gool, Fiddelaers and Moons [28], restated for operators indexed by pairs of scales, and each is motivated in turn below. Space has no arrow: an observer may look both ways, and a record may be reread. So no causality requirement arises, and the symmetry that does, reflection, is stated in §3.2.

A spatial measurement is an operator that smooths the signal. Signals are integrable, $L^1 = L^1(\mathbb{R}, dx)$, with positive cone L^1_+ and $\int f := \int_{\mathbb{R}} f(x) dx$. This and the other notation used informally here is fixed in §2. Because a record is itself a signal, and is measured further in the cascade of §3.4, the measurement operators are indexed by ordered pairs of scales,

$$\Phi_{s,t} : L^1 \rightarrow L^1, \quad 0 \leq s \leq t < \infty,$$

with $\Phi_{s,t}$ carrying the record at scale s to the record at the coarser scale t , and the signal itself as the record at scale 0. In the classical theory a single scale index suffices. Why two are

needed here is the point of §3.4. The apertures (the measures realizing these operators) are not postulated: that the operators are convolutions is recovered from the axioms in §4, and which convolutions in §7.

The formal theory is set on the closed half-line $[0, \infty)$ of scales, with the signal as the record at scale 0 and $\Phi_{t,t} = \text{Id}$ on the diagonal. Neither is a realizable measurement. Both enter as the limits the continuity requirement of §3.5 refers to, as the Dirac measure enters the extended-point axiom of the classical theory, $\phi_t \rightarrow \delta$ as $t \rightarrow 0$: as a limit the family approaches and never attains. Nothing is measured *at* scale 0. Every statement below is about the family and its limits.

3.1 Linearity, positivity, and mass

An uncommitted front end treats all intensity levels alike and prefers no signal feature over another. Superposition is the operational form of that indifference: a device that does not respect it responds to some combinations of inputs differently from the sum of its responses, and has thereby singled those combinations out before anything is known about the signals. So each $\Phi_{s,t}$ is linear. This is the assumption on which the convolution route of §4 rests. The nonlinear scale-space axiomatics [2], which give it up, are a different theory. Measurement is also stable (a small signal leaves a small record), which on L^1 is boundedness of each $\Phi_{s,t}$ (A1).

An intensity signal is nonnegative, and a measurement of nonnegative input that reports negative values has created structure of its own, structure of the apparatus rather than of the signal. Measurement preserves the positive cone (A4). This is the axiom whose role the paper singles out (§3.8, §7).

Smoothing an image may not change its average intensity (*gray-level invariance*), so the smoothing operator is conservative: it redistributes mass and neither creates nor absorbs any. On integrable signals the invariant quantity is the total mass $\int f$, and this is (A5). A device that loses mass models absorption rather than measurement, and is not pursued.

3.2 Symmetry: translation and reflection

A front end built before its signals has no preferred position and no preferred direction on the line.

No preferred position: each $\Phi_{s,t}$ commutes with the translations $(T_a f)(x) = f(x - a)$, $a \in \mathbb{R}$, which is (A2).

No preferred direction: each $\Phi_{s,t}$ commutes with the reflection $(Rf)(x) = f(-x)$, which is (A3). Pauwels et al. impose it as evenness of the kernel. Stated on the operator it is a symmetry, and it acts in two directions at once. It admits the even apertures, the Gaussian among them, and it excludes every one-sided aperture and every aperture whose centre is displaced from the origin, that is any net drift of the records across scale. The one aperture that (A3) admits and that measures nothing, the identity, is removed by (ND) below. So the class has no degenerate member, and its boundary member is the Gaussian itself (§7).

3.3 Covariance

A front end built before its signals has no preferred unit of length either: rescaling the line must carry the family into itself. The mass-preserving dilations are

$$(D_\lambda f)(x) = \lambda^{-1} f(x/\lambda), \quad \lambda > 0.$$

One point requires care before the requirement can be stated. Nothing so far ties the scale index t to a unit of length: the scale axis is, at this stage, only an ordered parametrization of the records (which is finer, which is coarser), and the labels themselves have no metric meaning. A rescaling of the line can therefore not be asked to fix each operator. It can only be asked to

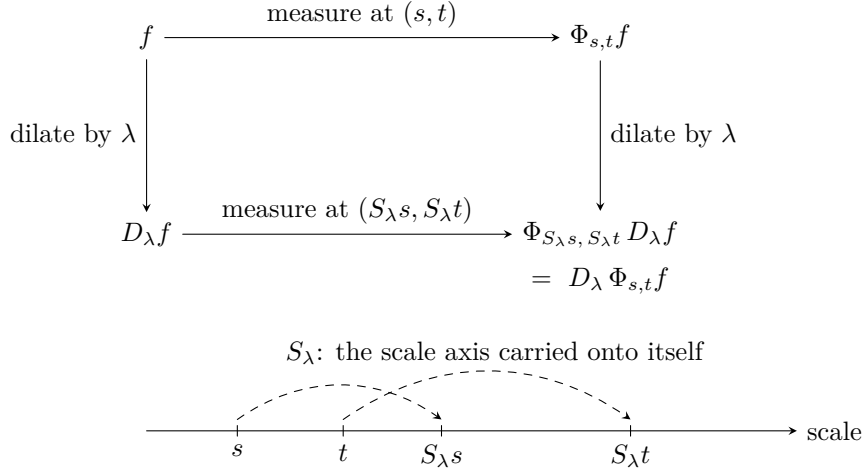


Figure 2: Scale covariance (A8). Dilating the signal by λ and then measuring is the same as measuring and then dilating, provided the measurement is taken at the transported scales $(S_\lambda s, S_\lambda t)$: the square commutes. The lower strip shows the relabelling. Nothing ties the scale labels to a unit of length at this stage, so (A8) can only ask that the family be mapped onto itself, with the labels rearranged by some increasing bijection S_λ . §6 shows that the axioms determine S_λ and thereby a canonical parametrization of the scale axis.

map the family onto itself, with the scale labels rearranged. Formally: there is a family $(S_\lambda)_{\lambda>0}$ of increasing bijections of $[0, \infty)$ with

$$D_\lambda \Phi_{s,t} = \Phi_{S_\lambda s, S_\lambda t} D_\lambda \quad \text{for all } 0 \leq s \leq t, \lambda > 0,$$

which is (A8) (Figure 2). The axioms, with the nondegeneracy of §3.5, determine each S_λ uniquely, and §6 recovers a canonical scale parametrization from the action. This is the free rescaling function of [28], and the difference is only where it enters: posited there as part of the kernel's form, it is here recovered from a covariance requirement in which it is left free. The requirement is *full* covariance, every $\lambda > 0$. The discrete form, $\lambda \in \beta^{\mathbb{Z}}$ for a fixed ratio β between successive scales, is a weaker requirement with a theory of its own, and it is outside this paper.

3.4 The cascade property

A record is itself a signal and can be measured further. A record at a coarser scale must then be computable from the record at any finer scale, without returning to the raw signal: a measurement of a measurement is a measurement, and the family closes under composition,

$$\Phi_{s,t} \Phi_{r,s} = \Phi_{r,t}, \quad r \leq s \leq t, \quad \Phi_{t,t} = \text{Id},$$

which is (A6).

This is the axiom the paper changes. In [28], as throughout the scale-space literature, the cascade is a one-parameter *semigroup*, $K_s K_t = K_{s+t}$ in their notation, and this single equation asserts two things at once (Figure 1). The first is the closure just motivated: composites of measurements are measurements. The second is tacit: the composite depends only on the total *amount* of scale crossed. Equivalently, the elementary smoothing applied between fine scales equals the one applied between coarse scales. The increments are stationary in the scale parameter (the operators commute in either reading). Nothing in the measurement picture supports the second assertion. A stage that carries a fine record to a slightly coarser one and a stage that does the same deep in the hierarchy act on different records, and there is no a priori reason the two operations should coincide.

The hemigroup form (A6) keeps the first assertion and drops the second, and nothing else: scale becomes an ordered interval structure rather than an additive quantity. Two consequences follow immediately. The family must be indexed by pairs, since an increment $\Phi_{s,t}$ is no longer determined by the difference $t - s$. And the scale axis acquires a gauge freedom (any increasing reparametrization of $[0, \infty)$ yields the same family relabelled), which is why (A8) was stated with the relabelling S_λ left free. §6 resolves the second: covariance determines the gauge up to normalization. And the semigroup case reappears when stationarity of the increments is reimposed: Corollary 7.4 returns the symmetric stable family, the positive part of the extended class.

3.5 Continuity and nondegeneracy

Measurements at nearby scales must be comparable: the record depends continuously on the pair of scales, jointly, in the norm of L^1 (A7). Together with the diagonal condition $\Phi_{t,t} = \text{Id}$, continuity is the extended point in its two-parameter form: as $t \downarrow s$ the record at scale t recovers the record at scale s .

Finally, the scale axis carries no idle stretches: strictly coarser means strictly more smoothed, $\Phi_{s,t} \neq \text{Id}$ for $s < t$ (ND). Nondegeneracy is a normalization rather than a restriction. A family that stalls on an interval of scales is the same family on a smaller scale axis.

3.6 The axiom system

Definition 3.1 (symmetric cascade measurement family). A family of linear operators $\Phi_{s,t} : L^1 \rightarrow L^1$, $0 \leq s \leq t < \infty$, is a *symmetric cascade measurement family* if:

- (A1) *Boundedness*. Each $\Phi_{s,t}$ is a bounded operator on L^1 .
- (A2) *Translation covariance*. $\Phi_{s,t} T_a = T_a \Phi_{s,t}$ for every $a \in \mathbb{R}$.
- (A3) *Reflection symmetry*. $\Phi_{s,t} R = R \Phi_{s,t}$.
- (A4) *Positivity*. $\Phi_{s,t} L_+^1 \subseteq L_+^1$.
- (A5) *Unit mass*. $\int \Phi_{s,t} f = \int f$ for every $f \in L_+^1$.
- (A6) *Cascade (hemigroup)*. $\Phi_{t,t} = \text{Id}$, and $\Phi_{s,t} \Phi_{r,s} = \Phi_{r,t}$ for all $r \leq s \leq t$.
- (A7) *Continuity*. For every $f \in L^1$ the map $(s, t) \mapsto \Phi_{s,t} f$ is continuous from $\{0 \leq s \leq t\}$ to L^1 .
- (A8) *Scale covariance*. There is a family $(S_\lambda)_{\lambda > 0}$ of increasing bijections of $[0, \infty)$ with $D_\lambda \Phi_{s,t} = \Phi_{S_\lambda s, S_\lambda t} D_\lambda$ for all $0 \leq s \leq t$ and all $\lambda > 0$.
- (ND) *Nondegeneracy*. $\Phi_{s,t} \neq \text{Id}$ whenever $s < t$.

Remark 3.2 (provenance: Pauwels et al.). The axioms of [28] are, for a family $\{K_t\}_{t \geq 0}$: a linear integral operator with an L^1 kernel, shift invariance, evenness of the kernel, unit mass, continuity of the kernel in x and in t , *recursivity* $K_s K_t = K_{s+t}$, and scale invariance with a free rescaling function, $k_t(x) = \psi(t)^{-1} \phi(x/\psi(t))$ with ψ continuous, strictly increasing, $\psi(0) = 0$ and $\psi(t) \rightarrow \infty$. Their linearity, integral form and shift invariance are (A1)–(A2) together with the representation of §4. Evenness is (A3), unit mass is (A5), and continuity in t is (A7). Recursivity is (A6) with the tacit assumption that the increments are stationary in the scale parameter, which is what this article drops. Their ψ is the relabelling S_λ of (A8), recovered rather than posited in Proposition 6.6. Two things they assume and this article does not: kernels that are continuous *functions*, and the semigroup. Measures are allowed here so that the identity is the diagonal limit, and nothing is lost, since by Proposition 7.9 every nondegenerate admissible

kernel has a density anyway. Positivity, which in their paper enters only at the end, to bound the stability index through a second moment, is an axiom here.

3.7 Two members

Two members are checked before the general theory is developed: the Gaussian family, with no jumps, and a family with jumps of every size and every moment finite. The axioms are checked on both, for the Gaussian family directly and for the second member through the exponents of its increments, which uses the admissibility criterion of the general theory (Figure 3).

Proposition 3.3 (two members). 1. (The Gaussian family.) *Let g_v be the centred Gaussian law of variance v , so that $\hat{g}_v(\omega) = e^{-v\omega^2/2}$, and put $\Phi_{s,t}f := g_{t^2-s^2} * f$ for $0 \leq s \leq t$. This family satisfies (A1)–(A8) and (ND), with $S_\lambda t = \lambda t$.*

2. (The Matérn family.) *For $\gamma > 0$ let $\rho_{s,t}$ be the symmetric probability measure with*

$$\hat{\rho}_{s,t}(\omega) = \left(\frac{1 + s^2\omega^2}{1 + t^2\omega^2} \right)^\gamma, \quad 0 \leq s \leq t,$$

*and put $\Phi_{s,t}f := \rho_{s,t} * f$. This family satisfies (A1)–(A8) and (ND), with $S_\lambda t = \lambda t$. Its kernel at scale t has mean displacement 0, variance $2\gamma t^2$, and every moment finite.*

The Gaussian clause is elementary at the kernel: variances add, which is (A6); the image of g_v under $x \mapsto \lambda x$ is $g_{\lambda^2 v}$, which is (A8); and g_v being a symmetric probability density gives (A1)–(A5). The Matérn clause is elementary in the same way once its kernels are known to be symmetric probability measures, which is the one non-obvious point: the exponent $\gamma \log(1 + \omega^2)$ is admissible, with the profile $2\gamma e^{-x}$, by Proposition A.3 in the appendix. Two clauses are not elementary at the kernel and are proved at the operator, as the proof shows: (A7), which asks for strong convergence in L^1 and not merely weak convergence of the kernels, and (ND), which asks that the operator is not the identity and not merely that the transform is not 1. Both members are machine-checked in full, every axiom including (A7), and so is the moment clause of (2), on Lean core: the mean and the variance directly, and the finiteness of every moment through the Gaussian-mixture route of Proposition A.3(3). The kernel itself is known in closed form, and the proposition does not state it: it is the symmetric variance-gamma density, cited ([12], the density (1.1) and the characteristic function (2.12)), which for $\gamma > \frac{1}{2}$ is the Matérn kernel of smoothness $\gamma - \frac{1}{2}$ and range t (the smoothness index governs the behaviour at the origin, the range is the length unit of the exponential tail; for $\gamma \leq \frac{1}{2}$ the kernel is unbounded at the origin and is a density but not a covariance function). Nothing in this paper uses the closed form.

Proof. (1) Each g_v with $v > 0$ is a symmetric probability density, and $g_0 := \delta_0$; so (A1)–(A5) hold by the converse clause of Lemma 4.1, (A3) because g_v is invariant under reflection. For (A6), $\hat{g}_{t^2-s^2} \hat{g}_{s^2-r^2} = \hat{g}_{t^2-r^2}$ and transforms determine measures (Proposition 2.7); $\Phi_{t,t} = \text{Id}$ is $g_0 = \delta_0$. For (A8) with $S_\lambda t = \lambda t$, the image of g_v under $x \mapsto \lambda x$ is $g_{\lambda^2 v}$, and $\lambda^2(t^2 - s^2) = (\lambda t)^2 - (\lambda s)^2$. For (ND): a probability measure whose convolution operator fixes every $f \in L^1$ is δ_0 — convolve a bump attaining a strict maximum at the origin, and read the resulting identity there — and $g_{t^2-s^2} = \delta_0$ only when $s = t$.

For (A7) the semigroup does the work. For $0 \leq v \leq w$ one has $g_w = g_v * g_{w-v}$, so $g_w * f - g_v * f = g_v * (g_{w-v} * f - f)$, and convolution by a probability measure is a contraction of L^1 :

$$\|g_w * f - g_v * f\|_1 \leq \|g_{w-v} * f - f\|_1.$$

Continuity in (s, t) therefore reduces to continuity of $v \mapsto g_v * f$ at $v = 0$, and that follows from the elementary estimate

$$\|\mu * f - f\|_1 \leq \int_{\mathbb{R}} \Theta_f(y) \mu(dy), \quad \Theta_f(y) := \|T_y f - f\|_1,$$

valid for every probability measure μ , which is Fubini applied to $(\mu * f - f)(x) = \int (f(x - y) - f(x)) \mu(dy)$. The function Θ_f is continuous, bounded by $2\|f\|_1$, and vanishes at $y = 0$, by strong continuity of translation on L^1 ; and g_v is the image of g_1 under $z \mapsto \sqrt{v}z$, so $\int \Theta_f dg_v = \int \Theta_f(\sqrt{v}z) g_1(dz) \rightarrow 0$ as $v \rightarrow 0$ by dominated convergence.

(2) By Proposition A.3 the function $F(\omega) = \gamma \log(1 + \omega^2)$ is admissible in the sense of Theorem 7.3, with Gaussian coefficient 0 and folded profile $k(x) = 2\gamma e^{-x}$, and $\hat{\rho}_{s,t}(\omega) = \exp[-(F(t\omega) - F(s\omega))]$; so $\rho_{s,t}$ is the increment kernel of the admissible family attached to F in the canonical gauge. That the kernel at scale t is the Matérn kernel of smoothness $\gamma - \frac{1}{2}$ and range t is the Fourier pair of the symmetric variance-gamma law ([12], (1.1) and (2.12)), cited; the variance and the finiteness of every moment are Proposition A.3(3).

The axioms themselves are read off the transform and need neither of those. (A1)–(A5) are again Lemma 4.1, the kernels being symmetric probability measures; (A6) is $\hat{\rho}_{r,s} \hat{\rho}_{s,t} = \hat{\rho}_{r,t}$ together with $\hat{\rho}_{t,t} \equiv 1$ and Proposition 2.7; (A8) with $S_\lambda t = \lambda t$ is $\hat{\rho}_{s,t}(\lambda\omega) = \hat{\rho}_{\lambda s, \lambda t}(\omega)$; and (ND) is $\hat{\rho}_{s,t}(1) = ((1 + s^2)/(1 + t^2))^\gamma < 1$ for $s < t$.

(A7) costs more than in (1), because a hemigroup has no single increment parameter: $\Phi_{s,t}$ and $\Phi_{s',t'}$ are compared through the intermediate $\Phi_{s',t}$, one endpoint at a time. Moving the left endpoint leaves an increment applied *first*, which the contraction bound absorbs as in (1); moving the right endpoint leaves one applied *last*, to $\Phi_{s',t}f$ rather than to f , so the estimate above is read at that element — which loses nothing, since translation commutes with convolution and hence $\Theta_{\mu*f} \leq \Theta_f$. Both remainders are increments whose two endpoints come together, and for those $\hat{\rho}_{a,b} \rightarrow 1$ pointwise, so $\rho_{a,b} \rightarrow \delta_0$ weakly (Proposition 2.8) and $\int \Theta_f d\rho_{a,b} \rightarrow \Theta_f(0) = 0$. On the diagonal $s = t$ the intermediate is unavailable and unnecessary: the limit operator is the identity, and one collapsing increment suffices.

Both families are verified in the machine-checked development, every axiom including (A7); the appeal to Proposition A.3 carries the *existence* of the Matérn kernels, and the cited closed form is not used in the verification of the axioms. $\square$

The Gaussian family is the member with no jumps, its profile being zero, and it is an extreme ray of the cone of admissible exponents (Remark 7.7). The Matérn family has jumps of every size. Theorem 7.3 characterizes all families satisfying the axioms.

3.8 Without positivity: an affine family through the Gaussian

In the semigroup theory positivity only trims the class. The next proposition shows that under the hemigroup it decides whether a classification exists at all: without (A4) the hemigroup axioms admit an infinite-dimensional affine family of exponents through the Gaussian, of which exactly the members of the form (2.3) are positive, so no classification in the style of Theorem 7.3 is possible without (A4). No attempt is made to describe the non-positive class beyond this.

Proposition 3.4 (without positivity: an affine family through the Gaussian). *Let $A(\mathbb{R}) := \{\hat{h} : h \in L^1\}$ be the Wiener algebra and let $\phi \in A(\mathbb{R})$ be real and even with $\phi(0) = 0$. Put $F(\omega) := \omega^2 + \phi(\omega)$ and define $\mu_{s,t}$, $0 \leq s \leq t$, by $\hat{\mu}_{s,t}(\omega) = e^{-(F(t\omega) - F(s\omega))}$. Then each $\mu_{s,t}$ is a finite symmetric signed measure, an integrable function when $s < t$, and $\Phi_{s,t} := \mu_{s,t} * \text{ satisfies (A1)–(A3), (A5)–(A8) with } S_\lambda t = \lambda t, \text{ and (ND). It satisfies (A4) if and only if } F \text{ is of the form (2.3). In particular, for } q \in L^1 \text{ even, supported in } \{|x| \geq 1\}, \text{ of mean zero and not almost everywhere nonnegative, and } \phi_\varepsilon(\omega) := \varepsilon \int_{\mathbb{R}} (1 - \cos \omega x) q(x) dx, \text{ the family violates (A4) for every } \varepsilon \neq 0.$*

Every step is elementary, and two of the three clauses are machine-checked: the positivity equivalence, forwards by reading (A4) at an approximate identity and backwards by identifying the constructed law through the Jordan decomposition of the signed kernel into its positive and negative parts, and the final sentence, that the family built from q violates (A4) for every $\varepsilon \neq 0$. What is not machine-checked is the first clause, the construction of $h_{s,t}$ in the Wiener algebra

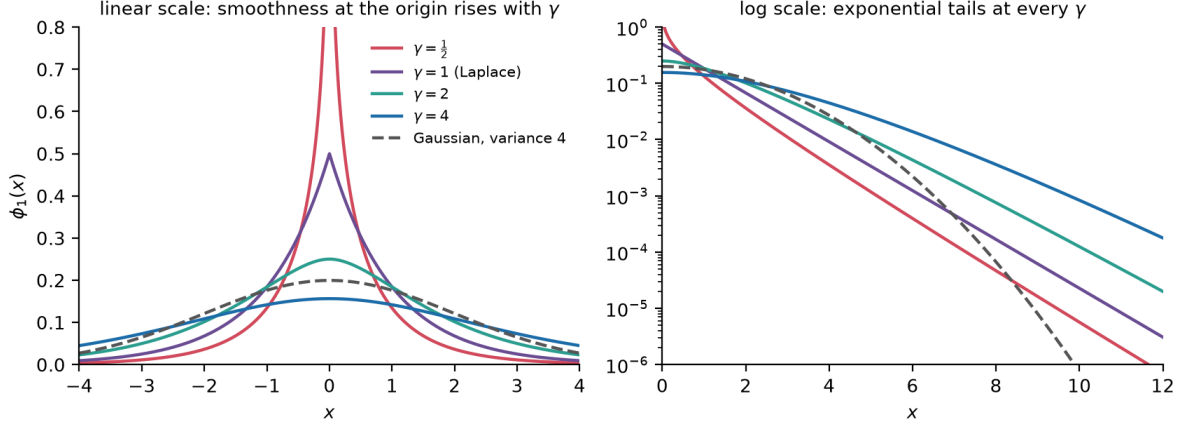


Figure 3: The two members of Proposition 3.3 at unit canonical scale, computed from their transforms. Left, linear scale: the variance-gamma densities with transform $(1 + \omega^2)^{-\gamma}$ for $\gamma = \frac{1}{2}, 1, 2, 4$, and a Gaussian of variance 4, the variance the $\gamma = 2$ member has by Proposition A.3(3); the variances of the four densities differ, 2γ each. At $\gamma = \frac{1}{2}$ the density is unbounded at the origin and the curve is clipped; at $\gamma = 1$ it is the Laplace density $e^{-|x|}/2$, with a cusp. Right, the same curves on a logarithmic scale: the four densities fall off exponentially, the Gaussian as $e^{-x^2/8}$.

$A(\mathbb{R})$ of transforms of integrable functions and the verification of the axioms for the signed family: the library used carries the L^1 Fourier transform but not $A(\mathbb{R})$ as a Banach algebra, and a route through the convolution operators in place of the functions has been estimated and not carried out. That clause rests on the printed proof alone.

Proof. The family. $A(\mathbb{R})$ is a Banach algebra under pointwise multiplication, invariant under dilation, so $\phi_{s,t} := \phi(t \cdot) - \phi(s \cdot)$ lies in it, and $e^{-\phi_{s,t}} = 1 + \hat{h}_{s,t}$ with $h_{s,t} \in L^1$, the exponential series converging in the algebra. (The map $b \mapsto \phi(b \cdot)$ is *not* norm-continuous into $A(\mathbb{R})$ at $b = 0$: for $b > 0$ it is the transform of the dilate $h_b := b^{-1}h(\cdot/b)$, of norm $\|h\|_1$, while $\phi(0 \cdot) = \phi(0) = 0$. So $h_{s,t}$ is not norm-continuous up to the edge $s = 0$, and (A7) is argued below through operators.) With $g_{2(t^2-s^2)}$ the Gaussian kernel of variance $2(t^2 - s^2)$, whose transform is $e^{-(t^2-s^2)\omega^2}$ in the normalization $\hat{g}_v = e^{-v\omega^2/2}$ of Proposition 3.3 — a density for $s < t$, and δ_0 on the diagonal — we get $\mu_{s,t} = g_{2(t^2-s^2)} + g_{2(t^2-s^2)} * h_{s,t}$: a finite signed measure, symmetric because its transform is real and even, and an integrable function for $s < t$.

The axioms. (A1) holds with $\|\mu_{s,t}\| \leq 1 + \|h_{s,t}\|_1$; (A2) because convolution commutes with translation; (A3) by symmetry; (A5) from $\hat{\mu}_{s,t}(0) = 1$; (A6) because the exponents add and transforms determine measures (Proposition 2.7); (A8) with $S_\lambda t = \lambda t$, because $F(\lambda t\omega) - F(\lambda s\omega)$ is $F(t \cdot) - F(s \cdot)$ evaluated at $\lambda\omega$; and (ND) because $F(t\omega) - F(s\omega) = (t^2 - s^2)\omega^2 + \phi_{s,t}(\omega)$ is unbounded, $\phi_{s,t}$ being bounded. For (A7), let h_e be the even part of h and $H_b f := h_b * f$ for $b > 0$, with h_b the dilate of h_e , and $H_0 := 0$. Each H_b is a bounded operator on L^1 of norm at most $\|h\|_1$, the H_b commute with one another and with the Gaussian convolutions, and $b \mapsto H_b f$ is continuous on $[0, \infty)$ for every $f \in L^1$: at $b > 0$ by strong continuity of dilation, and at $b = 0$ because h_b is an approximate identity of total mass $\int h_e = \phi(0) = 0$, so $h_b * f \rightarrow 0$ in L^1 . Then $\mu_{s,t} * f = g_{2(t^2-s^2)} * \exp(-(H_t - H_s))f$, the exponential series converging in operator norm uniformly in (s, t) , its n -th term being bounded by $(2\|h\|_1)^n/n!$. A product of uniformly bounded, strongly continuous operator families is strongly continuous, so each partial sum is, hence so is the sum, and composing with the strongly continuous Gaussian family gives (A7). The route is strong continuity throughout, and it does not need $h_{s,t}$ to depend continuously on (s, t) in norm, which it does not at $s = 0$.

Positivity. Suppose (A4) holds. The kernels $\mu_{s,t}$ are so far only integrable functions, and Theorem 7.3 is a statement about probability kernels, so the first step is to see that (A4) makes them nonnegative. Convolve with an approximate identity: $\mu_{s,t} * \rho_\varepsilon \geq 0$ for every $\varepsilon > 0$ because $\rho_\varepsilon \geq 0$, and $\mu_{s,t} * \rho_\varepsilon \rightarrow \mu_{s,t}$ in L^1 ; the positive cone of L^1 is closed, so $\mu_{s,t} \geq 0$ almost everywhere. Its total mass is $\hat{\mu}_{s,t}(0) = 1$, so $\mu_{s,t} dx$ is a probability measure, the family satisfies every axiom, and Theorem 7.3 applies. It returns a gauge χ and a profile P with $F(t\omega) - F(0) = P(\chi(t)\omega)$; reading that at $t = 1$ and using $F(0) = \phi(0) = 0$ gives $F(\omega) = P(\chi(1)\omega)$, and $\chi(1) > 0$, so F is P dilated and is itself of the form (2.3).

Conversely, suppose F has that form. Each increment exponent $\omega \mapsto F(t\omega) - F(s\omega)$ is then a member of ND_s by Lemma 7.1, hence the exponent of a symmetric probability law ν (Proposition 2.3); and ν has the same cosine transform as $\mu_{s,t}$, which is even and integrable. The Jordan parts of $\mu_{s,t}$ split it as a difference of two finite measures whose characteristic functions are determined by that cosine transform, evenness making both sine transforms vanish, so $\mu_{s,t}^+ dx = \nu + \mu_{s,t}^- dx$ by Proposition 2.7. Testing this on the set where $\mu_{s,t}$ is negative makes its left side vanish, so $\mu_{s,t}^- = 0$ almost everywhere: $\mu_{s,t} \geq 0$, and (A4) is positivity of an integral.

The example. Write $g(\omega) := \int (1 - \cos \omega x) q(x) dx$, so that $F = \omega^2 + \varepsilon g$. Since $\int q = 0$ we have $g = -\hat{q}$, so g is bounded and $g(0) = 0$; hence $\phi_\varepsilon = -\varepsilon \hat{q}$ lies in $A(\mathbb{R})$, real and even because q is, and the family is one of those described. (Mean zero is not cosmetic: without it ϕ_ε tends to $\varepsilon \int q \neq 0$ at infinity, which no member of the Wiener algebra does.)

Suppose F were of the form (2.3), with Gaussian coefficient $a \geq 0$ and Lévy measure ν on $(0, \infty)$ subject to $\int (1 \wedge x^2) \nu(dx) < \infty$. Average the two readings of F over $[0, T]$. The elementary primitive $\int_0^T (1 - \cos \omega x) d\omega = T(1 - \frac{\sin Tx}{Tx})$ and Tonelli give, for every $T > 0$,

$$\frac{T^2}{3} + \varepsilon \int \left(1 - \frac{\sin Tx}{Tx}\right) q(x) dx = \frac{aT^2}{3} + \int \left(1 - \frac{\sin Tx}{Tx}\right) \nu(dx),$$

both integrals converging absolutely because $0 \leq 1 - \frac{\sin u}{u} \leq \min(2, u^2/6)$.

Read this identity at rate T^2 . Divided by T^2 , the perturbation term vanishes in the limit because its integral is bounded by $2\|q\|_1$, and the Lévy term vanishes by dominated convergence, the integrand $T^{-2}(1 - \frac{\sin Tx}{Tx})$ being at most $2(1 \wedge x^2)$ for $T \geq 1$ and tending to 0 pointwise. Hence $a = 1$, and the identity reduces to $\int (1 - \frac{\sin Tx}{Tx}) \nu(dx) = \varepsilon \int (1 - \frac{\sin Tx}{Tx}) q(x) dx$.

Now read it at rate 1. The right-hand side tends to $\varepsilon \int q = 0$ as $T \rightarrow \infty$, by dominated convergence against $|q|$. On the left, $1 - \frac{\sin Tx}{Tx} \rightarrow 1$ at every $x \neq 0$ and ν gives the origin no mass, so Fatou forces $\nu((0, \infty)) = 0$. With $a = 1$ and $\nu = 0$, (2.3) reads $F(\omega) = \omega^2$, that is $\varepsilon g \equiv 0$, that is $\hat{q} \equiv 0$ because $\varepsilon \neq 0$. An even integrable function whose cosine transform vanishes identically is null: its positive and negative parts are two finite measures whose cosine transforms agree and whose sine transforms both vanish, evenness being what kills the latter, so Proposition 2.7 identifies them. Thus $q = 0$ almost everywhere, against the hypothesis that q is not almost everywhere nonnegative — and both signs of ε are covered, neither the identity nor either reading of it seeing the sign. So F is not of the form (2.3), and (A4) fails by the previous paragraph. $\square$

Remark 3.5 (the route through signed uniqueness). The last clause of the proposition, that the family built from q violates (A4), is usually argued the other way round: $\omega^2 + \phi_\varepsilon$ is a signed Lévy–Khinchine representation, with Gaussian coefficient 1 and signed measure $\varepsilon q(x) dx$, and such representations are unique among signed measures m with $\int (1 \wedge x^2) |m|(dx) < \infty$. Since εq is not a nonnegative measure, F is not of the form (2.3). The uniqueness is the averaging identity

$$\frac{1}{2h} \int_{\omega-h}^{\omega+h} \psi(u) du - \psi(\omega) = \frac{ah^2}{3} + \int \cos(\omega x) \left(1 - \frac{\sin hx}{hx}\right) m(dx)$$

for $\psi(\omega) = a\omega^2 + \int (1 - \cos \omega x) m(dx)$, whose right-hand side is a constant plus the Fourier–Stieltjes transform of the finite signed measure $(1 - \frac{\sin hx}{hx}) m(dx)$; so ψ determines that measure

for every $h > 0$, hence m off the origin, hence a . This is the more informative statement, because it is about the representation rather than about one F . It is also strictly more than the proposition needs, and it is **not machine-checked**: the proof of record above replaces it by the two readings of the averaged identity, at the rates T^2 and 1, which identify a and ν for the single exponent at hand and never mention a signed measure.

Under recursivity the same setting forces $F(\omega) = a|\omega|^\alpha$ with $\alpha > 0$: a one-parameter class whether or not positivity is imposed, positivity entering only to bound $\alpha \leq 2$ ([28], Prop. 1). Under the hemigroup, dropping positivity admits the affine family $\omega^2 + A(\mathbb{R})_{\text{even},0}$ through the Gaussian, and the same family through every admissible exponent unbounded at infinity. The positive members of the first are exactly those of the form (2.3), and the others have no description in the form of the theorem.

Remark 3.6 (relation to the causal theory). (A1), (A2), (A4)–(A8) and (ND) are the axioms of the time-causal article [8] up to the change of letters, and (A3) replaces causality by reflection symmetry: the axiom that removed the Gaussian there is the axiom that admits it here. Three consequences are visible already. The convolution representation yields symmetric probability measures on $\mathbb{R}$ rather than probability measures on a half-line, and the Fourier transform can vanish where the Laplace transform cannot, so a nonvanishing lemma (Lemma 4.2) replaces the positivity of the causal integrand. The drift slot of the causal profile representation is occupied here by the Gaussian coefficient, so the member that stands where the pure delay stood is the Gaussian scale space itself, admissible, extreme and nondegenerate. And the gauge argument, which used the causal vanishing lemma twice, is re-routed through a smoothed transform (Corollary 5.6) and a dilation argument (Lemma 6.5), the exclusion of lattice laws following from covariance (Lemma 6.2) rather than from the function class.

4 Convolution representation

This section derives the form of the measurement operators from the axioms that do not involve the cascade or covariance. The aim is Lemma 4.1: under (A1)–(A5) every operator $\Phi_{s,t}$ is a convolution by a symmetric probability measure $\mu_{s,t}$ on $\mathbb{R}$, so the apertures of §3 are measures. One operator $\Phi_{s,t}$, the measurement that carries the record at scale s to the record at scale t , is called a *stage* below, and $\mu_{s,t}$ is its kernel. The lemma has a converse, that every symmetric probability measure defines an operator satisfying (A1)–(A5), and the constructive half of the characterization in §7 builds its families through that converse. Whether the kernels have densities is settled later, once covariance and nondegeneracy are imposed (Proposition 7.9).

Read at the kernel, the lemma gives the picture the rest of the paper uses. A stage reports the signal at a randomly displaced position, its kernel is the law of that displacement, and the transform of the kernel is the stage's modulation transfer function (Remark 4.3). The second result of the section, Lemma 4.2, adds the cascade (A6) and continuity (A7), drops positivity, and shows that the transforms of the kernels do not vanish. That is what lets the rest of the article work with one function of each kernel, its exponent $-\log \hat{\mu}_{s,t}$.

The device. The characters $e_\omega(x) := e^{-i\omega x}$ are bounded on all of $\mathbb{R}$, so $e_\omega \in L^\infty$ and complex-valued functionals on the real space L^1 may be used freely, $\langle w, f \rangle := \int_{\mathbb{R}} w f$. Equivalently one pairs with $\cos \omega x$ and $\sin \omega x$. For a finite measure μ and $f \in L^1$, Fubini gives

$$\langle e_\omega, \mu * f \rangle = \hat{\mu}(\omega) \hat{f}(\omega), \quad \omega \in \mathbb{R}. \quad (4.1)$$

With f the standard Gaussian density, $\hat{f}(\omega) = e^{-\omega^2/2} > 0$, so a single test function reads off the transform of the kernel at every frequency.

Lemma 4.1 (representation). *Under (A1), (A2), (A4), (A5), for each pair $0 \leq s \leq t$ there is a unique probability measure $\mu_{s,t}$ on $\mathbb{R}$ with*

$$\Phi_{s,t} f = \mu_{s,t} * f, \quad f \in L^1;$$

under (A3) it is symmetric, $R\mu_{s,t} = \mu_{s,t}$. Conversely every symmetric probability measure on $\mathbb{R}$ defines an operator satisfying (A1)–(A5).

The lemma states symmetry as a separate clause, under (A3), rather than assuming (A1)–(A5) at once, because the representation is also of interest without positivity. Proposition 3.4 needs a representation of operators satisfying (A1)–(A3) and (A5) but not (A4). That is not what the lemma gives, in either of its forms; it is what Wendel’s theorem gives (Remark 4.4).

Proof. Fix (s, t) and write $\Phi = \Phi_{s,t}$.

Convolution as a vector-valued integral. For $f, g \in L^1$, $f * g = \int_{\mathbb{R}} f(y) T_y g dy$ as a Bochner integral in L^1 : the integrand is strongly measurable, since $y \mapsto T_y g$ is continuous because translation acts continuously on L^1 , and it is dominated by $|f(y)| \|g\|_1$. An L^1 -valued integral commutes with every bounded operator, so (A1) and (A2) give the interchange identity $\Phi(f * g) = f * \Phi g$.

The approximants. Let $\rho_\varepsilon := (2\varepsilon)^{-1} \mathbf{1}_{(-\varepsilon, \varepsilon)}$ and $h_\varepsilon := \Phi \rho_\varepsilon$. By (A4) and (A5), $h_\varepsilon \geq 0$ with $\int h_\varepsilon = 1$, so $\nu_\varepsilon := h_\varepsilon dx$ is a probability measure on $\mathbb{R}$.

Tightness. By the interchange identity $\rho_1 * h_\varepsilon = \Phi(\rho_1 * \rho_\varepsilon)$, and $\rho_1 * \rho_\varepsilon \rightarrow \rho_1$ in L^1 as $\varepsilon \downarrow 0$, so $\rho_1 * h_\varepsilon$ converges in L^1 ; and a convergent sequence in L^1 has uniformly small tails. Convolution with a nonnegative unit-mass density carried by $[-1, 1]$ moves mass by at most 1: for $|y| > R + 1$ the translate $\rho_1(\cdot - y)$ is carried by $\{|x| > R\}$, so by Tonelli

$$\int_{|x| > R} (\rho_1 * h_\varepsilon)(x) dx = \int_{\mathbb{R}} h_\varepsilon(y) \int_{|x| > R} \rho_1(x - y) dx dy \geq \int_{|y| > R+1} h_\varepsilon(y) dy = \nu_\varepsilon(\{|y| > R+1\}).$$

Along $\varepsilon_n = 1/(n + 1)$ the tails of ν_{ε_n} are therefore small uniformly in n : the family is tight, with compacts $[-R - 1, R + 1]$.

The limit. By Prokhorov’s theorem a subsequence $\nu_{\varepsilon_{n_k}}$ converges weakly to a probability measure μ on $\mathbb{R}$.

Identification. Let $\Psi \in L^\infty$ and $f \in L^1$. The map $y \mapsto \Psi(T_y f)$ is continuous and bounded by $\|\Psi\| \|f\|_1$, hence a legitimate test function for weak convergence. Reading $f * h_\varepsilon = \int h_\varepsilon(y) T_y f dy$ with h_ε in the density slot,

$$\Psi(f * h_{\varepsilon_{n_k}}) = \int_{\mathbb{R}} \Psi(T_y f) \nu_{\varepsilon_{n_k}}(dy) \xrightarrow{k \rightarrow \infty} \int_{\mathbb{R}} \Psi(T_y f) \mu(dy) = \Psi(\mu * f),$$

the last equality because $\mu * f = \int T_y f \mu(dy)$ is again an L^1 -valued Bochner integral. On the other hand $f * h_\varepsilon = \Phi(f * \rho_\varepsilon) \rightarrow \Phi f$ in L^1 , by the interchange identity and (A1). Hence $\Psi(\Phi f) = \Psi(\mu * f)$ for every $\Psi \in L^\infty$, and $\Phi f = \mu * f$ for every $f \in L^1$ at once.

Uniqueness. Take f the standard Gaussian density and pair against e_ω : by (4.1), $\langle e_\omega, \Phi f \rangle = \hat{\mu}(\omega) e^{-\omega^2/2}$, so Φ determines $\hat{\mu}$ on all of $\mathbb{R}$, and a finite measure on $\mathbb{R}$ is determined by its transform (Proposition 2.7). One test function suffices.

Symmetry. Under (A3), $R(\mu * f) = (R\mu) * (Rf)$ gives $(R\mu) * f = R\Phi Rf = \Phi f = \mu * f$ for every f , so $R\mu = \mu$ by uniqueness.

Converse. A probability measure μ on $\mathbb{R}$ makes $f \mapsto \mu * f$ bounded with norm at most 1 by Young’s inequality (A1), translation-commuting because convolution is (A2), positive (A4), and mass-preserving by Tonelli (A5); if μ is symmetric it commutes with R (A3). $\square$

The rest of the article works with one function of the measures, the *exponent* $g_{s,t}(\omega) := -\log \hat{\mu}_{s,t}(\omega)$, the quantity the cascade makes additive (§5). For it to exist the transform must be positive, which on the line is not automatic: a symmetric probability measure can have a negative or vanishing transform, as the uniform density on $[-1, 1]$ does, with $\hat{\mu}(\omega) = \sin \omega / \omega$. Excluding this stage by stage needs the cascade and continuity beyond (A1)–(A5).

Lemma 4.2 (nonvanishing of the transforms). *Assume (A1)–(A3), (A5)–(A7), and let $\Phi_{s,t}f = \mu_{s,t} * f$ be the representation of Lemma 4.1. Then $\hat{\mu}_{s,t}(\omega) > 0$ for every $\omega \in \mathbb{R}$ and every $s \leq t$. Consequently $g_{s,t}(\omega) := -\log \hat{\mu}_{s,t}(\omega) \in [0, \infty)$ is well defined and even, is continuous in ω and continuous in (s, t) on $\{0 \leq s \leq t\}$, and satisfies $g_{t,t} = 0$ and $g_{s,t}(0) = 0$.*

The lemma is elementary but two-axiom: the multiplicativity of (A6) and the continuity of (A7) are both used, and no clause of Proposition 2.3 is. Positivity enters only through the hypothesis: the lemma takes the kernels of Lemma 4.1 as given, and those are probability measures, which is what (A4) supplies. Without that the conclusion fails: a signed family of unit mass can have transforms above 1 and negative exponents, as the signed compound Poisson family $e^h \sum_n (-h)^n \sigma^{*n} / n!$ with $\sigma = \frac{1}{2}(\delta_{-1} + \delta_1)$ and $h = t - s$ shows, its transform being $e^{h(1-\cos \omega)}$. What the lemma does not use is (A4) beyond that. It is the hands-on precursor of the classical fact that an infinitely divisible law has a zero-free characteristic function ([31], Lemma 7.5, p. 32), which Theorem 5.2 establishes for the stages. The order of dependence cannot be reversed, since that theorem’s proof manipulates the exponents this lemma defines.

Proof. By (A3) and the uniqueness clause of Lemma 4.1, $R\mu_{s,t} = \mu_{s,t}$, so $\hat{\mu}_{s,t}$ is real and even.

Continuity in (s, t) . Let ρ be the standard Gaussian density, with $\hat{\rho}(\omega) = e^{-\omega^2/2} > 0$. If $(s_n, t_n) \rightarrow (s, t)$ then $\mu_{s_n, t_n} * \rho \rightarrow \mu_{s,t} * \rho$ in L^1 by (A7), and

$$|\hat{\mu}_{s_n, t_n}(\omega) \hat{\rho}(\omega) - \hat{\mu}_{s,t}(\omega) \hat{\rho}(\omega)| \leq \|\mu_{s_n, t_n} * \rho - \mu_{s,t} * \rho\|_1;$$

divide by $\hat{\rho}(\omega)$. The test density must have a zero-free transform, which is why an indicator — whose transform vanishes on $2\pi\mathbb{Z} \setminus \{0\}$ — would not do.

Nonvanishing. Fix s and ω . The function $t \mapsto \hat{\mu}_{s,t}(\omega)$ is continuous on $[s, \infty)$, equals 1 at $t = s$, and is multiplicative over adjacent intervals by (A6): $\hat{\mu}_{s,t} = \hat{\mu}_{t',t} \hat{\mu}_{s,t'}$ for $s \leq t' \leq t$. Let $Z := \{t \geq s : \hat{\mu}_{s,t}(\omega) = 0\}$, a closed set. If $Z \neq \emptyset$, put $t_* := \min Z$, which exceeds s . For $t' \in (s, t_*)$ we have $0 = \hat{\mu}_{s,t_*}(\omega) = \hat{\mu}_{t',t_*}(\omega) \hat{\mu}_{s,t'}(\omega)$ with the second factor nonzero by minimality, so $\hat{\mu}_{t',t_*}(\omega) = 0$; letting $t' \uparrow t_*$, joint continuity gives $\hat{\mu}_{t_*,t_*}(\omega) = 0$, against $\hat{\mu}_{t_*,t_*} = \hat{\delta}_0 = 1$. So $Z = \emptyset$: the function is continuous and zero-free on the interval $[s, \infty)$ and equals 1 at its left end, hence is positive throughout, by the intermediate value theorem.

The same conclusion follows without an extremal element, and this is the form the formalisation takes. Fix $s \leq t$. The triangle $\{(a, b) : s \leq a \leq b \leq t\}$ is compact and $(a, b) \mapsto \hat{\mu}_{a,b}(\omega)$ is continuous on it, hence uniformly continuous; since the function is 1 on the diagonal, there is a single $\delta > 0$ with $\hat{\mu}_{a,b}(\omega) > \frac{1}{2}$ whenever $s \leq a \leq b \leq t$ and $b - a < \delta$. Put $r_n := \min(s + n\delta/2, t)$. Then $r_0 = s$, $r_n = t$ for n large, consecutive steps are shorter than δ , and $\hat{\mu}_{s, r_{n+1}} = \hat{\mu}_{r_n, r_{n+1}} \hat{\mu}_{s, r_n}$, so induction on n carries positivity from s to t . Both arguments use exactly the joint continuity of the first step and nothing else.

The clauses on g . $g \geq 0$ since $|\hat{\mu}| \leq \|\mu\| = 1$; $g_{s,t}(0) = 0$ by (A5); continuity in ω by dominated convergence; evenness and continuity in (s, t) from the corresponding properties of $\hat{\mu}$; and $g_{t,t} = 0$ from $\mu_{t,t} = \delta_0$. $\square$

Remark 4.3 (modulation transfer; no contrast reversal). On the line the probe e_ω is a wave and ω a wavenumber, so $\hat{\mu}_{s,t}(\omega) = \mathbb{E} \cos(\omega X_{s,t})$ is the stage’s *modulation transfer function*, the contrast a sinusoidal grating of frequency ω retains after the random displacement $X_{s,t}$, and the exponent is minus its logarithm. Lemma 4.2 then says, in the language of imaging: no stage in a continuous cascade reverses contrast, none extinguishes a frequency as the box blur does at the zeros of $\sin \omega/\omega$, and none amplifies. The uniform blur is thereby excluded as a stage of any family of probability kernels satisfying (A6)–(A7), by an argument that does not yet know the stages are infinitely divisible. The lemma says nothing about the sign of the *kernel*. The extended semigroup class of [28] beyond $\alpha = 2$ has positive transfer functions and sign-changing kernels. Those members are excluded by positivity (A4), through Theorem 7.3, and not by this lemma. Transfer functions multiply along the cascade, so optical densities add

(§5), and the family is self-similar (§6). The reading of the density's two parts, a Gaussian part and a catalogue of jump displacements, is Remark 2.6.

Remark 4.4 (what the representation is, and the relation to the causal theory). Wendel's theorem states that the translation-invariant bounded operators on the L^1 of a locally compact group are exactly the convolutions by finite measures [36]. Lemma 4.1 is that step specialised to $\mathbb{R}$ and sharpened by (A3)–(A5) to a symmetric probability measure. It is proved rather than cited. The only imported fact is Proposition 2.7. Against the causal version in [8], two devices have fallen away and one has been added. The clamped exponential is gone, since characters are bounded on all of $\mathbb{R}$, which is why (4.1) needs no support hypothesis. The causal support clause is gone, and with it the one-sided tail bound of the tightness step. Its replacement, that convolution with a density carried by $[-1, 1]$ moves mass by at most 1, is simpler and would have served there too. What has been added is Lemma 4.2, which on the half-line was a one-line consequence of the positivity of the integrand. Its continuity step tests against a Gaussian where the causal proof could test against an indicator.

5 The cascade: additivity and infinite divisibility

With the representation in hand, the cascade axiom (A6) becomes a statement about kernels: composition of operators is convolution of measures, so the exponent of §4 is additive over adjacent intervals of the scale axis. The optical densities of stacked stages add (Remark 4.3). Set

$$G(t, \omega) := g_{0,t}(\omega),$$

the density accumulated from scale 0 to scale t . The exponent of any increment is a difference of its values, so $G(\cdot, \omega)$ is nondecreasing; every increment has the symmetric Lévy–Khintchine form (2.2), which is the hemigroup form of infinite divisibility; and, under nondegeneracy, $G(\cdot, \omega)$ is strictly increasing for all but countably many ω . The coordinate that §6 uses on the scale axis is therefore not $G(\cdot, \omega_0)$ but a smoothed transmittance, Corollary 5.6, which is strictly monotone with no exceptions. No covariance is used anywhere in this section: everything below holds for every family satisfying (A1)–(A7), with (ND) added only where a statement says so, including the lattice family of Remark 5.4.

Lemma 5.1 (additivity). *Under (A1)–(A7), $\mu_{r,t} = \mu_{s,t} * \mu_{r,s}$ for $r \leq s \leq t$, hence*

$$g_{r,t}(\omega) = g_{r,s}(\omega) + g_{s,t}(\omega), \quad g_{t,t} = 0.$$

Consequently $g_{s,t}(\omega) = G(t, \omega) - G(s, \omega)$, where $G(0, \cdot) = 0$; $G(\cdot, \omega)$ is nondecreasing and continuous for each ω ; and $G(t, \cdot)$ is even and continuous with $G(t, 0) = 0$.

Proof. (A6) gives $\Phi_{s,t}\Phi_{r,s} = \Phi_{r,t}$, which by Lemma 4.1 reads $\mu_{s,t} * (\mu_{r,s} * f) = \mu_{r,t} * f$ for every $f \in L^1$, so $\mu_{r,t} = \mu_{s,t} * \mu_{r,s}$ by the uniqueness clause there. Fourier transforms turn convolution into multiplication and hence $-\log$ into addition, which is legitimate because $\hat{\mu} > 0$ (Lemma 4.2); and $\Phi_{t,t} = \text{Id}$ gives $\mu_{t,t} = \delta_0$ and $g_{t,t} = 0$. The remaining clauses are Lemma 4.2, together with $g_{s,t} \geq 0$ for the monotonicity of $G(\cdot, \omega)$. $\square$

In the semigroup case the next step is trivial, $\mu_t = \mu_{t/n}^{*n}$ exhibiting every kernel as a convolution power. In the hemigroup case there is no such power, and what replaces it is a null-array limit. The idea is the following. Split the increment from s to t into n consecutive stages. By additivity the exponent $g_{s,t}$ is the sum of the n small exponents, and by continuity every one of them is close to zero once n is large. For a small exponent g_i the kernel μ_i is close to a point mass at the origin, and $g_i(\omega) \approx 1 - \hat{\mu}_i(\omega) = \int (1 - \cos \omega x) \mu_i(dx)$. Summing over the stages, $g_{s,t}(\omega)$ is the limit of $\int (1 - \cos \omega x) \Pi_n(dx)$, where Π_n is the sum of the n kernels. If the measures Π_n converged to a measure ν , the limit would be of the form (2.2). They do not:

Π_n has total mass n , and that mass piles up at the origin. The remedy is to move the factor $1 \wedge x^2$ from the integrand to the measure, writing $\int (1 - \cos \omega x) \Pi_n(dx) = \int \frac{1 - \cos \omega x}{1 \wedge x^2} \varrho_n(dx)$ with $\varrho_n(dx) := (1 \wedge x^2) \Pi_n(dx)$. The new integrand is bounded, and near the origin it tends to $\omega^2/2$, so the mass that piles up there produces the Gaussian term $a\omega^2$ rather than a divergence. The measures ϱ_n have bounded mass, and a limit of them along a subsequence gives the pair (a, ν) .

One obstacle remains. The new integrand oscillates as $x \rightarrow \infty$ instead of converging, so it does not extend continuously to $x = \infty$, and mass of ϱ_n escaping to infinity would be lost from the limit rather than accounted for. So it has to be shown, before the limit is taken, that no mass escapes: the family ϱ_n is tight. The tool is a truncation inequality, and its only input is that $g_{s,t}$ is continuous at $\omega = 0$ with $g_{s,t}(0) = 0$.

Theorem 5.2 (increments are symmetric Lévy exponents). *Under (A1)–(A7), $g_{s,t} \in \text{LE}_s$ for every $s \leq t$: there are a Gaussian coefficient $a(s,t) \geq 0$ and a measure $\nu_{s,t}$ on $(0, \infty)$ with $\int_{(0,\infty)} (1 \wedge x^2) \nu_{s,t}(dx) < \infty$ and*

$$g_{s,t}(\omega) = a(s,t)\omega^2 + \int_{(0,\infty)} (1 - \cos \omega x) \nu_{s,t}(dx), \quad (5.1)$$

the pair being unique by Proposition 2.3(3). In particular every kernel $\mu_{s,t}$ is infinitely divisible.

The representation is extracted directly, and the closure clause Proposition 2.3(4) is never used. What Proposition 2.3(3) supplies is uniqueness of the pair and, for the last sentence, the converse direction. The null-array limit itself, including the truncation inequality that excludes escape of mass, is proved below, and Prokhorov's theorem is the one classical fact it stands on. It is a theorem of the library, on Lean core, and not a cited interface. The proof is the machine-checked route: its two constants, u^2 and $\frac{2}{3\pi^2}$, are those of the verified inequalities rather than the sharp $u^2/2$ and $\frac{2}{15}$, which are recorded at the end of the second step and are not machine-checked. In the Lean development the representation itself rests on Lean core alone, and only the closing sentence uses two clauses of the toolbox, Proposition 2.3(1) and the converse half of (3).

Proof. Fix $s < t$ and the uniform partition $s_i = s + \frac{i}{n}(t - s)$, $i = 0, \dots, n$; write $\mu_i := \mu_{s_i, s_{i+1}}$ and $g_i := g_{s_i, s_{i+1}}$, so that $g_{s,t} = \sum_{i < n} g_i$ for every n , by Lemma 5.1.

The null array. Fix ω . $G(\cdot, \omega)$ is uniformly continuous on $[s, t]$, so $m_n := \max_i g_i(\omega) \rightarrow 0$. For $u \geq 0$ we have $0 \leq u - (1 - e^{-u}) \leq u^2$: the left inequality is $e^v \geq 1 + v$ at $v = -u$, and the right one is the same inequality at $v = u$ followed by $e^{-u} \leq (1 + u)^{-1} \leq 1 - u + u^2$. Hence

$$\left| g_{s,t}(\omega) - \sum_i (1 - e^{-g_i(\omega)}) \right| \leq \sum_i g_i(\omega)^2 \leq m_n g_{s,t}(\omega) \rightarrow 0.$$

On the other hand, with $\Pi_n := \sum_i \mu_i$, a symmetric finite measure of mass n ,

$$\sum_i (1 - e^{-g_i(\omega)}) = \sum_i (1 - \hat{\mu}_i(\omega)) = \int_{\mathbb{R}} (1 - \cos \omega x) \Pi_n(dx) = \int_{(0,\infty)} (1 - \cos \omega x) \tilde{\Pi}_n(dx),$$

where $\tilde{\Pi}_n$ is the image of Π_n under $x \mapsto |x|$, the integrand being even; the origin carries no weight, since $1 - \cos 0 = 0$, so $\tilde{\Pi}_n$ may be taken on $(0, \infty)$.

Two bounds, uniform in n . Both come from one inequality. For $\eta > 0$, Tonelli and the identity $\eta^{-1} \int_0^\eta (1 - \cos \omega x) d\omega = 1 - \frac{\sin \eta x}{\eta x}$ give

$$\int_{(0,\infty)} \left(1 - \frac{\sin \eta x}{\eta x}\right) \tilde{\Pi}_n(dx) = \frac{1}{\eta} \int_0^\eta \sum_i (1 - \hat{\mu}_i(\omega)) d\omega \leq \frac{1}{\eta} \int_0^\eta g_{s,t}(\omega) d\omega =: \bar{g}(\eta), \quad (5.2)$$

using $1 - e^{-u} \leq u$ and Lemma 5.1. The function $1 - \frac{\sin u}{u}$ is nonnegative and satisfies

$$1 - \frac{\sin u}{u} \geq \frac{2}{3\pi^2} (1 \wedge u^2) \quad (u \in \mathbb{R}), \quad (5.3)$$

with the convention $\frac{\sin 0}{0} = 1$. Both sides being even, take $u > 0$. For $0 < u \leq \pi$, integrate the elementary bound $1 - \cos v \geq \frac{2}{\pi^2} v^2$, valid for $|v| \leq \pi$, over $[0, u]$ against the identity $u - \sin u = \int_0^u (1 - \cos v) dv$; this gives $u - \sin u \geq \frac{2}{3\pi^2} u^3$, that is $1 - \frac{\sin u}{u} \geq \frac{2}{3\pi^2} u^2 \geq \frac{2}{3\pi^2} (1 \wedge u^2)$. For $u > \pi$ the crude bound $\frac{\sin u}{u} \leq \frac{1}{u} < \frac{1}{\pi}$ leaves $1 - \frac{\sin u}{u} > 1 - \frac{1}{\pi} > \frac{2}{3} > \frac{2}{3\pi^2} \geq \frac{2}{3\pi^2} (1 \wedge u^2)$. *Mass:* with $\eta = 1$, (5.2) and (5.3) bound the weighted measures $\varrho_n := (1 \wedge x^2) \tilde{\Pi}_n$ on $(0, \infty)$ by $\varrho_n((0, \infty)) \leq \frac{3\pi^2}{2} \bar{g}(1)$. *Tightness:* for $x \geq 1/\eta$ the argument $u = \eta x$ has $1 \wedge u^2 = 1$, so (5.2) and (5.3) give $\frac{2}{3\pi^2} \tilde{\Pi}_n([1/\eta, \infty)) \leq \bar{g}(\eta)$, and $\bar{g}(\eta) \rightarrow 0$ as $\eta \downarrow 0$ because $g_{s,t}$ is continuous with $g_{s,t}(0) = 0$ (Lemma 4.2); since $\varrho_n \leq \tilde{\Pi}_n$ on $[1, \infty)$, the tails of ϱ_n are small uniformly in n . This is the truncation inequality of characteristic-function theory, proved here in the few lines it takes; it is where continuity at $\omega = 0$ does quantitative work.

Sharper constants are available and are not used: Taylor's theorem gives $u - (1 - e^{-u}) \leq u^2/2$ and, through $\sin u \leq u - u^3/6 + u^5/120$, $1 - \frac{\sin u}{u} \geq \frac{2}{15} (1 \wedge u^2)$. Neither is worth its derivative argument here — the first constant multiplies a quantity that is going to zero, the second one that has only to be finite — and neither is machine-checked.

The limit. A family of finite measures on $[0, \infty)$ of uniformly bounded mass and uniformly small tails has a weakly convergent subsequence: pass first to a subsequence along which the total masses converge; if their limit is 0 the measures converge weakly to the zero measure, and otherwise Prokhorov's theorem applies to the normalised measures, which are tight. Let $\varrho_{n_k} \rightarrow \varrho$, a finite measure on $[0, \infty)$, the subsequence chosen once and independently of ω . The function

$$k_\omega(x) := \frac{1 - \cos \omega x}{1 \wedge x^2}, \quad k_\omega(0) := \frac{1}{2} \omega^2,$$

is continuous on $[0, \infty)$ — the singularity is removable, since $1 - \cos u = \frac{1}{2} u^2 + O(u^4)$ — and bounded by $\max(\frac{1}{2} \omega^2, 2)$, so $\int k_\omega d\varrho_{n_k} \rightarrow \int k_\omega d\varrho$, while $\int k_\omega d\varrho_n = \int (1 - \cos \omega x) \tilde{\Pi}_n(dx) \rightarrow g_{s,t}(\omega)$ by the first step. Reading ϱ in two pieces, the atom at 0 and the rest,

$$g_{s,t}(\omega) = \frac{1}{2} \varrho(\{0\}) \omega^2 + \int_{(0, \infty)} (1 - \cos \omega x) \frac{\varrho(dx)}{1 \wedge x^2},$$

which is (5.1) with $a(s, t) := \frac{1}{2} \varrho(\{0\})$ and $\nu_{s,t}(dx) := (1 \wedge x^2)^{-1} \varrho(dx)$ on $(0, \infty)$, for which $\int (1 \wedge x^2) \nu_{s,t} = \varrho((0, \infty)) < \infty$. The atom at 0 is the Gaussian coefficient, as in the causal theory it was the drift; there is no atom at infinity to exclude, because tightness has already forbidden it, and correspondingly no killing term. The case $s = t$ is $g_{t,t} = 0$, with the pair $(0, 0)$.

Infinite divisibility. For each n , $g_{s,t}/n$ is of the form (2.2) with pair $(a/n, \nu/n)$, so by the converse half of Proposition 2.3(3) and by Proposition 2.3(1), $e^{-g_{s,t}/n}$ is the transform of a symmetric probability measure, whose n -fold convolution has transform $e^{-g_{s,t}} = \hat{\mu}_{s,t}$ and hence is $\mu_{s,t}$ by Proposition 2.7. $\square$

Note the order of dependence. Infinite divisibility implies a zero-free transform, so Lemma 4.2 is now also a corollary of the theorem. But the theorem's proof used the lemma to define the exponents it manipulates, and the order cannot be reversed.

Corollary 5.3 (monotonicity, and where it is strict). *Assume in addition (ND). For $s < t$ the zero set $N_{s,t} := \{\omega : g_{s,t}(\omega) = 0\}$ is a closed proper subgroup of $\mathbb{R}$, hence $\{0\}$ or a lattice $c\mathbb{Z}$ with $c > 0$, and it is a lattice exactly when $\mu_{s,t}$ is a lattice law. Consequently $G(\cdot, \omega)$ is strictly increasing on $[0, \infty)$ for every ω outside the countable set $E := \bigcup N_{p,q}$, the union over rational $0 \leq p < q$.*

Proof. $N_{s,t} = N(\mu_{s,t})$ in the notation of Lemma 2.11, a closed subgroup, proper because $\mu_{s,t} \neq \delta_0$ by (ND) and the uniqueness clause of Lemma 4.1. If $[p, q] \subseteq [s, t]$ then $g_{s,t} = g_{s,p} + g_{p,q} + g_{q,t}$ with all three terms nonnegative (Lemmas 5.1 and 4.2), so $N_{s,t} \subseteq N_{p,q}$; every interval $[s, t]$ with $s < t$ contains one with rational endpoints, so every $N_{s,t}$ is contained in some $N_{p,q}$ with $p < q$ rational, and E is a countable union of lattices, hence countable, and it contains 0, every $N_{p,q}$ being a subgroup. For $\omega \notin E$ and $s < t$ we get $g_{s,t}(\omega) > 0$, and $G(t, \omega) - G(s, \omega) = g_{s,t}(\omega)$ is Lemma 5.1. $\square$

This is as much as the line gives before covariance: the exceptional frequencies are exactly those at which some increment is a lattice law, and strict monotonicity at every frequency becomes true again only after the classification, Proposition 7.8(2). The exceptional set can be larger than $\{0\}$.

Remark 5.4 (the exceptional set is not $\{0\}$). For the symmetric compound Poisson family on $\mathbb{Z}$ with $\hat{\mu}_{s,t}(\omega) = e^{-(t-s)(1-\cos \omega)}$ (a hemigroup, indeed a semigroup, of unit steps), every axiom except (A8) holds, every kernel is a lattice law, $E = 2\pi\mathbb{Z}$, and $G(\cdot, 2\pi n) \equiv 0$ for every integer n . Covariance fails because $D_\lambda \mu_{s,t}$ has exponent $(t-s)(1-\cos \lambda\omega)$, which is of the form $c(1-\cos \omega)$ only for $\lambda = 1$. So the exclusion of lattice kernels is a consequence of covariance rather than a property of the function class. That is Lemma 6.2.

Remark 5.5 (relation to the causal theory). The first two statements of this section port from the causal article [8], the additivity lemma verbatim and the increments theorem with the weight $1 - e^{-x}$ replaced by $1 \wedge x^2$ and the compactification of the half-line replaced by the truncation inequality (5.2), which is the one new idea. The atom at the origin that was the drift there is the Gaussian coefficient here, and the atom at infinity that had to be excluded there cannot form here, because tightness has been established first. The third statement does not port verbatim. The causal corollary says that the accumulated exponent is injective in scale at every value of the transform variable, and that this is what makes the relabelling of the scale axis rigid. On the line that sentence is false before covariance, Remark 5.4 being the counterexample, and Corollary 5.3 states what survives. The rigidity argument of §6 is therefore run on a different coordinate, the smoothed transmittance below, which has no causal counterpart because none was needed.

The coordinate is one number attached to each scale, strictly monotone in it, and obtained without covariance. It is the transform of the kernel averaged against a window on the frequency axis. Any strictly positive integrable window would serve: the average is strictly monotone because a nondegenerate increment cannot have transform 1 on a set of positive measure, whereas a single frequency can fall in the zero set of a lattice increment. The Gaussian is chosen because it is its own transform and because its dilates are dominated by one Gaussian, which is what the continuity estimate of §6 uses.

Corollary 5.6 (the smoothed transmittance). *Assume (A1)–(A7) and let ρ be the standard Gaussian density. Put*

$$\Theta(t) := \int_{\mathbb{R}} \hat{\mu}_{0,t}(\omega) \rho(\omega) d\omega = \int_{\mathbb{R}} e^{-x^2/2} \mu_{0,t}(dx) \in (0, 1].$$

Then Θ is continuous on $[0, \infty)$, $\Theta(0) = 1$, and Θ is nonincreasing; under (ND) it is strictly decreasing.

Proof. The two expressions agree by Fubini, since $\int e^{-i\omega x} \rho(\omega) d\omega = e^{-x^2/2}$ exactly for $\rho(\omega) = (2\pi)^{-1/2} e^{-\omega^2/2}$. Continuity in t is dominated convergence, with the bound $|\hat{\mu}| \leq 1$ and the continuity of $\hat{\mu}_{0,t}(\omega)$ in t (Lemma 4.2); positivity is that $e^{-x^2/2} > 0$ at every x and $\mu_{0,t}$ is a probability measure, and $\Theta(0) = 1$ is $\mu_{0,0} = \delta_0$. For $s < t$, $\hat{\mu}_{0,t} = \hat{\mu}_{s,t} \hat{\mu}_{0,s}$ (Lemma 5.1) with $0 < \hat{\mu}_{s,t} \leq 1$, so

$$\Theta(t) - \Theta(s) = \int_{\mathbb{R}} (\hat{\mu}_{s,t} - 1) \hat{\mu}_{0,s} \rho d\omega \leq 0;$$

under (ND) the continuous integrand is negative on the open set $\{\hat{\mu}_{s,t} < 1\}$, which is nonempty because $\mu_{s,t} \neq \delta_0$ (Lemma 4.1), while $\hat{\mu}_{0,s} \rho > 0$, so the inequality is strict. $\square$

Probabilistically $\Theta(t) = \mathbb{E} e^{-X_t^2/2}$ with $X_t := X_{0,t}$: the transmittance of the kernel from the origin averaged over a Gaussian spectrum of probe wavenumbers, or equally the probability that $X_t^2/2$ falls short of an independent standard exponential threshold. It is what Lemma 6.4(3) inverts. Note that the positivity of Θ is not Lemma 4.2 but the positivity of the integrand $e^{-x^2/2}$. The positivity of $\hat{\mu}$ is used only on the two monotonicity clauses.

6 Scale covariance: the similarity form

Covariance turns the free relabelling S_λ of (A8) into a coordinate. The action is rigid, and its orbit through the point 1 parametrizes the scale axis. That parametrization is the gauge. In it the accumulated exponent is a dilate of a single function, $G(t, \omega) = F(\chi(t)\omega)$, and the exponent of any increment is a difference of two dilates of F . In the reading of Remark 4.3, covariance makes the family of transfer functions *self-similar*: descending the scale axis is indistinguishable from probing at a finer wavenumber. The rigidity lemma says that no scale is distinguished by the action, and the gauge is the coordinate in which self-similarity becomes homogeneity, one transfer function F read through the dimensionless product $\chi(t)\omega$. The first lemma only translates (A8) into the language of kernels and exponents.

Lemma 6.1 (Fourier form of covariance). *Under (A1)–(A5) the following are equivalent: (1) axiom (A8); (2) $D_\lambda \mu_{s,t} = \mu_{S_\lambda s, S_\lambda t}$ for all $0 \leq s \leq t$ and $\lambda > 0$; (3) $g_{S_\lambda s, S_\lambda t}(\omega) = g_{s,t}(\lambda\omega)$ for all $0 \leq s \leq t$, $\lambda > 0$ and $\omega \in \mathbb{R}$, whenever the exponents exist. In particular, under (A8) together with (A6)–(A7),*

$$G(S_\lambda t, \omega) = G(t, \lambda\omega) \quad \text{for all } t \geq 0, \lambda > 0, \omega \in \mathbb{R}. \quad (6.1)$$

The proviso in clause (3) is there because (A1)–(A5) alone do not give Lemma 4.2. Under (A6)–(A7) it is vacuous, and every later use is in that setting.

Proof. The compatibility $D_\lambda(\mu * f) = (D_\lambda \mu) * (D_\lambda f)$ turns (A8) into the statement that $D_\lambda \mu_{s,t}$ and $\mu_{S_\lambda s, S_\lambda t}$ represent the same operator, and the uniqueness clause of Lemma 4.1 makes that an identity of measures; this is (1) if and only if (2). For (2) if and only if (3), $\widehat{D_\lambda \mu}(\omega) = \hat{\mu}(\lambda\omega)$, so taking $-\log$ (Lemma 4.2) gives $g_{S_\lambda s, S_\lambda t}(\omega) = g_{s,t}(\lambda\omega)$. Equation (6.1) is the case $s = 0$ of that identity together with $S_\lambda 0 = 0$, which holds because an increasing bijection of $[0, \infty)$ fixes 0. $\square$

The first thing covariance buys on the line is the exclusion of the lattice laws that Lemma 2.11 left open.

Lemma 6.2 (the kernels from the origin are not lattice laws). *Under (A1)–(A8) and (ND), for every $t > 0$ the kernel $\mu_{0,t}$ is carried by no lattice $p\mathbb{Z}$ with $p > 0$; equivalently $G(t, \omega) > 0$ for every $\omega \neq 0$.*

Beyond the cascade the proof uses two things, covariance at a single non-integer ratio and nondegeneracy, which excludes the trivial subgroup $\mathbb{R}$. The argument does not extend to the increments $\mu_{s,t}$ with $s > 0$, because S_λ moves both endpoints and the increments do not form a chain. Strict positivity for every increment is recovered after the classification, in Proposition 7.8(2). That covariance is the operative axiom is shown by Remark 5.4, a family satisfying (A1)–(A7) and (ND) all of whose kernels are lattice laws.

Proof. Write $N_t := N(\mu_{0,t})$ in the notation of Lemma 2.11, a closed subgroup of $\mathbb{R}$. For $s \leq t$, $g_{0,t} = g_{0,s} + g_{s,t}$ with both terms nonnegative (Lemmas 5.1 and 4.2), so $N_t \subseteq N_s$: the family $(N_t)_{t>0}$ is a chain under inclusion, and $N_t \neq \mathbb{R}$ for $t > 0$ by (ND) and Lemma 2.11. By Lemma 6.1, $\hat{\mu}_{0,S_\lambda t}(\omega) = \hat{\mu}_{0,t}(\lambda\omega)$, so $N_{S_\lambda t} = \lambda^{-1}N_t$, and $S_\lambda t > 0$. Suppose $N_t = c\mathbb{Z}$ for some $t > 0$ and $c > 0$. Then for every $\lambda > 0$ the two chain members $\lambda^{-1}c\mathbb{Z}$ and $c\mathbb{Z}$ are comparable under inclusion, which happens if and only if λ or λ^{-1} is a positive integer; $\lambda = 3/2$ is neither. So $N_t = \{0\}$, which is the statement. $\square$

The next lemma is the fact that a function invariant under one dilation of its argument is constant, so that an exponent invariant under one dilation vanishes. It is used below and in §7, and it is stated once.

Lemma 6.3 (a function fixed by one dilation vanishes). *Let $h : \mathbb{R} \rightarrow \mathbb{R}$ be continuous at 0 with $h(0) = 0$, and suppose $h(\omega) = h(\varkappa\omega)$ for every ω and one fixed $\varkappa > 0$ with $\varkappa \neq 1$. Then $h \equiv 0$.*

Proof. Iterating, $h(\omega) = h(\varkappa^n \omega)$ for every $n \in \mathbb{Z}$. Choose the sign of n so that $\varkappa^n \rightarrow 0$; then $\varkappa^n \omega \rightarrow 0$ and continuity at 0 gives $h(\omega) = h(0) = 0$. $\square$

Nothing in (A8) says that the relabellings S_λ are unique, that they compose, or that they move every point. The next lemma says all three. The natural engine would be the injectivity of the accumulated exponent in scale at one fixed value of the transform variable, which Corollary 5.3 does not deliver. Two weaker facts replace it and suffice: the accumulated exponent is injective in scale *as a function of ω* , which (ND) gives, and the smoothed transmittance is injective in scale at a single number.

Lemma 6.4 (rigidity of the action). *Assume (A1)–(A8) and (ND). Then:*

1. *If $G(t_1, \cdot) = G(t_2, \cdot)$ as functions on $\mathbb{R}$, then $t_1 = t_2$; consequently, for each λ , S_λ is uniquely determined by (6.1).*
2. *$S_\lambda S_\kappa = S_{\lambda\kappa}$ for all $\lambda, \kappa > 0$, and $S_1 = \text{Id}$.*
3. *$\lambda \mapsto S_\lambda t$ is continuous for each $t > 0$.*
4. *The action has no fixed point in $(0, \infty)$, in the strong form: if $S_\lambda t^* = t^*$ for one $\lambda \neq 1$, then $t^* = 0$.*

Three of the four clauses are the function-injectivity of (1) and nothing else. Clause (3) needs more: a single strictly monotone number attached to each scale, which is what Θ is. Clause (4) is where (ND) does its most visible work (clauses (1) and (3) use it too, through the identity increment and the strictness of Θ): a fixed point of the action would be a scale at which the family is the identity, an idle stretch, and the orbit below could stall there. The strong form, one λ rather than all, costs nothing and is used in Proposition 6.6.

Proof. (1). If $t_1 < t_2$ then $G(t_2, \cdot) - G(t_1, \cdot) = g_{t_1, t_2}$ by Lemma 5.1; if this vanishes identically then $\hat{\mu}_{t_1, t_2} \equiv 1$, so $\mu_{t_1, t_2} = \delta_0$ and $\Phi_{t_1, t_2} = \text{Id}$ by Lemma 4.1, against (ND). If S and S' both satisfy (6.1) then $G(S_\lambda t, \cdot) = G(t, \lambda \cdot) = G(S'_\lambda t, \cdot)$, so $S_\lambda t = S'_\lambda t$.

(2). Applying (6.1) twice, $G(S_\lambda S_\kappa t, \omega) = G(S_\kappa t, \lambda\omega) = G(t, \lambda\kappa\omega) = G(S_{\lambda\kappa} t, \omega)$ for every ω , and (1) gives $S_\lambda S_\kappa t = S_{\lambda\kappa} t$. Taking $\lambda = 1$ in (6.1) gives $G(S_1 t, \cdot) = G(t, \cdot)$, whence $S_1 = \text{Id}$.

(3). Fix $t > 0$ and let ρ be the standard Gaussian density. By (6.1) and the definition of Θ in Corollary 5.6,

$$\Theta(S_\lambda t) = \int_{\mathbb{R}} \hat{\mu}_{0, S_\lambda t}(\omega) \rho(\omega) d\omega = \int_{\mathbb{R}} \hat{\mu}_{0, t}(\lambda\omega) \rho(\omega) d\omega = \int_{\mathbb{R}} \hat{\mu}_{0, t}(u) \rho(u/\lambda) \lambda^{-1} du =: \theta_t(\lambda).$$

The last integrand is continuous in λ and, for λ in a compact subinterval $[\lambda_0/2, 2\lambda_0]$ of $(0, \infty)$, is dominated by $\frac{2}{\lambda_0} (2\pi)^{-1/2} e^{-u^2/(8\lambda_0^2)}$, which is integrable; so θ_t is continuous by dominated

convergence. Under (ND), Θ is continuous and strictly decreasing on $[0, \infty)$ (Corollary 5.6), hence has a continuous inverse on its image, and $\theta_t(\lambda) = \Theta(S_\lambda t)$ lies in that image; so $S_\lambda t = \Theta^{-1}(\theta_t(\lambda))$ is continuous in λ .

(4). Suppose $S_\lambda t^* = t^*$ for some $\lambda \neq 1$. Then (6.1) gives $G(t^*, \omega) = G(t^*, \lambda\omega)$ for every ω , and $G(t^*, \cdot)$ is continuous with $G(t^*, 0) = 0$ (Lemma 5.1), so $G(t^*, \cdot) \equiv 0$ by Lemma 6.3; that is, $\mu_{0,t^*} = \delta_0$ and $\Phi_{0,t^*} = \text{Id}$, which by (ND) forces $t^* = 0$. $\square$

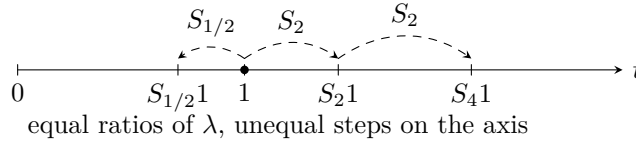
The direction of the orbit coordinate needs one more elementary fact, the Fourier form of “a law whose expansions $\lambda_n X$, $\lambda_n \rightarrow \infty$, converge to a point mass is a point mass”. The restriction to expansions matters. Contractions converge to δ_0 for every law.

Lemma 6.5 (dilation to infinity keeps only the atom at the origin). *Let μ be a probability measure on $\mathbb{R}$ and let $\lambda_n \rightarrow \infty$. If $\hat{\mu}(\lambda_n \omega) \rightarrow 1$ for every $\omega \in \mathbb{R}$, then $\mu = \delta_0$.*

Proof. With ρ the standard Gaussian density, Fubini gives $\int \hat{\mu}(\lambda_n \omega) \rho(\omega) d\omega = \int e^{-\lambda_n^2 x^2 / 2} \mu(dx)$. The left side tends to $\int \rho = 1$ by dominated convergence, since $|\hat{\mu}| \leq 1$; the right side tends to $\mu(\{0\})$ by dominated convergence, the integrand being bounded by 1 and tending pointwise to the indicator of $\{0\}$. So $\mu(\{0\}) = 1$. $\square$

The proof needs no weak convergence, only two applications of dominated convergence with the constant bound 1. The coordinate itself is the orbit (Figure 4). Fix the point 1 on the scale axis and follow it under the action: $\lambda \mapsto S_\lambda 1$ is continuous and injective, and it has no fixed point to stall at. Two things remain to be settled, which way it sweeps and that it sweeps the whole axis, and the proof of Proposition 6.6 settles both.

(a) the scale axis as given: the orbit of 1 under S_λ



(b) the canonical gauge $\tilde{t} = \chi(t)$: the same action is multiplication

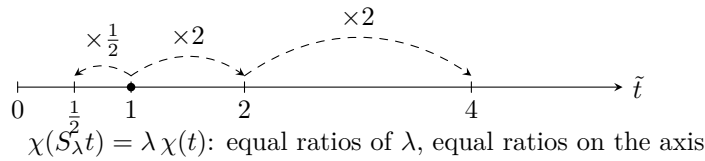


Figure 4: The orbit coordinate. (a) On the scale axis as the family presents it, the action S_λ of (A8) is some increasing bijection. The orbit of the point 1, $\lambda \mapsto S_\lambda 1$, is continuous, strictly increasing and has no fixed point, so it sweeps all of $t > 0$ (Lemma 6.4, Proposition 6.6). The given scale labels need not preserve dilation ratios, and the canonical coordinate does: equal ratios of λ are unequal steps on the axis. (b) Labelling each point by the λ that reaches it, $\tilde{t} = \chi(t)$, the same action becomes multiplication by λ : the canonical gauge. In this gauge every accumulated exponent is a dilate of one function, $G(t, \omega) = F(\tilde{t}\omega)$.

Proposition 6.6 (orbit coordinate; the gauge). *Assume (A1)–(A8) and (ND). Then $c(\lambda) := S_\lambda 1$ is a continuous strictly increasing bijection of $(0, \infty)$ onto $(0, \infty)$ with $c(1) = 1$. Consequently*

$$\chi := c^{-1} : (0, \infty) \rightarrow (0, \infty), \quad \chi(0) := 0,$$

is an increasing bijection of $[0, \infty)$, the canonical gauge, satisfying $\chi(S_\lambda t) = \lambda \chi(t)$; the maps $\lambda \mapsto S_\lambda t$ are strictly increasing for every $t > 0$; and

$$G(t, \omega) = F(\chi(t) \omega), \quad F := G(1, \cdot) \in \text{LE}_s, \quad F \not\equiv 0, \quad (6.2)$$

with F even and continuous, $F(0) = 0$, and with the pair (a, ν) of (2.2).

The proposition derives the gauge from the axioms. It is the semigroup literature's free rescaling function, and the hemigroup case differs from the semigroup case only by the generality of the reparametrization and the freedom in F . In the canonical gauge every increment reads $g_{s,t}(\omega) = F(t\omega) - F(s\omega)$, which is the form §7 classifies. The proof below is the machine-checked route at its two delicate steps, the direction of the orbit and its surjectivity: both use the monotonicity of S_κ , never its continuity in the scale variable, which this section does not prove.

Proof. c is continuous by Lemma 6.4(3), and $c(1) = S_1 1 = 1$ by Lemma 6.4(2).

The action expands above 1: for every $\varkappa > 1$ and every $u > 0$, $u < S_\varkappa u$. Suppose not. Equality is impossible, because Lemma 6.4(4) leaves $S_\varkappa$ no fixed point but 0 and $u > 0$; so $S_\varkappa u < u$ for some such pair. Write $w_n := S_{\varkappa^n} u$ for the orbit of u . The group law gives $w_{n+1} = S_\varkappa w_n$, and $S_\varkappa$ is increasing, so $w_1 < w_0 = u$ propagates by induction to $w_{n+1} < w_n$ for every n . The orbit is nonnegative, hence decreases to $L := \inf_n w_n \geq 0$, and that infimum is a fixed point of $S_\varkappa$ by monotonicity alone: $S_\varkappa L \leq S_\varkappa w_n = w_{n+1}$ for every n , so $S_\varkappa L \leq L$; and $S_{\varkappa^{-1}} L \leq S_{\varkappa^{-1}} w_{n+1} = w_n$ for every n , so $S_{\varkappa^{-1}} L \leq L$ and therefore $L = S_\varkappa(S_{\varkappa^{-1}} L) \leq S_\varkappa L$. Thus $S_\varkappa L = L$, and Lemma 6.4(4) gives $L = 0$: the orbit of u decreases to 0. By (6.1), $\hat{\mu}_{0,u}(\varkappa^n \omega) = \hat{\mu}_{0,w_n}(\omega)$, which tends to $\hat{\mu}_{0,0}(\omega) = 1$ for every ω , by continuity of $\hat{\mu}_{0,\cdot}(\omega)$ in scale (Lemma 4.2). Then Lemma 6.5 gives $\mu_{0,u} = \delta_0$, against (ND).

c is strictly increasing, and so is $\lambda \mapsto S_\lambda t$ for every $t > 0$. Let $0 < a < b$ and $t > 0$. The group law reads $S_b t = S_{b/a}(S_a t)$ with $b/a > 1$ and $S_a t > 0$, so the previous paragraph gives $S_a t < S_b t$. Taking $t = 1$ is the assertion for c ; injectivity is contained in it, and nothing here needs c to be continuous.

c is onto $(0, \infty)$. Fix $u > 0$. Since c is continuous and strictly increasing on $(0, \infty)$, it is enough to exhibit one value of c below u and one above it, the intermediate value theorem then supplying a λ with $c(\lambda) = u$. Both are obtained from the same two lines, and neither uses any continuity of S_κ in the scale variable.

Suppose $c \geq u$ on $(0, \infty)$, and let $m := \inf_{\lambda > 0} c(\lambda)$, so $m \geq u > 0$. For every $\kappa > 0$ and every $\lambda > 0$ the group law gives $c(\lambda) = S_\kappa(c(\lambda/\kappa)) \geq S_\kappa m$, because S_κ is nondecreasing on $[0, \infty)$ and $c(\lambda/\kappa) \geq m$; taking the infimum over λ , $S_\kappa m \leq m$ for every $\kappa > 0$. Reading that at κ^{-1} and applying the nondecreasing S_κ to it, $m = S_\kappa(S_{\kappa^{-1}} m) \leq S_\kappa m$, so $S_\kappa m = m$. Taking $\kappa = 2$, Lemma 6.4(4) gives $m = 0$, contradicting $m \geq u > 0$.

Suppose instead $c \leq u$ on $(0, \infty)$, and let $M := \sup_{\lambda > 0} c(\lambda)$, which is finite and at least $c(1) = 1$. The same computation with the inequalities reversed gives $c(\lambda) = S_\kappa(c(\lambda/\kappa)) \leq S_\kappa M$ and hence $M \leq S_\kappa M$ for every $\kappa > 0$; reading that at κ^{-1} and applying S_κ , $S_\kappa M \leq S_\kappa(S_{\kappa^{-1}} M) = M$, so $S_\kappa M = M$ and Lemma 6.4(4) gives $M = 0$, contradicting $M \geq 1$.

The gauge relation. By the group law, $S_\lambda t = S_\lambda S_{\chi(t)} 1 = S_{\lambda \chi(t)} 1 = c(\lambda \chi(t))$, so $\chi(S_\lambda t) = \lambda \chi(t)$, and $\lambda \mapsto S_\lambda t$ is strictly increasing because c is.

The similarity form. For $t > 0$, writing $t = S_{\chi(t)} 1$ and using (6.1), $G(t, \omega) = G(S_{\chi(t)} 1, \omega) = G(1, \chi(t) \omega) = F(\chi(t) \omega)$. Here $F = g_{0,1} \in \text{LE}_s$ by Theorem 5.2, even and continuous with $F(0) = 0$ by Lemma 4.2, and $F \not\equiv 0$ by (ND). The case $t = 0$ is $G(0, \cdot) = 0 = F(0)$. $\square$

Remark 6.7 (gauge freedom). In the hemigroup setting the scale coordinate carries no intrinsic parametrization: replacing t by any increasing bijection of $[0, \infty)$ yields an equivalent family. Proposition 6.6 fixes the *canonical gauge* $\tilde{t} := \chi(t)$, in which S_λ acts by multiplication and $G(t, \omega) = F(\tilde{t} \omega)$. In it scale carries the dimension of length. Since ω is dual to position, $\tilde{t} \omega$

must be dimensionless for $F(\tilde{t}\omega)$ to make sense, so t in this article is a length, the standard deviation of the Gaussian member, where the scale-space literature's t is the variance. The other homogeneous gauges are $c_0 \tilde{t}^\beta$ with $c_0, \beta > 0$. The one that recurs is the *variance gauge* $\tilde{t}^2$, in which the Gaussian member is a one-parameter semigroup and the diffusion equation has its textbook form. It is named at each use and never written $\tilde{t}$.

For the Gaussian member, $G(t, \omega) = \frac{1}{2}t^2\omega^2$ in the coordinate of §3, so $F(\omega) = \frac{1}{2}\omega^2$ and that coordinate already is the canonical gauge: it is the standard deviation. In the semigroup case the straightening is the classical reparametrization: in the semigroup's own additive parameter v the accumulated exponent is $v|\omega|^\alpha$, the action is $S_\lambda v = \lambda^\alpha v$, and the gauge is $\chi(v) = v^{1/\alpha}$, which is the rescaling function of [28]. Their functional equations for the exponent are the Cauchy step of Corollary 7.4.

Remark 6.8 (relation to the causal theory). This section reruns the covariance chapter of the causal article [8] with three substitutions, at the three places where its proof used the vanishing lemma that Lemma 2.11 says has no Fourier analogue. The exclusion of lattice kernels, free there, is Lemma 6.2 here, bought by covariance at a single non-integer ratio. The continuity of the action, read there off the accumulated exponent at one frequency, is read here off the smoothed transmittance, Lemma 6.4(3). And the direction of the orbit coordinate, oriented there by the monotonicity of Bernstein functions, is oriented here by the dilation argument of Lemma 6.5. Monotonicity of F in ω is unavailable before the classification and recovered from it in Proposition 7.8(1). Everything else (the Fourier form of covariance, the group law and the fixed-point clause of the rigidity lemma, the gauge and the similarity form) ports verbatim, with the Laplace transform replaced by the Fourier transform.

7 Self-decomposability and the characterization theorem

This section turns the similarity form of §6 into a condition on one function F , solves that condition, and states the classification (Theorem 7.3). By Lemma 5.1 the exponent of a kernel is a difference of accumulated exponents, and by Proposition 6.6 the accumulated exponent is a dilate of one function. So every kernel is read off F by stretching its argument and subtracting,

$$g_{s,t}(\omega) = F(\tilde{t}\omega) - F(\tilde{s}\omega), \quad \tilde{s} = \chi(s) \leq \tilde{t} = \chi(t),$$

and since χ is onto, $(\tilde{s}, \tilde{t})$ ranges over all pairs $0 \leq \tilde{s} \leq \tilde{t}$. Theorem 5.2 requires each of these differences to be a symmetric Lévy exponent. The pairs with $\tilde{s} = 0$ ask only that the dilates $F(\tilde{t}\cdot)$ lie in LE_s , which they do. The content is in the pairs $0 < \tilde{s} \leq \tilde{t}$. That is a condition on F alone, that every stretched difference is a Lévy exponent, and it is the condition this section solves.

Canonical-gauge scales are written $s \leq t$, plain, when no other family is in play, so that a is free for the Gaussian coefficient.

The condition is self-decomposability of the law of the displacement at unit scale. Let X_1 be the random displacement whose law is the kernel at canonical scale 1, so that $\mathbb{E} \cos(\omega X_1) = e^{-F(\omega)}$. In the canonical gauge the kernel at scale t is the law of tX_1 . That $F(\omega) - F(c\omega)$ is a Lévy exponent for $c \in (0, 1)$ says that there is a random variable R_c , independent of X_1 , with $X_1 \stackrel{d}{=} cX_1 + R_c$. That is self-decomposability (Definition 2.12): covariance turns a shrinking of the scale axis into a shrinking of the displacement, and the cascade requires what remains to be a measurement in its own right. The lemma below is the analytic form of this statement, and Figure 5 draws what it says about the profile. Its proof of (1) implies (3) uses two facts about a nonincreasing profile that are stated in the appendix, because they belong to the geometry of the cone: the tail measure $\varpi = -dk$ it attaches, and the growth of Cin at infinity (Lemma A.1).

Lemma 7.1 (symmetric self-decomposable exponents). *For $F \in \text{LE}_s$ with pair (a, ν) the following are equivalent.*

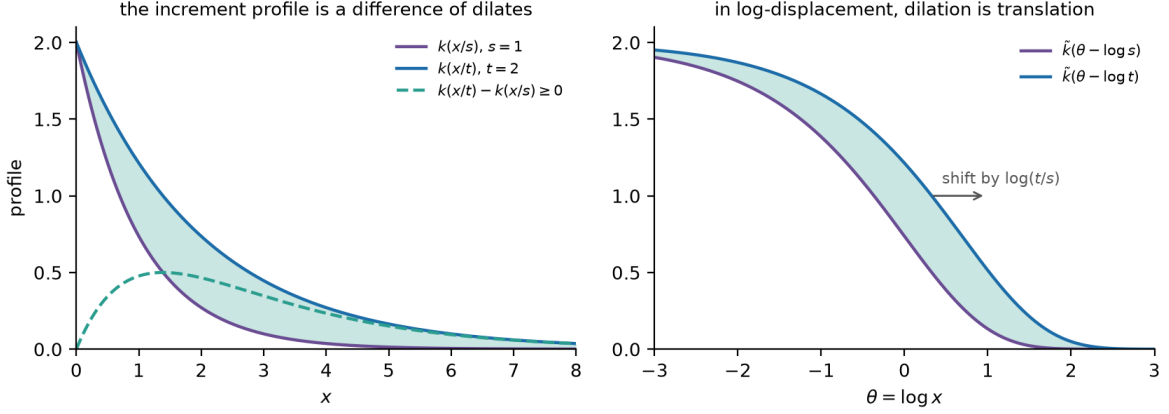


Figure 5: The condition of Lemma 7.1, drawn for the profile $k(x) = 2e^{-x}$ of the second member of §3. Left: the profile at scales $s = 1$ and $t = 2$ is the same function dilated, $k(x/s)$ and $k(x/t)$, and the increment from s to t has profile $k(x/t) - k(x/s)$, the shaded difference. The difference is nonnegative for every $s < t$ exactly when k is nonincreasing, which is what makes the increment a symmetric Lévy exponent. Right: in the log-displacement coordinate $\theta = \log x$ the dilates are translates of one function $\tilde{k}(\theta) = k(e^\theta)$, shifted by $\log(t/s)$, and the condition says that the profile can only rise as one moves to finer scales.

1. $F(t \cdot) - F(s \cdot) \in \text{LE}_s$ for all $0 < s \leq t$.
2. F is continuously differentiable on $(0, \infty)$ and the symbol $B(\omega) := \omega F'(\omega)$ lies in LE_s .
3. F has the representation (2.3): $\nu(dx) = k(x) x^{-1} dx$ with $k : (0, \infty) \rightarrow [0, \infty)$ nonincreasing, $\int_0^1 x k(x) dx < \infty$ and $\int_1^\infty k(x) x^{-1} dx < \infty$.

In (3) the profile k may be taken right-continuous with $k(\infty) = 0$, and then $\varpi := -dk$ is a positive measure on $(0, \infty)$ with $k(x) = \varpi((x, \infty))$, $\int (1 \wedge u^2) \varpi(du) < \infty$ and $\int \log_+ u \varpi(du) < \infty$; in (2) the pair of B is $(2a, \varpi)$.

The three conditions are not equally deep, and the asymmetry organizes what follows. (3) implies (1) is elementary (a change of variables in the representation, using no property of the positive definite class), and it is all the constructive direction of Theorem 7.3 uses. (1) implies (3) reads the dilation identity backwards and then argues about monotone measures, elementary but genuine. It is the analysis direction of the main theorem. (2) is here a consequence rather than a stepping stone, because the line offers no a priori differentiability. What the superposition (7.3) buys instead is that every admissible exponent is C^1 off the origin with a symbol that is again a Lévy exponent. The whole lemma uses the uniqueness clause of Proposition 2.3(3) and nothing else, twice: once in (1) implies (3), to read the dilation identity backwards, and once in (2) implies (3), to match the pair of F with the pair the profile produces. The lemma is the one-dimensional case of the classical description of self-decomposable laws by their Lévy measures, Proposition 2.13, which is therefore cited for orientation and not used. The tail measure ϖ with its logarithmic moment is likewise classical: it is the Lévy measure of the background driving Lévy process Y of Jurek and Vervaat, whose integral $\int_0^\infty e^{-s} dY(s)$ represents every self-decomposable law and converges exactly when $\mathbb{E} \log(1 + |Y(1)|) < \infty$ ([20], p. 247 and Thm. 2.3, pp. 250–251).

Proof. (3) implies (1): the dilation identity. Write $F(c\omega) = ac^2\omega^2 + \int_0^\infty (1 - \cos c\omega x) k(x) \frac{dx}{x}$ for $c > 0$ and substitute $u = cx$. The measure dx/x is invariant under dilations, so

$$F(c\omega) = ac^2\omega^2 + \int_0^\infty (1 - \cos \omega u) k(u/c) \frac{du}{u} :$$

dilating the argument by c dilates the profile and rescales the Gaussian coefficient, and does nothing else. Subtracting the identity at $c = s$ from the one at $c = t$,

$$F(t\omega) - F(s\omega) = a(t^2 - s^2)\omega^2 + \int_0^\infty (1 - \cos \omega u) (k(u/t) - k(u/s)) \frac{du}{u}, \quad (7.1)$$

which is of the form (2.2): the Gaussian coefficient $a(t^2 - s^2)$ is nonnegative since $s \leq t$, and the density is nonnegative since $u/t \leq u/s$ and k is nonincreasing. For the Lévy condition, the density is dominated by $k(u/t)$, so it is enough to bound the $(1 \wedge u^2)$ -integral of $k(u/t) u^{-1} du$; and the same substitution $u = tx$ turns that integral into $\int_0^\infty (1 \wedge t^2 x^2) k(x) x^{-1} dx$, at most $\max(1, t^2) \int_0^\infty (1 \wedge x^2) k(x) x^{-1} dx$, which is finite by Lemma 2.15; the comparison $1 \wedge t^2 x^2 \leq \max(1, t^2)(1 \wedge x^2)$ holds in each of the two regimes of the truncation separately. So the difference is itself of the form (2.2), hence in LE_s ; the class being named by its representation, this direction makes no appeal to Proposition 2.3.

(1) implies (3): the dilation identity read backwards, and the monotone density. Fix $c \in (0, 1)$. By hypothesis $F(\omega) - F(c\omega) = a'\omega^2 + \int (1 - \cos \omega x) \nu'(dx)$ for some $a' \geq 0$ and some Lévy measure $\nu' \geq 0$. On the other hand, with $D_c \nu$ the image of ν under $x \mapsto cx$, the first display of this proof reads $F(c\omega) = ac^2\omega^2 + \int (1 - \cos \omega x) (D_c \nu)(dx)$. Hence

$$a'\omega^2 + \int (1 - \cos \omega x) (\nu' + D_c \nu)(dx) = a\omega^2 + \int (1 - \cos \omega x) \nu(dx) \quad \text{for every } \omega,$$

an identity between two representations of the form (2.2) with *positive* data; so by the uniqueness clause of Proposition 2.3(3), $a' = a(1 - c^2)$ and $\nu' + D_c \nu = \nu$. Thus

$$D_c \nu \leq \nu \quad \text{for every } c \in (0, 1). \quad (7.2)$$

Pass to the log-displacement coordinate $\theta = \log x$: let $\tilde{\nu}$ be the image of ν under $\log$, a measure on $\mathbb{R}$ with $\tilde{\nu}((\theta, \infty)) = \nu((e^\theta, \infty)) < \infty$ for every θ , finiteness coming from $\int (1 \wedge x^2) \nu < \infty$. Dilation by c becomes translation by $\log c < 0$, so (7.2) reads $\tilde{\nu}(A + h) \leq \tilde{\nu}(A)$ for every Borel A and every $h > 0$: the measure decreases under every right shift. Put

$$\tilde{k}(\theta) := \sup_{n \geq 0} 2^n \tilde{\nu}((\theta, \theta + 2^{-n}]).$$

Each term is nonincreasing in θ , by the shift property; and the sequence is nondecreasing in n , because splitting $(\theta, \theta + 2\delta]$ at its midpoint and applying the shift property to the right half gives $\tilde{\nu}((\theta, \theta + 2\delta]) \leq 2 \tilde{\nu}((\theta, \theta + \delta])$. So $\tilde{k}$ is nonincreasing, and it is a supremum of a nondecreasing sequence. Fix $a < b$. By monotone convergence and then Tonelli,

$$\int_a^b \tilde{k} = \sup_n 2^n \int_a^b \tilde{\nu}((\theta, \theta + \varepsilon_n]) d\theta = \sup_n 2^n \int |(a, b] \cap [x - \varepsilon_n, x)| \tilde{\nu}(dx), \quad \varepsilon_n = 2^{-n},$$

and the inner length is ε_n for $x \in (a + \varepsilon_n, b]$, at most ε_n always, and 0 outside $(a, b + \varepsilon_n]$. So the n -th term lies between $\tilde{\nu}((a + \varepsilon_n, b])$ and $\tilde{\nu}((a, b + \varepsilon_n])$, and both converge to $\tilde{\nu}((a, b])$ — the first by continuity of a measure from below, the second by continuity from above, which is where the finiteness of the rays is used. Hence $\tilde{\nu}((a, b]) = \int_a^b \tilde{k}$ for every $a < b$, so $\tilde{\nu}(d\theta) = \tilde{k}(\theta) d\theta$. Returning to x , $\nu(dx) = k(x) x^{-1} dx$ with $k(x) := \tilde{k}(\log x)$ nonincreasing; and $k(\infty) = 0$, since $\tilde{k}$ is nonincreasing and nonnegative with $\int_\theta^\infty \tilde{k} = \tilde{\nu}((\theta, \infty)) < \infty$. The two integrability conditions are $\int (1 \wedge x^2) \nu(dx) < \infty$ read through Lemma 2.15.

The measure ϖ , and the right-continuous profile. A profile that is merely nonincreasing has no right-continuous representative to hand, so the measure is built first and the modification read off it. Take for ϖ the measure Lemma A.1(2) attaches to k : the image of Lebesgue measure on $(0, \infty)$ under the generalized inverse $y \mapsto \sup\{u > 0 : k(u) > y\}$, whose tails satisfy $\varpi((x, \infty)) = k(x)$ at every continuity point of k , hence at almost every $x > 0$. Every positive

tail is then finite, since a point of that identity below x bounds $\varpi((x, \infty))$ by a value of k ; so $k^*(x) := \varpi((x, \infty))$ is a finite nonincreasing function on $(0, \infty)$, equal to k almost everywhere and therefore inducing the same Lévy measure, right-continuous because (x, ∞) is the increasing union of the $(x + 1/n, \infty)$, and vanishing at infinity because k does. This is the modification the statement asserts, and its tail identity holds at *every* positive point rather than almost everywhere.

For the two integrability conditions, the layer cake formula against ϖ turns $\int G(u) \varpi(du)$, for $G(u) = \int_0^u g(t) dt$ with $g \geq 0$ continuous, into $\int_0^\infty k^*(t) g(t) dt$. Take $g(t) = 2t(1+t^2)^{-2}$, so that $G(u) = u^2/(1+u^2) \geq \frac{1}{2}(1 \wedge u^2)$, and note $g(t) \leq 2t$ and $g(t) \leq 2t^{-1}$; then $\int (1 \wedge u^2) \varpi(du) \leq 4 \int_0^1 t k(t) dt + 4 \int_1^\infty k(t) t^{-1} dt$. Take $g(t) = t(1+t^2)^{-1}$, so that $G(u) = \frac{1}{2} \log(1+u^2) \geq \log_+ u$, with the same two bounds on g ; then $\int \log_+ u \varpi(du)$ is bounded by the same two integrals. So both conditions on ϖ follow from $\int_0^1 x k(x) dx < \infty$ and $\int_1^\infty k(x) x^{-1} dx < \infty$.

(3) *implies* (2): the Cin superposition. Tonelli over $\{0 < x < u\}$ turns (2.3) into

$$F(\omega) = a\omega^2 + \int_{(0,\infty)} \text{Cin}(\omega u) \varpi(du), \quad \text{Cin}(z) := \int_0^z \frac{1 - \cos v}{v} dv, \quad (7.3)$$

since $\int_0^u (1 - \cos \omega x) x^{-1} dx = \text{Cin}(\omega u)$; the measure ϖ is the one supplied above, and only its almost-everywhere tail identity is used. The integrand $v \mapsto (1 - \cos v)/v$ of Cin is continuous at the origin as well, its value there being 0 and $|(1 - \cos v)/v| \leq |v|/2$; so Cin is continuously differentiable on all of $\mathbb{R}$ with $\text{Cin}'(v) = (1 - \cos v)/v$, and $\partial_\omega \text{Cin}(\omega u) = u \text{Cin}'(\omega u) = (1 - \cos \omega u)/\omega$. For ω in the interval $(\omega_0/2, 2\omega_0)$ this is bounded by $(4\omega_0^{-1} + 2\omega_0)(1 \wedge u^2)$ — it is at most $2/\omega$ and at most $\omega u^2/2$, which are the two regimes of the truncation — and $1 \wedge u^2$ is ϖ -integrable, as the previous paragraph shows for every such ϖ . So differentiation under the integral sign applies at every $\omega > 0$, and

$$B(\omega) = \omega F'(\omega) = 2a\omega^2 + \int_{(0,\infty)} (1 - \cos \omega u) \varpi(du), \quad (7.4)$$

which is of the form (2.2) with pair $(2a, \varpi)$, hence in LE_s — the class being named by its representation, so exhibiting the pair is the conclusion and no appeal to Proposition 2.3 is made. That F is C^1 on $(0, \infty)$ and not merely differentiable is a second dominated-convergence argument: $1 - \cos \omega v$ is bounded by $(2 + (|\omega_0| + 1)^2)(1 \wedge v^2)$ for ω within 1 of ω_0 , so $\omega \mapsto \int (1 - \cos \omega v) \varpi(dv)$ is continuous on $\mathbb{R}$ and F' is continuous where $\omega \neq 0$. Both sides of (7.4) are stated at every real ω : the identity extends to $\omega < 0$ because F is even, hence F' odd, and holds at $\omega = 0$ because the factor ω annihilates whatever value the derivative is assigned there. (The half-line route, differentiating under the integral in the profile form and applying Tonelli, is not available here: $\omega \sin(\omega x)$ is not of one sign, and $\int u \varpi(du)$ may diverge, as it does for the stable profiles of index below 1. The Cin form has a positive integrand and needs no such integral. Nor is the profile form saved by the two admissibility conditions: $k(x) = (\log x)^{-2}$ at infinity satisfies $\int_1^\infty k(x) x^{-1} dx < \infty$ while $\int_1^\infty k(x) dx$ diverges, and that integral is what dominating $|\sin(\omega x)|k(x)$ would need.)

(2) *implies* (3). Let $B = \omega F'$ have pair (a_B, ϖ) and put $k(x) := \varpi((x, \infty))$, which is nonnegative and nonincreasing on $(0, \infty)$ and finite at every positive x , since $\int (1 \wedge u^2) \varpi(du) < \infty$ bounds $\varpi((x, \infty))$ by $(1 \wedge x^2)^{-1}$ times that integral. The claim is that F is (2.3) at the data $(\frac{1}{2}a_B, k)$, and the two integrability conditions on k come from two different places — which is the order the proof has to be read in, because the data is not admissible until both are in hand.

The first is the Lévy condition on ϖ alone. The layer cake formula of the previous paragraph, read from right to left at $g(t) = 2t(1+t^2)^{-2}$, turns $\int_0^\infty k(t) g(t) dt$ into $\int u^2(1+u^2)^{-1} \varpi(du)$, at most $\int (1 \wedge u^2) \varpi(du)$; and $g(t) \geq t/2$ on $(0, 1)$, so $\int_0^1 t k(t) dt \leq 2 \int (1 \wedge u^2) \varpi(du) < \infty$.

The second is not implied by the Lévy condition: $\varpi(du) = du/(u \log^2 u)$ on $(2, \infty)$ has finite truncated integral and divergent logarithmic one. What supplies it is the finiteness

of F . Since $B \geq 0$, $F' \geq 0$ on $(0, \infty)$, so the fundamental theorem of calculus on $[\varepsilon, 1]$ gives $\int_\varepsilon^1 F'(v) dv = F(1) - F(\varepsilon) \leq F(1)$ for every $\varepsilon \in (0, 1)$, and letting $\varepsilon \downarrow 0$ along a sequence — monotonically, the integrand being nonnegative — $\int_0^1 F'(v) dv \leq F(1) < \infty$. Now $F'(v) \geq v^{-1} \int (1 - \cos vu) \varpi(du)$, and Tonelli over $(0, 1) \times (0, \infty)$ with a positive integrand, together with the substitution $\int_0^1 (1 - \cos uv) v^{-1} dv = \text{Cin}(u)$, turns the integral of that lower bound into $\int \text{Cin}(u) \varpi(du)$. So $\int \text{Cin}(u) \varpi(du) \leq F(1) < \infty$. Since $\text{Cin} \geq 0$ everywhere and $\text{Cin}(u) = \log u + O(1)$ at infinity (Lemma A.1(1)), $\log_+ u \leq \text{Cin}(u) + C(1 \wedge u^2)$ for a fixed C , so $\int \log_+ u \varpi(du) < \infty$; the second layer cake, at $g(t) = t(1 + t^2)^{-1}$, which is at least $(2t)^{-1}$ beyond 1, converts that into $\int_1^\infty k(t) t^{-1} dt < \infty$.

So $(\frac{1}{2}a_B, k)$ is admissible data, and defines an exponent $\tilde{F}$ of the form (2.3). It is F , and the identification does not re-derive the representation: by (7.4) applied to $\tilde{F}$, whose profile has ϖ for its tail measure, $\omega \tilde{F}'(\omega) = a_B \omega^2 + \int (1 - \cos \omega u) \varpi(du) = B(\omega) = \omega F'(\omega)$ at every real ω , so $\tilde{F}$ and F have the same derivative at every $\omega > 0$; both are continuous and vanish at the origin, so they agree on $[0, \infty)$ — a function continuous on $[0, \infty)$ with vanishing derivative on $(0, \infty)$ is constant on every $[\varepsilon, \omega]$, and its value at the origin is the limit of those constants — and both are even, so they agree on $\mathbb{R}$. The uniqueness clause of Proposition 2.3(3), applied to the two representations of that one function, then identifies the pairs: $a = \frac{1}{2}a_B$ and $\nu(dx) = k(x) x^{-1} dx$, which is (3). $\square$

Remark 7.2 (the probabilistic reading). Together with the similarity structure, (2.3) says that the kernels are the marginal laws of a *symmetric self-similar additive process*: a process $(X_{\tilde{t}})_{\tilde{t} \geq 0}$ on $\mathbb{R}$ with independent, symmetric, not necessarily stationary increments satisfying $X_{\lambda \tilde{t}} \stackrel{d}{=} \lambda X_{\tilde{t}}$, the Sato process of exponent 1 attached to the self-decomposable law of X_1 . The paths oscillate, and the Gaussian coefficient a sets the variance $2a\tilde{t}^2$ of their continuous part. Nothing in the proofs uses this correspondence, which is stated as a reading and grounds no statement.

Theorem 7.3 (main theorem). *A family $(\Phi_{s,t})_{0 \leq s \leq t}$ satisfies (A1)–(A8) and (ND) if and only if there are an increasing bijection χ of $[0, \infty)$, the gauge, and a function F of the form (2.3) with $F \not\equiv 0$, such that*

$$\Phi_{s,t} f = \mu_{s,t} * f, \quad \hat{\mu}_{s,t}(\omega) = \exp[-(F(\chi(t)\omega) - F(\chi(s)\omega))].$$

In the canonical gauge the kernel at scale t is the symmetric probability measure ϕ_t with $\hat{\phi}_t = e^{-F(t \cdot)}$, and the pair (χ, F) is unique up to the normalization $\chi(1) = 1$.

Read against the axioms: the apertures are the marginal laws of one symmetric self-similar additive process, the kernel at canonical scale t being the law of tX_1 , so the whole family is generated by one law and the gauge. In the canonical gauge, what the axioms leave free is exactly the pair (a, k) of (2.3): a Gaussian variance and a nonincreasing displacement profile. The Gaussian scale space is $(a, 0)$. Everything else has a catalogue of jumps. The theorem is machine-checked in all three clauses (construction, analysis, uniqueness) and its trust base is Lean core together with two cited facts and no more: Bochner's theorem in the forward direction, and the symmetric Lévy–Khinchine representation at its converse and its uniqueness clauses (§1.1). The kernels have densities as well, unimodal about the origin, but that is Proposition 7.9 and not this theorem: the statement here is about measures, deliberately, absolute continuity resting on a cited interface that the theorem does not use. A reader who wants the density should read the two together.

Proof. Necessity. Lemma 4.1, Lemma 4.2, Lemma 5.1, Theorem 5.2, Lemma 6.1, Lemma 6.4 and Proposition 6.6 together give the representation with $F = G(1, \cdot) \in \text{LE}_s$ and $F \not\equiv 0$. Every increment $F(\chi(t) \cdot) - F(\chi(s) \cdot) = g_{s,t}$ lies in LE_s by Theorem 5.2, and χ is onto $(0, \infty)$ by Proposition 6.6, so condition (1) of Lemma 7.1 holds for every pair $0 < \tilde{s} \leq \tilde{t}$ of canonical

scales, and that lemma yields (2.3). The gauge so obtained is already *normalized*: it is the inverse of the orbit coordinate $c(\lambda) = S_\lambda 1$, which fixes 1 by Proposition 6.6, so $\chi(1) = 1$. What this direction returns is therefore a normalized representation — one of those the uniqueness clause compares — and not merely a representative of its orbit under $(\chi, F) \mapsto (c\chi, F(c^{-1}\cdot))$.

Sufficiency. Let F be of the form (2.3) with $F \not\equiv 0$, and work in the canonical gauge. For $0 \leq s \leq t$ the function $g_{s,t}(\omega) := F(t\omega) - F(s\omega)$ lies in LE_s by Lemma 7.1, (3) implies (1) — for $s = 0$ the increment is $F(t\cdot)$ itself, which is in LE_s by hypothesis and the dilation identity — so by the converse half of Proposition 2.3(3) and by Proposition 2.3(1), $e^{-g_{s,t}}$ is the transform of a unique probability measure $\mu_{s,t}$ on $\mathbb{R}$, symmetric because the transform is real and even. Put $\Phi_{s,t}f := \mu_{s,t} * f$. Then (A1)–(A5) hold by the converse direction of Lemma 4.1; (A6) because the exponents add and transforms determine measures (Proposition 2.7); (A8) with $S_\lambda t = \lambda t$, since $\hat{\mu}_{\lambda s, \lambda t}(\omega) = \hat{\mu}_{s,t}(\lambda\omega)$; and (ND) because $F(t\cdot) - F(s\cdot) \not\equiv 0$ for $s < t$: at $s = 0$ that is $F \not\equiv 0$, and for $s > 0$, if the increment vanished identically then $F(\omega) = F(c\omega)$ for every ω with $c = s/t \in (0, 1)$, whence $F \equiv 0$ by Lemma 6.3. Only that *some* frequency is moved is needed here, which is less than Proposition 7.8(2) asserts.

For (A7), a collapsing-increment estimate does the work. For a probability measure μ and $f \in L^1$ the L^1 distance from $\mu * f$ to f is at most $\int \Theta_f(y) \mu(dy)$, where $\Theta_f(y)$ is the L^1 distance from $T_y f$ to f ; and Θ_f is bounded, continuous, and vanishes at the origin, because the translation group acts continuously on L^1 . So if $(s_n, t_n) \rightarrow (c, c)$ along admissible pairs then $\hat{\mu}_{s_n, t_n} \rightarrow 1$ pointwise, by continuity of F and of χ ; hence $\mu_{s_n, t_n} \rightarrow \delta_0$ weakly by Proposition 2.8, hence $\int \Theta_f(y) \mu_{s_n, t_n}(dy) \rightarrow 0$: a collapsing increment tends to the identity in the strong operator topology. At a pair $s < t$ the two endpoints are then moved separately — there is no $t - s$ to move, this being a hemigroup: the left endpoint through an increment applied *first*, where the contraction property absorbs it, and the right endpoint through one applied *last*, where the same estimate is read at the image. No compact set, no density argument and no $\varepsilon/3$ are needed.

Finally, for an arbitrary gauge χ set $\Phi'_{s,t} := \Phi_{\chi(s), \chi(t)}$: (A1)–(A6) and (ND) hold verbatim, (A7) because an increasing bijection of $[0, \infty)$ is continuous — it is an order isomorphism of $[0, \infty)$ onto itself, and an order isomorphism between linear orders carrying the order topology is a homeomorphism — and (A8) with $S'_\lambda = \chi^{-1}(\lambda \chi(\cdot))$.

Uniqueness. The exponents $g_{0,t}$ are determined by the family: the kernels by Lemma 4.1, their transforms by Proposition 2.7, and $-\log$ is single-valued on them by Lemma 4.2. The normalization $\chi(1) = 1$ recovers $F = g_{0,1}$; without it the pair is determined up to $(\chi, F) \mapsto (c\chi, F(c^{-1}\cdot))$ with $c > 0$, which is the freedom the normalization removes. For each t , $\chi(t)$ is then the unique c with $g_{0,t} = F(c\cdot)$: such a c exists by the representation, and if $F(c\omega) = F(c'\omega)$ for every ω with $c/c' \neq 1$, then $F \equiv 0$ by Lemma 6.3, a contradiction. $\square$

Corollary 7.4 (recovering the semigroup case). *Assume in addition that the family is one-parameter, that is, $\Phi_{s,t}$ depends only on $t - s$. Then, after the normalization $g_{0,1}(1) = 1$,*

$$F(\omega) = |\omega|^\alpha \quad \text{for some } 0 < \alpha \leq 2 :$$

the kernels are exactly the symmetric stable densities, which are the members of the extended semigroup class of [28] with positive kernels. Here $\alpha = 2$ is the Gaussian, and $0 < \alpha < 2$ has Gaussian coefficient 0 and profile $k(x) = C_\alpha^{-1} x^{-\alpha}$ with $C_\alpha := \int_0^\infty (1 - \cos u) u^{-1-\alpha} du \in (0, \infty)$.

The semigroup case is thus recovered from inside the hemigroup theory. The bound $\alpha \leq 2$ comes from Lemma 2.10 rather than from a cited growth estimate.

Proof. The exponent is linear in the scale. One-parametricity makes $\Phi_{0,t}$ and $\Phi_{s,s+t}$ the same operator, so the uniqueness half of Lemma 4.1 makes $\mu_{0,t}$ and $\mu_{s,s+t}$ the same measure, and Lemma 5.1 then reads $G(s+t, \omega) = G(s, \omega) + G(t, \omega)$ for all $s, t \geq 0$: the accumulated exponent is additive in the scale. It is continuous there as well (Lemma 5.1), and an additive function

continuous on $[0, \infty)$ is linear on it — extend it oddly to $\mathbb{R}$, where it stays additive and continuous, and a continuous additive map between real vector spaces is $\mathbb{R}$ -linear. So $g_{0,t} = tg$ with $g := g_{0,1}$.

The exponent is multiplicative in the frequency. (6.1) at $t = 1$, read through the previous paragraph on the left-hand side, is $S_\lambda(1)g(\omega) = g(\lambda\omega)$ for every $\lambda > 0$ and every ω . At $\omega = 1$ the normalization $g(1) = 1$ gives $S_\lambda(1) = g(\lambda)$, so $g(\lambda\omega) = g(\lambda)g(\omega)$: the multiplicative function is g itself, and no separate homomorphism has to be identified.

Cauchy. g is continuous (Lemma 4.2) and strictly positive off the origin (Lemma 6.2), so $u \mapsto \log g(e^u)$ is additive and continuous on $\mathbb{R}$, hence linear by the same theorem as above, and $g(\omega) = \omega^\alpha$ for $\omega > 0$ with $\alpha := \log g(e)$. Evenness of the exponent extends the formula to $\omega < 0$, and $g(0) = 0$ covers the origin once $\alpha \neq 0$ is known.

The index. $\alpha > 0$ because g is continuous at 0 with $g(0) = 0$, while $\omega^\alpha \geq 1$ on $(0, 1]$ when $\alpha \leq 0$. For the upper bound, Lemma 2.10 applied to the Lévy pair of g , which Theorem 5.2 supplies, gives a constant C with $g(\omega) \leq C(1 + \omega^2)$; the normalization $g(1) = 1$ forces $C \geq 1/2$, and if $\alpha > 2$ then at $\omega := (2C + 1)^{1/(\alpha-2)} \geq 1$ the left side is $(2C + 1)\omega^2 > C(1 + \omega^2)$, which the bound forbids. So $\alpha \leq 2$.

The profile. When $0 < \alpha < 2$ the integral C_α converges, by the two regimes of the truncation: $1 - \cos u \leq u^2/2$ makes its integrand at most $u^{1-\alpha}/2$ near the origin, integrable exactly for $\alpha < 2$, and $1 - \cos u \leq 2$ makes it at most $2u^{-1-\alpha}$ beyond 1, integrable exactly for $\alpha > 0$. Those two comparisons are the two integrability conditions of (2.3) for $k(x) = C_\alpha^{-1}x^{-\alpha}$, which is nonincreasing, with $a = 0$, so no separate appeal is needed; and $C_\alpha > 0$ because its integrand is nonnegative and does not vanish on $(0, 1)$. The substitution $u = |\omega|x$, performed on the $[0, \infty]$ -valued integral so that integrability does not have to be revisited at each frequency, gives $\int_0^\infty (1 - \cos \omega x) x^{-1-\alpha} dx = C_\alpha |\omega|^\alpha$, hence the exponent $|\omega|^\alpha$. When $\alpha = 2$, $k \equiv 0$ and $a = 1$. $\square$

Remark 7.5 (the boundary, the Gaussian ray, and what is not there). The boundary $\alpha = 2$ of the semigroup slice is the Gaussian, nothing is degenerate, and the trace of the boundary in the general class is the Gaussian coefficient a , which adds an independent Gaussian displacement of variance $2at^2$ to the kernel at scale t and of variance $2a(t^2 - s^2)$ to every increment. The members of [28] beyond $\alpha = 2$ have positive transfer functions and sign-changing kernels. They are excluded by (A4) through Theorem 7.3, and by nothing in §§4 to 6. So positivity, rather than recursivity, narrows the class.

The admissible pairs (a, k) are closed under sums and nonnegative multiples, so the exponents, with the zero exponent adjoined, form a convex cone. What is read off it below is the position of the Gaussian ray and the places of the named members (Figure 6).

Lemma 7.6 (the admissible exponents form a convex cone). *If F_1, F_2 are admissible in the sense of Theorem 7.3, of the form (2.3) with Gaussian coefficients $a_1, a_2 \geq 0$ and nonincreasing profiles k_1, k_2 , then so are $F_1 + F_2$, with data $(a_1 + a_2, k_1 + k_2)$, and cF_1 for $c \geq 0$, with data (ca_1, ck_1) . So the admissible exponents form a convex cone, and the correspondence with the pairs (a, k) is linear.*

Proof. Each defining condition — nonnegativity of a , nonnegativity and monotonicity of k , and the two integrability conditions — is stable under sums and under multiplication by $c \geq 0$, the last two because the integrands $xk(x)$ and $k(x)/x$ are nonnegative on $(0, \infty)$, so that the integral of a sum is the sum of the integrals with no side condition. The same additivity, applied to (2.3) at a fixed ω , gives $F_1 + F_2$ and cF_1 as the exponents of the summed and scaled data, as identities in $[0, \infty]$; they are identities between real numbers because Lemma 2.15 makes each F_i the exponent of a symmetric Lévy pair and Lemma 2.10 makes that finite at every frequency. $\square$

Remark 7.7 (the Gaussian ray is extreme). The correspondence of Lemma 7.6 makes the Gaussian ray $\{(a, 0) : a \geq 0\}$ an extreme ray of the cone: if $(1, 0) = (a_1, k_1) + (a_2, k_2)$ with both

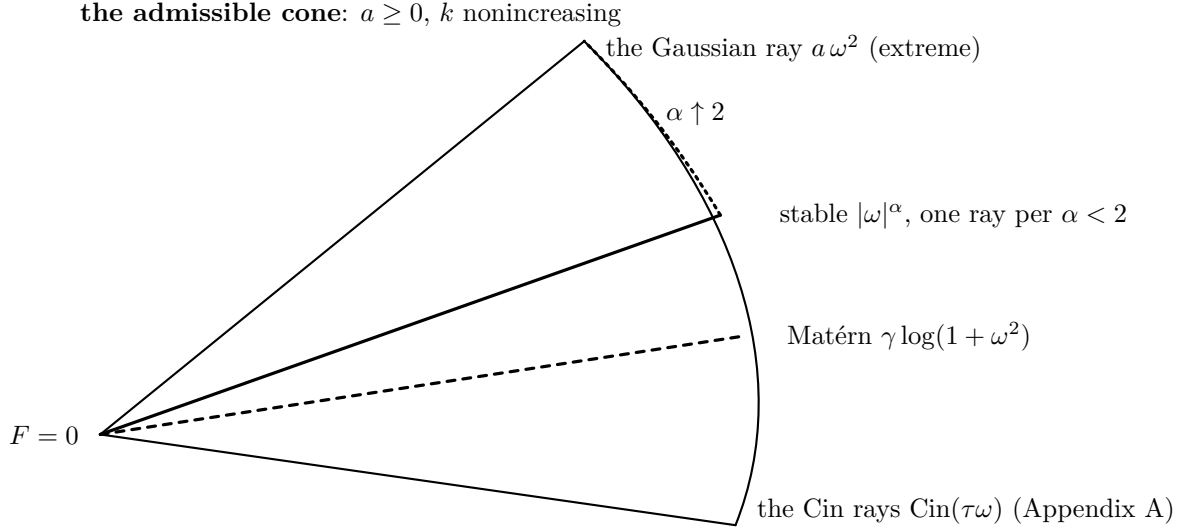


Figure 6: The admissible cone. Theorem 7.3 identifies the exponents F with pairs (a, k) , a Gaussian coefficient and a nonincreasing profile, and these form a convex cone (Lemma 7.6). The Gaussian ray is an extreme ray (Remark 7.7), and it is a smoothing member, not a degenerate boundary. The exponents of the unit-step profiles, the Cin rays of Appendix A, are the rays that generate the cone together with the Gaussian ray: every admissible exponent is a superposition of them ((A.1)). They are drawn along the far side only to keep them apart; whether they are extreme, which the superposition alone does not show, is outside this paper, and the drawing places nothing. The semigroup case cuts the slice of stable exponents $|\omega|^\alpha$ (Corollary 7.4), one ray for each $\alpha \in (0, 2)$, and the slice ends on the Gaussian ray at $\alpha = 2$. The Matérn member of §3 is one ray, with the profile $2\gamma e^{-x}$ (Proposition A.3). The drawing is schematic: the cone is infinite-dimensional.

summands admissible, then $k_1 + k_2 = 0$ with $k_1, k_2 \geq 0$ forces $k_1 = k_2 = 0$, so both summands lie on the ray. Every admissible exponent is a superposition of the Gaussian ray and the exponents of the unit-step profiles of Lemma A.1 ((A.1)). That does not by itself make the unit-step exponents extreme: a family of rays can represent every point of a cone without each of its members being indecomposable, as the points of a square's boundary represent every point of the square. That the unit-step exponents are extreme, and that no other rays are, is outside this paper. Figure 6 draws the cone with the members this paper names.

Two facts that the earlier sections could not supply are here consequences of the classification.

Proposition 7.8 (strict positivity and monotonicity of the admissible exponents). *Let F be of the form (2.3) with $F \not\equiv 0$.*

1. F is nondecreasing on $[0, \infty)$.
2. $F(t\omega) - F(s\omega) > 0$ for all $0 \leq s < t$ and every $\omega \neq 0$. Consequently, in the canonical gauge, no increment $\mu_{s,t}$ with $s < t$ is a lattice law, and $G(\cdot, \omega)$ is strictly increasing for every $\omega \neq 0$.

Clause (2) is what Corollary 5.3 could not give before covariance and Lemma 6.2 gave only for the kernels from the origin: after the classification the exceptional set is empty. So strict monotonicity in scale, and monotonicity in the transform variable, both hold here: for the exponents the axioms select, not for the class ND_s , in which Lemma 2.11 shows they are false.

Proof. (1) For $0 \leq \omega_1 \leq \omega_2$ the difference $F(\omega_2) - F(\omega_1)$ is the value at frequency 1 of the increment $F(\omega_2 \cdot) - F(\omega_1 \cdot)$, which lies in LE_s by Lemma 7.1, (3) implies (1) — at $\omega_1 = 0$ read

as the dilate $F(\omega_2 \cdot)$ itself — and a member of LE_s is nonnegative. So monotonicity is the dilation identity evaluated at one frequency, and no second change of variables is needed.

(2) By the same formula, for $\omega > 0$ — which suffices, F being even —

$$F(t\omega) - F(s\omega) = a(t^2 - s^2)\omega^2 + \int_0^\infty (1 - \cos u) [k(u/(t\omega)) - k(u/(s\omega))] \frac{du}{u},$$

with a nonnegative integrand, reading $k(u/(s\omega)) = k(\infty) = 0$ when $s = 0$. Suppose it vanishes. Then $a = 0$, because $t^2 > s^2$ and $\omega \neq 0$; and the integral vanishes, so its integrand vanishes for almost every $u > 0$, and since $1 - \cos u$ is positive off the countable set $2\pi\mathbb{Z}$, which is null, $k(u/(t\omega)) = k(u/(s\omega))$ for almost every $u > 0$. For $s = 0$ this says $k = 0$ almost everywhere, and $F \equiv 0$.

For $s > 0$ it says $k(\varkappa x) = k(x)$ for almost every $x > 0$, with $\varkappa := t/s > 1$, and the tail integral finishes it. For every $\varepsilon > 0$ the integral $J(\varepsilon) := \int_\varepsilon^\infty k(u) du/u$ is finite: bound $k(u)/u$ by $k(\varepsilon)/\varepsilon$ on $(\varepsilon, 1]$, using that k is nonincreasing, and use $\int_1^\infty k(u) u^{-1} du < \infty$ beyond 1. The change of variables $u = \varkappa x$ carries $J(\varepsilon)$ to $\int_\varepsilon^\infty k(\varkappa x) dx/x$, which is $J(\varepsilon)$ by the almost-everywhere relation — the measure dx/x being invariant under the multiplicative group. Splitting $J(\varepsilon)$ at $\varkappa\varepsilon$ and cancelling the finite tail gives $\int_\varepsilon^{\varkappa\varepsilon} k(u) du/u = 0$; on that interval k is bounded below by $k(\varkappa\varepsilon)$, which must therefore vanish; and since ε was arbitrary, reading the conclusion at $\varepsilon = x/\varkappa$ gives $k(x) = 0$ for every $x > 0$. So $F \equiv 0$ here too, which is excluded. No covering argument and no passage from almost everywhere to everywhere is needed.

In the canonical gauge $g_{s,t} = F(t \cdot) - F(s \cdot)$, so $G(\cdot, \omega)$ is strictly increasing for every $\omega \neq 0$; and if some increment $\mu_{s,t}$ were carried by a lattice $p\mathbb{Z}$ then its cosine transform at $2\pi/p$ would be the integral of the constant 1, hence 1, so $g_{s,t}(2\pi/p) = 0$ at a nonzero frequency, which the clause just proved forbids. $\square$

The regularity of the kernels is classical and is cited rather than reproven. A self-decomposable law on $\mathbb{R}$ that is not a point mass is absolutely continuous ([31], Thm. 27.13, p. 181, stated on $\mathbb{R}^d$ and used for $d = 1$). A law of class L (the classical name for the self-decomposable laws, Definition 2.12) is unimodal ([39], Thm. 1, p. 523), unimodality meaning there that the distribution function is convex to the left of the mode and concave to the right of it.

Proposition 7.9 (regularity of the kernels from the origin). *Let $(\Phi_{s,t})$ satisfy (A1)–(A8) and (ND). Then for every $t > 0$ the kernel $\mu_{0,t}$ is absolutely continuous, and its density is unimodal with mode at the origin.*

Proof. For a nondegenerate self-decomposable law on $\mathbb{R}$ the statement is the two cited facts: absolute continuity is [31], Thm. 27.13, and unimodality is [39], Thm. 1. What remains is the hypothesis check. That $\mu_{0,t}$ is self-decomposable is Proposition 6.6 with Lemma 5.1: the factor law Definition 2.12 asks for at ratio b is the increment $\mu_{t',t}$ at the scale t' with $\chi(t') = \chi(t)/b$, which exists because the gauge is onto $[0, \infty)$. That it is nondegenerate is the last clause of Proposition 6.6, that some frequency is moved. The mode is at the origin because the density is even: the set of modes of a unimodal law is an interval, and for an even density that interval is symmetric about 0, so it contains 0. $\square$

Two things the proposition does not say. It is made for the kernels *from the origin* only: the increments $\mu_{s,t}$ with $s > 0$ are infinitely divisible but not in general self-decomposable, and nothing is claimed about them. And “unimodal with mode at the origin” is a claim about the shape of the density, nonincreasing on $(0, \infty)$ and even, and not about its value there, where the density need not be bounded.

That an exponent vanishing at one nonzero frequency vanishes identically is false for ND_s (Lemma 2.11) but true, monotonicity in ω included, for the exponents the axioms select (Proposition 7.8). The axioms never produce a multimodal kernel from the origin. No truncation of

the Gaussian is admissible, for a different reason: the transform of an even density carried by $[-R, R]$ with $p(R-) > 0$ has the leading term $2p(R-)\sin(R\omega)/\omega$ as $\omega \rightarrow \infty$, by integration by parts, so it changes sign, which Lemma 4.2 forbids; or, more generally, an infinitely divisible law that is not a point mass has unbounded support ([31], Cor. 24.4, p. 149).

Remark 7.10 (the scale evolution, in one line). In the canonical gauge $\hat{\phi}_t(\omega) = e^{-F(t\omega)}$, so on the Fourier side the family solves $\partial_t \hat{u} = -t^{-1}B(t\omega) \hat{u}$ with $B = \omega F'$ the symbol of Lemma 7.1(2). By (7.4) that symbol is the generator of the symmetric Lévy process with Gaussian part $2a$ and jump measure ϖ . For the Gaussian, $B(\omega) = 2a\omega^2$ and the equation is the heat equation in the variance gauge; for the Matérn member, $F = \gamma \log(1 + \omega^2)$ and $B(\omega) = 2\gamma\omega^2/(1 + \omega^2)$, a *bounded* symbol. That the symbol be a polynomial, so that the evolution is a differential equation in x , forces $\varpi = 0$ by (7.4). The equation and the observation are stated here without proof; the evolution in generator form and the locality theorem they lead to are outside this paper.

Remark 7.11 (relation to the causal theory). The characterization is the twin of the causal article's [8], Laplace replaced by Fourier and the drift slot by the Gaussian slot, and three things differ in how it is reached. The analysis direction of Lemma 7.1 is proved here, from the dilation identity and the monotone density on the log-displacement axis, where the causal development cites the classical characterization of self-decomposable Lévy measures and carries it in its trust base. At this point the trust base of the present article is the narrower of the two. The two facts of Proposition 7.8, strict positivity of every increment off the origin and monotonicity of F in the transform variable, were available on the half-line before the gauge argument, from the vanishing lemma and the monotonicity of Bernstein functions, and are here consequences of the classification. This is the last of the three substitutions announced in Remark 6.8. And the continuity axiom in the sufficiency half is verified by a collapsing-increment estimate in place of the $\varepsilon/3$ argument on a compact set, the machine-checked route in both articles. One axiom does not transfer: the additional axiom that removed the drift in the causal theory has a counterpart here, $\lim_{\omega \rightarrow \infty} F(\omega)/\omega^2 = 0$, which says $a = 0$. Unlike its causal twin it would exclude the Gaussian, and this article does not impose it.

8 Concluding remarks

This paper weakened one axiom of [28], with two smaller changes recorded in Remark 3.2. Recursivity, the one-parameter semigroup in scale, carries the tacit assumption that an increment depends on the scale difference alone, the stationarity of the increments in the scale parameter; it was replaced by the two-parameter hemigroup. Under the remaining axioms the admissible measurement families on the line are exactly the symmetric self-decomposable laws, each determined by a Gaussian coefficient and a nonincreasing displacement profile, with the gauge and the exponent unique up to one normalization (Theorem 7.3). The semigroup case is the symmetric stable family with $\alpha \leq 2$ (Corollary 7.4), so the heavy tails of the stable members below $\alpha = 2$ are a consequence of stationarity and not of covariance. Positivity is the narrowing axiom: without it the hemigroup axioms admit an infinite-dimensional affine family through the Gaussian (Proposition 3.4, whose construction is the one step of this paper not machine-checked), and with it the class is a convex cone in which the Gaussian scale space is the extreme ray without jumps. The kernels from the origin are absolutely continuous and unimodal (Proposition 7.9). No truncation of the Gaussian is admissible, since its transform changes sign (Lemma 4.2).

On scope: this paper delivers the classification on the line, in the L^1 signal model, and nothing more. It makes no empirical claims about natural signals. The geometry of the cone, the families at its corners, the scale evolution in generator form, locality, non-creation of local extrema, the exact implementation of the cascade, higher dimension and the discrete covariance strata are outside it.

In relation to the axiom systems of §1: the Gaussian axiomatics, from Iijima's conditions [18, 35] to the non-creation and non-enhancement principles [21, 24], keep the semigroup and add a requirement on how structure may change across scale. Within the class characterized here, the semigroup selects the stable slice and such a requirement selects its end $\alpha = 2$. The extended class of [28] and the α -scale spaces of [6] are that slice, and the Poisson scale space of [11] is its member $\alpha = 1$. The Bessel scale space of [5], composed along the range, is the variance-gamma member, with jumps of every size and every moment finite. The time-causal classification of [8] is the corresponding theorem under the Laplace transform, with a drift term where the Gaussian term is here; the axiom that differs, causality there and reflection symmetry here, is what excludes the Gaussian on the half-line and admits it on the line.

One methodological remark. Several of the proofs printed above are not the ones first written, and in each case the machine-checked route is shorter or assumes less. The null-array limit of Theorem 5.2 extracts the representation of an increment through a truncation inequality proved in a few lines, and never uses the closure of the class under pointwise limits. The analysis direction of Lemma 7.1 first argued through the convexity of a tail function and a right derivative. The checked proof defines the profile as a supremum of dyadic difference quotients, which is nonincreasing by construction, and uses no convexity at all. The continuity axiom in the sufficiency half of Theorem 7.3 was first verified by an $\varepsilon/3$ argument on a compact set carrying most of the mass. The checked proof bounds the distance of a convolution from its argument by the average of the modulus of continuity of translation, and needs no compact set and no density argument. The strict positivity of the admissible exponents (Proposition 7.8) first covered the half-line by intervals and passed from almost everywhere to everywhere. The checked proof reads a dilation-invariant tail integral at one point. And the bound $\alpha \leq 2$ of the semigroup case is the three-line quadratic growth estimate rather than a cited one. The net effect is that the characterization rests on the two cited facts of §1.1, and that the printed proofs are the verified ones. The two further clauses the classical route uses, the closure of the class under pointwise limits and the existence half of the Lévy–Khintchine representation, are used nowhere.

Four questions are left open. First, the construction in Proposition 3.4 (the signed family built from an even integrable transform) is the one step of this paper that is not machine-checked. A route through the convolution operators has been estimated and not carried out. Second, the non-positive hemigroup class is shown to admit no description in the form of the theorem, and it is not described: what it contains beyond the affine families through the admissible exponents is not known. Third, the closed form of the Matérn member's kernel is cited in §3 and not claimed, and a self-contained derivation of the densities of the members with a completely monotone profile would make it a statement of the paper. Fourth, the classification is on the line. Which of it survives rotation invariance in dimension d , and which of it survives covariance under a discrete group of dilations only, are the questions the hemigroup form of the axioms raises next. The probability classes that would answer them are known: the self-decomposable laws and self-similar additive processes in $\mathbb{R}^d$ ([30]), and for a discrete group the semi-selfdecomposable laws and semi-selfsimilar processes of Maejima and Sato ([27], Def. 1, p. 348, and Thm. 6, p. 360). What is open is the derivation from measurement axioms, and its formalization.

A The Cin rays and the Matérn exponent

Two computations that the body of the paper uses are collected here: the exponents of the unit-step profiles, which the proof of Lemma 7.1 uses through the tail measure they attach to a nonincreasing profile, and the exponent of the Matérn member of §3. Both belong to material outside this paper (the geometry of the admissible cone and the families at its corners) and are stated here in the form the body uses and in no more generality. Both are machine-checked and

rest on Lean core alone.

Lemma A.1 (the Cin rays and the superposition). *Write $\text{Cin}(z) := \int_0^z (1 - \cos v) v^{-1} dv$ for $z \geq 0$, extended evenly to $\mathbb{R}$.*

1. (The rays.) *For $\tau > 0$ the profile $k = \mathbf{1}_{(0,\tau)}$ is admissible, with Gaussian coefficient 0, and its exponent is $\omega \mapsto \text{Cin}(\tau\omega)$. The function Cin is even and nondecreasing on $[0, \infty)$, with $\text{Cin}(z) \leq z^2/4$ and $\text{Cin}(z) = \frac{1}{4}z^2 + O(z^4)$ as $z \rightarrow 0$; and there is a finite constant C with $\text{Cin}(z) = \log z + C + O(1/z)$ as $z \rightarrow \infty$.*
2. (The superposition.) *Every admissible F , with Gaussian coefficient a and profile k , satisfies*

$$F(\omega) = a\omega^2 + \int_{(0,\infty)} \text{Cin}(\tau\omega) \varpi(d\tau), \quad \omega \in \mathbb{R}, \quad (\text{A.1})$$

where ϖ is any positive measure on $(0, \infty)$ with $\varpi((x, \infty)) = k(x)$ for almost every $x > 0$; such a ϖ exists.

Proof. (1) Evenness of Cin needs no stipulation: the integrand $v \mapsto (1 - \cos v)/v$ is odd, so the integral from 0 is even, and $\text{Cin}(-z) = \text{Cin}(z)$. Monotonicity on $[0, \infty)$ is the positivity of that integrand there. From $1 - \cos v \leq v^2/2$ we get $\text{Cin}(z) \leq \int_0^z v/2 dv = z^2/4$ for $z \geq 0$, and for $z < 0$ by evenness, so the bound holds on $\mathbb{R}$. For the expansion at the origin, $|\cos v - (1 - v^2/2)| \leq \frac{5}{96}v^4$ for $|v| \leq 1$ gives $|(1 - \cos v)/v - v/2| \leq \frac{5}{96}v^3$ on $(0, 1]$, and integrating over $[0, z]$,

$$\left| \text{Cin}(z) - \frac{1}{4}z^2 \right| \leq \frac{5}{96}z^4, \quad |z| \leq 1,$$

which is the asserted $\text{Cin}(z) = \frac{1}{4}z^2 + O(z^4)$. For the expansion at infinity, split at $v = 1$:

$$\text{Cin}(z) = \text{Cin}(1) + \log z - \int_1^z \frac{\cos v}{v} dv, \quad z \geq 1,$$

using $(1 - \cos v)/v = v^{-1} - v^{-1} \cos v$ on $[1, z]$; and integrating by parts against the primitive $\sin v/v$, $\int_1^z v^{-1} \cos v dv = \sin z/z - \sin 1 + \int_1^z v^{-2} \sin v dv$. The last integrand is dominated by v^{-2} , so $\int_1^\infty v^{-2} \sin v dv$ converges absolutely and, with

$$C := \text{Cin}(1) + \sin 1 - \int_1^\infty \frac{\sin v}{v^2} dv,$$

one gets $\text{Cin}(z) - \log z - C = -\sin z/z + \int_z^\infty v^{-2} \sin v dv$, whence $|\text{Cin}(z) - \log z - C| \leq 2/z$ for $z \geq 1$. That is the expansion, with an explicit constant in the error.

For the ray, $k = \mathbf{1}_{(0,\tau)}$ is nonnegative and nonincreasing on $(0, \infty)$ and satisfies both integrability conditions of Lemma 2.15 at once, being bounded by 1 and supported in $(0, \tau)$: $\int_0^1 x k(x) dx \leq \frac{1}{2}$ and $\int_1^\infty k(x) x^{-1} dx \leq \tau$. So $(0, k)$ is a pair of the form (2.3), and its exponent is $\int_0^\tau (1 - \cos \omega x) x^{-1} dx = \text{Cin}(\tau\omega)$ by the substitution $v = \omega x$, valid for every real ω because both sides vanish at $\omega = 0$ and Cin is even.

(2) A positive measure ϖ on $(0, \infty)$ with $\varpi((x, \infty)) = k(x)$ for almost every $x > 0$ is the image of Lebesgue measure on $(0, \infty)$ under the generalised inverse $y \mapsto \sup\{u > 0 : k(u) > y\}$, restricted to $(0, \infty)$ to discard the atom the transform leaves at the origin when k is bounded. Only two properties of k are used: it is nonincreasing, and it tends to 0 at infinity — and the second is forced rather than assumed, since k nonincreasing and bounded below by a positive constant would make $\int_1^\infty k(x) x^{-1} dx$ diverge like the harmonic integral. What comes out is the sandwich $k(x+) \leq \varpi((x, \infty)) \leq k(x)$, whose two sides agree off the countably many discontinuities of k ; that is why the tail identity is asserted almost everywhere and not everywhere, and a right-continuous version of k is neither available nor needed.

Given such a ϖ , (A.1) is the layer-cake exchange

$$\int_{(0,\infty)} \text{Cin}(\tau\omega) \varpi(d\tau) = \int_0^\infty \frac{1 - \cos \omega x}{x} \varpi((x, \infty)) dx = \int_0^\infty (1 - \cos \omega x) k(x) \frac{dx}{x},$$

in which $\text{Cin}(\tau\omega)$ is read as $\int_0^\tau (1 - \cos \omega x) x^{-1} dx$ and the exchange is Tonelli, for which ϖ is σ -finite: the tail measure of an unbounded profile puts infinite mass near the origin, but its restriction to $[\varepsilon, \infty)$ is finite for every $\varepsilon > 0$. Adding $a\omega^2$ gives (A.1). $\square$

Remark A.2 (the constant). The constant of Lemma A.1(1) is Euler's constant: $\text{Cin}(z) = \gamma_E + \log z - \text{Ci}(z)$ with Ci the cosine integral and $\text{Ci}(z) = O(1/z)$ ([1], §6.2). The identification is not used anywhere in this paper, which is why Lemma A.1 asserts only that the constant is finite.

(A.1) writes every admissible exponent as a superposition of the Gaussian ray and the unit-step exponents. That the unit-step exponents are extreme rays of the cone does not follow from the superposition and is outside this paper; only the two clauses above are used in it.

In clause (3) of the next proposition B denotes standard Brownian motion, taken independent of the random time T ; the letter is the statement's, and it is not the symbol $B = \omega F'$ of Lemma 7.1.

Proposition A.3 (the Matérn family: exponent, transforms, moments). *For $\gamma > 0$:*

1. (Exponent.) $F(\omega) = \gamma \log(1 + \omega^2)$ is admissible, with Gaussian coefficient $a = 0$ and folded profile $k(x) = 2\gamma e^{-x}$; the profile at range θ is $k(x) = 2\gamma e^{-x/\theta}$, with exponent $\gamma \log(1 + \theta^2 \omega^2)$.
2. (Transforms.) $e^{-F(t\omega)} = (1 + t^2 \omega^2)^{-\gamma}$, and the increments have transfer functions $((1 + s^2 \omega^2)/(1 + t^2 \omega^2))^\gamma$, rational in ω for integer γ .
3. (Moments.) Every moment is finite: $X_t \stackrel{d}{=} B_T$ with T Gamma-distributed of shape γ and scale $2t^2$, so $\mathbb{E}X_t^{2n} = (2n - 1)!! (2t^2)^n \Gamma(\gamma + n)/\Gamma(\gamma)$, and in particular $\mathbb{E}X_t^2 = 2\gamma t^2$.

Finiteness of every moment follows from the even moments by $|x|^n \leq 1 + x^{2n}$, and the Gaussian mixture is constructed rather than cited.

Proof. (1) Fix $\theta > 0$ and write $p = 1/\theta$. The integrand $(1 - \cos \omega x) e^{-px} x^{-1}$ is nonnegative on $(0, \infty)$ and, since $1 - \cos u \leq u^2/2$, is bounded there by $\frac{1}{2} \omega^2 x e^{-px}$, so it is integrable for every ω ; its derivative in ω is $\sin(\omega x) e^{-px}$, whose modulus is at most e^{-px} uniformly in ω . Differentiation under the integral sign is therefore legitimate at every ω with one and the same dominating function, and gives

$$\frac{d}{d\omega} \int_0^\infty (1 - \cos \omega x) e^{-px} \frac{dx}{x} = \int_0^\infty \sin(\omega x) e^{-px} dx = \frac{\omega}{p^2 + \omega^2},$$

which is also the derivative of $\frac{1}{2} \log(1 + \omega^2/p^2)$. The two functions of ω therefore differ by a constant, and both vanish at $\omega = 0$, so

$$\int_0^\infty (1 - \cos \omega x) e^{-x/\theta} \frac{dx}{x} = \frac{1}{2} \log(1 + \theta^2 \omega^2).$$

Hence $F(\omega) = \gamma \log(1 + \theta^2 \omega^2)$ is of the form (2.3) with $a = 0$ and $k(x) = 2\gamma e^{-x/\theta}$, which is non-negative and nonincreasing on $(0, \infty)$. Its two integrability conditions are the two dominating functions above read on the two halves of the range: $\int_0^1 x k(x) dx \leq 2\gamma$ and $\int_1^\infty k(x) x^{-1} dx \leq 2\gamma \int_1^\infty e^{-x/\theta} dx < \infty$. Exhibiting the pair $(0, 2\gamma e^{-1/\theta})$ is what admissibility asks for, so the argument stops there; the range θ is carried by the computation and needs no separate dilation step.

(2) is (1) read through the similarity form of Theorem 7.3: the increment transfer function is the ratio of the endpoint transforms, the denominators being nonvanishing.

(3) Let $\Gamma_{\gamma, 2t^2}$ be the Gamma law of shape γ and scale $2t^2$, that is the law on $(0, \infty)$ with density $u^{\gamma-1}e^{-u/2t^2}/(\Gamma(\gamma)(2t^2)^\gamma)$. Its Laplace transform is $\int_0^\infty e^{-\sigma u} \Gamma_{\gamma, 2t^2}(du) = (1 + 2t^2\sigma)^{-\gamma}$ for $\sigma \geq 0$, one Gamma integral; so the Gaussian mixture $\int_0^\infty g_u \Gamma_{\gamma, 2t^2}(du)$ — the law of B_T with T so distributed — has cosine transform $(1 + 2t^2 \cdot \omega^2/2)^{-\gamma} = (1 + t^2\omega^2)^{-\gamma}$, which by (2) is the transform of the kernel at scale t . Both are symmetric probability measures, so Proposition 2.7 identifies them and $X_t \stackrel{d}{=} B_T$. Conditioning on T , $\mathbb{E}X_t^{2n} = \mathbb{E}[T^n] \mathbb{E}[B_1^{2n}] = (2n-1)!! \mathbb{E}T^n$ with $\mathbb{E}[B_v^{2n}] = (2n-1)!! v^n$ and $\mathbb{E}T^n = (2t^2)^n \Gamma(\gamma+n)/\Gamma(\gamma)$; the two are the same Gamma integral read at half-integer and at integer argument. Every moment is then finite, since $|x|^n \leq 1 + x^{2n}$, and $n = 1$ with $\Gamma(\gamma+1) = \gamma\Gamma(\gamma)$ gives $\mathbb{E}X_t^2 = 2\gamma t^2$.

Remark. Finiteness of every moment also follows from the moment criterion for infinitely divisible laws — $\mathbb{E}|X|^n < \infty$ exactly when the Lévy measure integrates $|x|^n$ off a neighbourhood of the origin ([31], Thm. 25.3, p. 159) — the profile decaying exponentially. That route is machine-checked too, uses that cited fact, and is independent of this one; it does not reach the general even moment. $\square$

Author contribution and use of AI

This article was written in close collaboration with generative AI, and the collaboration went beyond editing assistance at every stage. I write this section in the first person because it is my own declaration; the article itself is written in the neutral third person.

Roles. The research programme and the question are mine. The axioms of [28] were the template for my semigroup theory of temporal scale space [7], and after the time-causal article had been drafted I asked what the hemigroup weakening would give on the line, naming that paper as the point of departure. Every decision of scope, method and policy is mine: the method of drafting, one section per exchange in dialogue; the notation; the reference policy toward the companion article; the split of the material into three modules and the decision to release this one first; the cut of this version at the characterization theorem; the rule that the proof of record follows the machine-checked route; and the voice of the text. I reviewed the article section by section and set the scope of every claim.

The prose and the code were written by AI agents throughout, in dialogue with me; I worked almost entirely from the conversation rather than from the editor. The Lean 4 development is theirs essentially in full, and none of the proof terms are mine. So is mathematical content: the results this article rests on were identified by an agent on reading the source, before any plan existed, and during the work the agents found and repaired errors in the mathematics of record that no human had seen. The technical decisions of the formalization were, for the most part, the agents' recommendations approved by me rather than my own drafts; I read each with its consequences and followed all of them.

Verification and review. What is machine-checked, and the axioms on which the verified development rests, are set out in §1.1. Review by agents that had no part in the writing was a working method throughout: the statement skeleton before any proof was attempted, the finished development against the article in a fidelity review, and the sections blind against the writing standard. Before the final revision the manuscript was read by AI systems of two other vendors, in two rounds: a referee-style review returning a verdict of major revision and a presentation review, then, on the revised text, a second presentation review and a referee report the reviewing system left unfinished. Every finding was verified against the text and the formal development before it was acted on, and the errors the first referee report found, in

the one construction that is not machine-checked and in claims stated beyond their results, are corrected in this version. The reviews are archived verbatim in the repository beside the plan that records the disposition of each finding. Beyond me, no human reader has yet reviewed the article.

The record. The sessions were archived as the work went on, so the division of labour can be stated rather than estimated, and it is stated, with its dates and its caveats, in the repository's process note (`notes/PROCESS.md`), together with the chronology of the work and the places where the archive corrected my own recollection of it. The short form: the sustained work took eight days, the agents' words outnumber mine by about twenty-two to one, which is the ratio of the companion article, and the difference from that article is that the formalization ran as parallel agents with me approving their recommendations. Those figures are lower bounds. I would not, unaided and as a spare-time activity, have written this article.

Responsibility. I directed the work throughout, have satisfied myself of the mathematical content, and have checked every citation. Responsibility for this article, including for its errors, is mine alone; the validity of the machine-checked results depends on the Lean kernel and not on how, or by whom, the proof terms were produced.

References

- [1] NIST digital library of mathematical functions. <https://dlmf.nist.gov/>, 2026. Release 1.2.7 of 2026-06-15.
- [2] Luis Alvarez, Frédéric Guichard, Pierre-Louis Lions, and Jean-Michel Morel. Axioms and fundamental equations of image processing. *Archive for Rational Mechanics and Analysis*, 123(3):199–257, 1993.
- [3] Jean Babaud, Andrew P. Witkin, Michel Baudin, and Richard O. Duda. Uniqueness of the Gaussian kernel for scale-space filtering. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 8(1):26–33, 1986.
- [4] Christian Berg and Gunnar Forst. *Potential Theory on Locally Compact Abelian Groups*, volume 87 of *Ergebnisse der Mathematik und ihrer Grenzgebiete*. Springer, Berlin, Heidelberg, 1975.
- [5] Bernhard Burgeth, Stephan Didas, and Joachim Weickert. The Bessel scale-space. In *Deep Structure, Singularities, and Computer Vision*, volume 3753 of *Lecture Notes in Computer Science*, pages 84–95. Springer, Berlin, Heidelberg, 2005.
- [6] Remco Duits, Luc Florack, Jan de Graaf, and Bart ter Haar Romeny. On the axioms of scale space theory. *Journal of Mathematical Imaging and Vision*, 20(3):267–298, 2004.
- [7] Daniel Fagerström. Temporal scale spaces. *International Journal of Computer Vision*, 64(2-3):97–106, 2005.
- [8] Daniel Fagerström. Time-causal scale space from hemigroup axioms: Characterization of the kernels, September 2026. Version 1.0.0.
- [9] William Feller. *An Introduction to Probability Theory and Its Applications, Volume II*. John Wiley & Sons, New York, 2nd edition, 1971.
- [10] M. Felsberg and G. Sommer. The Monogenic Scale-Space: A Unifying Approach to Phase-Based Image Processing in Scale-Space. *Journal of Mathematical Imaging and Vision*, 21(1-2):5–26, 2004.

- [11] Michael Felsberg and Gerald Sommer. Scale adaptive filtering derived from the Laplace equation. In *Pattern Recognition*, volume 2191 of *Lecture Notes in Computer Science*, pages 124–131. Springer, Berlin, Heidelberg, 2001.
- [12] Adrian Fischer, Robert E. Gaunt, and Andrey Sarantsev. The variance-gamma distribution: A review. *Statistical Science*, 40(2), 2025.
- [13] L. M. J. Florack. *Image Structure*. Kluwer Academic Publishers, Dordrecht, 1997.
- [14] Luc M. J. Florack, Bart M. ter Haar Romeny, Jan J. Koenderink, and Max A. Viergever. Scale and the differential structure of images. *Image and Vision Computing*, 10(6):376–388, 1992.
- [15] Peter Guttorp and Tilmann Gneiting. Studies in the history of probability and statistics XLIX: On the Matérn correlation family. *Biometrika*, 93(4):989–995, 2006.
- [16] Wilfried Hazod and Eberhard Siebert. *Stable Probability Measures on Euclidean Spaces and on Locally Compact Groups*. Number 531 in Mathematics and Its Applications. Kluwer, Dordrecht, 2001.
- [17] Herbert Heyer. *Probability Measures on Locally Compact Groups*. Number 94 in Ergebnisse der Mathematik und ihrer Grenzgebiete. Springer, Berlin, Heidelberg, 1977.
- [18] Taizo Iijima. Basic theory on normalization of pattern (in case of typical one-dimensional pattern). *Bulletin of the Electrotechnical Laboratory*, 26:368–388, 1962. In Japanese.
- [19] Niels Jacob. *Pseudo Differential Operators and Markov Processes, Volume I: Fourier Analysis and Semigroups*. Imperial College Press, London, 2001.
- [20] Zbigniew J. Jurek and Wim Vervaat. An integral representation for selfdecomposable Banach space valued random variables. *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete*, 62(2):247–262, 1983.
- [21] Jan J. Koenderink. The structure of images. *Biological Cybernetics*, 50:363–370, 1984.
- [22] John Lamperti. Semi-stable stochastic processes. *Transactions of the American Mathematical Society*, 104(1):62–78, 1962.
- [23] Tony Lindeberg. Scale-space for discrete signals. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 12(3):234–254, 1990.
- [24] Tony Lindeberg. *Scale-Space Theory in Computer Vision*. Kluwer Academic Publishers, Dordrecht, 1994.
- [25] Tony Lindeberg. Generalized Gaussian scale-space axiomatics comprising linear scale-space, affine scale-space and spatio-temporal scale-space. *Journal of Mathematical Imaging and Vision*, 40(1):36–81, 2011.
- [26] Dilip B. Madan and Eugene Seneta. The variance gamma (V.G.) model for share market returns. *The Journal of Business*, 63(4):511–524, 1990.
- [27] Makoto Maejima and Ken-iti Sato. Semi-Selfsimilar Processes. *Journal of Theoretical Probability*, 12(2):347–373, 1999.
- [28] Eric J. Pauwels, Luc J. Van Gool, Peter Fiddelaers, and Theo Moons. An extended class of scale-invariant and recursive scale space filters. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 17(7):691–701, 1995.

- [29] Emilio Porcu, Moreno Bevilacqua, Robert Schaback, and Chris J. Oates. The Matérn model: A journey through statistics, numerical analysis and machine learning. *Statistical Science*, 39(3), 2024.
- [30] Ken-iti Sato. Self-similar processes with independent increments. *Probability Theory and Related Fields*, 89(3):285–300, 1991.
- [31] Ken-iti Sato. *Lévy Processes and Infinitely Divisible Distributions*. Number 68 in Cambridge Studies in Advanced Mathematics. Cambridge University Press, Cambridge, 1999.
- [32] René L. Schilling, Renming Song, and Zoran Vondraček. *Bernstein Functions: Theory and Applications*. De Gruyter, Berlin, 2nd edition, 2012.
- [33] Eberhard Siebert. Continuous hemigroups of probability measures on a Lie group. In Herbert Heyer, editor, *Probability Measures on Groups*, volume 928 of *Lecture Notes in Mathematics*, pages 362–402. Springer, Berlin, Heidelberg, 1982.
- [34] Fred W. Steutel and Klaas van Harn. *Infinite Divisibility of Probability Distributions on the Real Line*. Number 259 in Monographs and Textbooks in Pure and Applied Mathematics. Marcel Dekker, New York, 2004.
- [35] Joachim Weickert, Seiji Ishikawa, and Atsushi Imiya. Linear scale-space has first been proposed in Japan. *Journal of Mathematical Imaging and Vision*, 10(3):237–252, 1999.
- [36] James G. Wendel. Left centralizers and isomorphisms of group algebras. *Pacific Journal of Mathematics*, 2(2):251–261, 1952.
- [37] Peter Whittle. On stationary processes in the plane. *Biometrika*, 41(3-4):434–449, 1954.
- [38] Andrew P. Witkin. Scale-space filtering. In *Proceedings of the Eighth International Joint Conference on Artificial Intelligence*, pages 1019–1022, San Francisco, CA, 1983. Morgan Kaufmann.
- [39] Makoto Yamazato. Unimodality of infinitely divisible distribution functions of class L . *The Annals of Probability*, 6(4):523–531, 1978.
- [40] Alan L. Yuille and Tomaso Poggio. Scaling theorems for zero crossings. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 8(1):15–25, 1986.